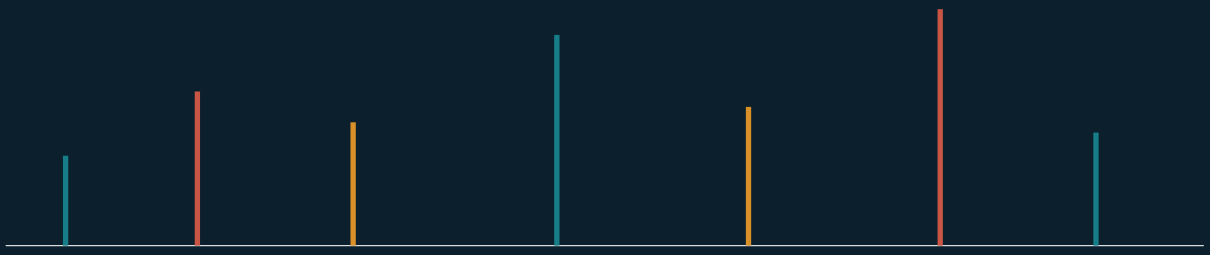
LINE ATLAS

A Field Guide to Astrophysical Spectroscopy



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Preface

Spectroscopy turns small departures from a smooth continuum into measurements of temperature, density, composition, radiation field, motion, and geometry. That power comes with a practical difficulty: the meaning of a line is never contained in its wavelength alone. A transition becomes useful only when its atomic or molecular physics, excitation mechanism, optical depth, environment, instrumental window, and degeneracies are considered together.

Line Atlas is designed as a bridge between a line list and a textbook. Each chapter develops the reasoning behind common measurements, while the reference atlas provides a compact route back to individual transitions. The companion website uses the same catalogue so that wavelength and redshift questions can be explored interactively. Values in the atlas are starting points, not substitutes for the critically evaluated source data and modelling appropriate to a particular analysis.

This first edition emphasizes extragalactic work and the baryon cycle: H II regions, the neutral and molecular interstellar medium, stellar feedback, the circumgalactic and intergalactic media, and active galactic nuclei. Radio, infrared, ultraviolet, optical, and X-ray diagnostics are treated as connected views of the same evolving gas.

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Part I

Reading Light

Chapter 1

How to Read the Line Atlas

Spectroscopy is a translation problem. The detector records counts as a function of coordinate; the astronomer wants temperature, density, chemical composition, radiation field, velocity, geometry, and evolutionary history. A spectral transition connects those descriptions, but never by itself. Its meaning depends on how the relevant state was populated, how photons propagated, which material shared the resolution element, and how the instrument transformed the incoming radiation. This opening chapter establishes the conventions used throughout the atlas and a disciplined way to move from line identification to physical inference.

The central rule

A line is evidence for a transition of a species. Calling it a tracer of a temperature, phase, or process is a model-dependent inference. Preserve the distinction between what the spectrum measures and what a physical model adds.

1.1 The information carried by a line

An isolated feature may provide several nearly independent observables:

1. its centroid identifies a candidate transition and line-of-sight velocity;
2. its integrated flux or equivalent width measures strength relative to a calibrated continuum;
3. its width and profile encode thermal motion, turbulence, rotation, outflow, opacity, unresolved components, and the instrumental response;
4. a resolved doublet or multiplet tests saturation, density, optical depth, and line identification;
5. ratios to other transitions constrain excitation, ionization, attenuation, abundance, and the shape of the radiation field.

These observables do not all imply the same spatial weighting. Emissivity often scales as a product of two densities, while absorption optical depth is linear in the population of the lower state. Bright emission therefore favors dense structures; absorption can detect diffuse material along a long path.

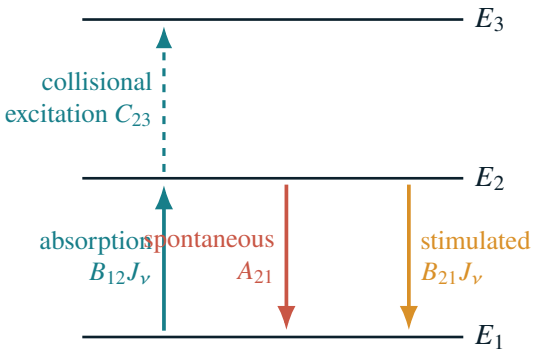
For a transition between levels u and l,

$$h\nu_0 = E_u - E_l, \qquad \lambda_0 = \frac{c}{\nu_0}. \tag{1.1}$$

The rest coordinate is a laboratory property, whereas the observed coordinate is altered by cosmological redshift and peculiar velocity. To first order in v/c, a small velocity displacement from a systemic wavelength obeys

$$\Delta v \simeq c\,\frac{\lambda - \lambda_{\rm sys}}{\lambda_{\rm sys}}. \tag{1.2}$$

For high precision or large velocities one must state and use a relativistic Doppler convention rather than mixing radio, optical, and relativistic velocity definitions.



The same observed transition can be populated by radiative or collisional routes.

Statistical equilibrium balances every route into and out of each level; a line is not assigned a unique formation condition by wavelength alone.

Figure 1.1: The principal bound–bound population and depopulation routes. The Einstein coefficients describe radiative transitions; collisional rates depend on collider density and the velocity-averaged cross section.

1.2 Transition families

The atlas groups lines by mechanism as well as wavelength. The mechanism determines the relevant selection rules, characteristic lifetime, and sensitivity to density or radiation.

The adjective “forbidden” means forbidden by the leading electric-dipole selection rules, not impossible. In a sufficiently low-density plasma the upper state survives long enough to radiate. At high collider density, de-excitation may occur collisionally before a photon escapes; this is the origin of critical density diagnostics. Likewise, a molecular line with a large dipole moment is not automatically a unique dense-gas tracer: radiative trapping, subthermal excitation, abundance, infrared pumping, and beam dilution all matter.

1.3 Reading species and transition labels

Astronomical ion notation counts the neutral atom as stage I, so O VI is O^{5+} rather than O^{6+}. Square brackets denote a forbidden transition, as in [O III] λ5007; a closing bracket alone conventionally marks a semi-forbidden transition, as in C III]. A wavelength following λ is usually in Angstrom unless another unit is shown. Molecular notation instead specifies rotational, vibrational, inversion, or hyperfine quantum numbers, for example CO($J = 1 \rightarrow 0$) or H_2 1–0 S(1).

The ionization threshold needed to create an ion is not the energy of the listed line. The former describes removal of an electron; the latter is the separation of two bound levels. Confusing them can lead to incorrect claims about the hardness of the radiation required to produce a feature. Ion fractions also depend on recombination and charge exchange, not only threshold energy (Osterbrock and Ferland, 2006; Draine, 2011).

1.4 Coordinates, units, and conventions

1.4.1 Vacuum and standard air wavelengths

This edition stores atomic wavelengths in vacuum unless an entry explicitly says otherwise. Familiar ground-based optical names often use standard air values: Hα is commonly called 6562.8 Å in air but is approximately 6564.6 Å in vacuum. The difference is scientifically significant for precision velocities. Conversion depends on a refractive-index prescription and its validity range, so an analysis must record both the input convention and the formula used. NIST distinguishes observed and Ritz wavelengths and documents

Table 1.1: A physical taxonomy of common astronomical transitions. The final column gives a tendency, not an exclusive rule.

Family	Physical origin	Typical examples	Often constrains
Permitted resonance	Electric-dipole transition from a populated lower level	Lyα, C IV, Mg II	column density, covering fraction, kinematics
Recombination	Cascades following radiative recombination	Hα, Hβ, He II, radio recombination lines	ionizing-photon rate, attenuation, emission measure
Forbidden or semi-forbidden	Long-lived state populated collisionally	[O III], [S II], C III]	temperature, density, abundance, excitation source
Fine structure	Splitting within a ground electronic term	[C II] 158 μm, [O I] 63 μm	gas cooling, collider density, PDR or ionized-gas conditions
Rotational or inversion	Molecular rotation or tunnelling between configurations	CO, HCN, NH_3	molecular excitation, mass proxies, temperature, dense gas
Hyperfine	Coupling of nuclear and electronic angular momentum	H I 21 cm, OH 18 cm	neutral gas, spin temperature, magnetic field
Inner-shell or nuclear	Atomic K-shell rearrangement or nuclear decay	Fe Kα, 511 keV, ^{26}Al	irradiation, compact objects, nucleosynthesis

the medium used by its tabulations (Kramida, A. and Ralchenko, Yu. and Reader, J. and NIST ASD Team, 2024).

1.4.2 Use the native coordinate

At centimetre and millimetre wavelengths, observers tune to a rest frequency and should retain it rather than relying on a rounded wavelength conversion. X-ray features are often reported in energy, while optical and ultraviolet features are normally reported in wavelength. The equivalent representations obey

$$E = h\nu = \frac{hc}{\lambda}, \qquad \left|\frac{\Delta E}{E}\right| = \left|\frac{\Delta \nu}{\nu}\right| = \left|\frac{\Delta \lambda}{\lambda}\right| \tag{1.3}$$

for small intervals. Significant digits should follow the authoritative laboratory uncertainty and component definition, not the numerical precision of a unit conversion.

1.4.3 Flux-density transformations

Flux density per unit frequency and per unit wavelength are related by a Jacobian,

$$f_\lambda = f_\nu \left|\frac{\mathrm{d}\nu}{\mathrm{d}\lambda}\right| = f_\nu \frac{c}{\lambda^2}, \qquad \lambda f_\lambda = \nu f_\nu. \tag{1.4}$$

A spectrum that is flat in f_ν is therefore not flat in f_λ. Integrated line flux remains an energy flux after integrating over a consistently defined coordinate, but equivalent widths and continuum slopes must retain their coordinate convention (Rybicki and Lightman, 1979).

1.5 From identification to inference

A credible line identification is a network of consistency checks rather than a nearest-wavelength match. The following sequence is used throughout this book.

Step 1: **Calibrate the coordinate.** Establish the wavelength solution, barycentric or local-standard-of-rest correction, air/vacuum convention, and uncertainty.

Step 2: **Generate candidate transitions.** Use the allowed redshift range, spectral coverage, expected source class, and a current laboratory database.

Step 3: **Demand companions.** Search for doublet partners, other members of the same series, adjacent ionization stages, and physically expected continuum features.

Step 4: **Test the profile.** Compare component velocities and widths after applying the line-spread function; inspect blends, sky residuals, detector defects, and telluric absorption.

Step 5: **Separate measurement from model.** Report flux, equivalent width, centroid, width, covariance, and limits before translating them into abundance, density, temperature, or mass.

Step 6: **Stress-test alternatives.** Vary continuum placement, component count, attenuation law, atomic data, ionization mechanism, geometry, and priors.

A non-detection is not zero

A non-detection constrains a line only through the noise model, assumed profile, searched velocity interval, and selection procedure. Report an upper limit with its confidence convention and assumed width; preserve the likelihood or sensitivity curve when it will enter population inference.

1.6 How each atlas entry is constructed

Each generated entry records an operational rest coordinate, physical mechanism, common environments, useful tracers, an indicative temperature regime, density guidance, diagnostic companions, observing cautions, and source keys. The source hierarchy is deliberate: critically evaluated atomic or molecular databases supply rest data; primary papers and reviews supply interpretation; observatory documents supply current instrumental limits. The prose is a guide to discovery, not a replacement for a live database query or a forward model.

The phrase “temperature regime” is especially easy to overread. A quoted collisional-ionization-equilibrium peak is an orientation point. Photoionization, non-equilibrium cooling, shocks, mixing, charge exchange, optical depth, and an unresolved multiphase sightline can produce the same ion under different kinetic conditions. Similarly, a critical density depends on the collider and temperature; an effective excitation density also depends on opacity and column.

1.7 A compact reporting standard

For a result to be reproducible, a line measurement should state at least:

- transition and component, adopted rest coordinate, source database and version, and air/vacuum convention;

- reference frame, redshift or velocity convention, and systemic-redshift uncertainty;
- instrument, configuration, resolving power, sampling, aperture, and line-spread-function treatment;
- continuum model, integration or fitting window, blend model, and attenuation correction;
- flux, equivalent width, centroid, width, uncertainties and covariance, or the complete upper-limit prescription;
- physical model, radiation field, abundance pattern, atomic or molecular data, priors, and sensitivity tests used for interpretation.

This standard may look longer than a table cell, but most fields belong in a machine-readable companion product. The purpose of the atlas is to make those dependencies visible early, before a memorable line name hardens into an overconfident physical claim.

Chapter 2

Radiation, Matter, and the Making of a Spectral Line

Spectral lines are not labels pasted onto otherwise featureless spectra. They are the observable end point of a chain of physical events: particles are placed in quantum states, photons are created or destroyed, radiation propagates through a structured medium, and an instrument averages the emerging field over angle, wavelength, and time. A line atlas is most useful when every entry can be read in both directions along this chain. From an observed feature one would like to infer density, temperature, abundance, velocity, and radiation field; from a proposed physical model one would like to predict a measurable flux profile.

This chapter establishes the vocabulary needed for both directions. It emphasizes the assumptions hidden in familiar formulae and the combinations of quantities that spectra actually constrain. More complete treatments are given by Rybicki and Lightman (1979), Mihalas (1978), Osterbrock and Ferland (2006), and Draine (2011).

2.1 What a spectrograph measures

The fundamental radiometric quantity is the specific intensity $I_\nu(\boldsymbol{x}, \hat{\boldsymbol{n}}, t)$, defined so that the energy crossing an area element $\mathrm{d}A$ in time $\mathrm{d}t$, within solid angle $\mathrm{d}\Omega$ and frequency interval $\mathrm{d}\nu$, is

$$\mathrm{d}E = I_\nu \cos\theta \, \mathrm{d}A \, \mathrm{d}t \, \mathrm{d}\Omega \, \mathrm{d}\nu. \tag{2.1}$$

Unlike flux, intensity is conserved along a ray in empty Euclidean space. The monochromatic flux is the first angular moment,

$$F_\nu = \int_{4\pi} I_\nu(\hat{\boldsymbol{n}}) \cos\theta \, \mathrm{d}\Omega, \qquad J_\nu = \frac{1}{4\pi} \int_{4\pi} I_\nu \, \mathrm{d}\Omega, \tag{2.2}$$

where J_ν is the mean intensity. An unresolved source at luminosity distance D_L is normally reported through a flux density f_ν or f_λ. The two are not numerically interchangeable:

$$f_\lambda = f_\nu \left| \frac{\mathrm{d}\nu}{\mathrm{d}\lambda} \right| = f_\nu \frac{c}{\lambda^2}, \qquad \lambda f_\lambda = \nu f_\nu. \tag{2.3}$$

This Jacobian is the reason that a spectrum flat in f_ν slopes as λ^{-2} in f_λ.

For a source at redshift z, $\nu_{\mathrm{obs}} = \nu_{\mathrm{em}}/(1+z)$ and $\lambda_{\mathrm{obs}} = (1+z)\lambda_{\mathrm{em}}$. Specific intensity obeys the Liouville invariant I_ν/ν^3 along a null geodesic in the absence of interactions. Consequently surface brightness dims as $(1+z)^{-4}$ after integrating over frequency. Line luminosity and observed integrated line flux are related by

$$F_{\mathrm{line}} = \int f_\lambda \, \mathrm{d}\lambda_{\mathrm{obs}} = \frac{L_{\mathrm{line}}}{4\pi D_L^2}. \tag{2.4}$$

No extra factor of $(1+z)$ belongs in this bolometric line relation; such factors enter relations between spectral densities or between velocity-integrated radio brightness and luminosity.

Definition: line profile. The line profile $\phi(\nu)$ is a probability density in frequency, normalized as $\int \phi(\nu)\,\mathrm{d}\nu = 1$. It therefore has units of inverse frequency. Omitting this unit is a common route to dimensionally incorrect opacity formulae.

2.2 The transfer equation and its formal solution

Along path length s, emission adds radiation and extinction removes it:

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = j_\nu - \alpha_\nu I_\nu, \tag{2.5}$$

where j_ν is the emission coefficient per steradian and α_ν is the extinction coefficient. True absorption and scattering may both contribute to α_ν; only true absorption transfers radiant energy to internal energy. Define optical depth increasing away from the observer by $\mathrm{d}\tau_\nu = \alpha_\nu \mathrm{d}s$ and source function $S_\nu = j_\nu/\alpha_\nu$. For a slab of total optical depth τ_ν, the emergent intensity is

$$I_\nu(0) = I_\nu(\tau_\nu)e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t)e^{-t}\,\mathrm{d}t. \tag{2.6}$$

For a homogeneous slab this becomes

$$I_{\nu,\mathrm{out}} = I_{\nu,\mathrm{bg}}e^{-\tau_\nu} + S_\nu\left(1 - e^{-\tau_\nu}\right). \tag{2.7}$$

Equation (2.7) is one of the most useful diagnostic equations in spectroscopy. A feature appears in emission when the line source function exceeds the incident background intensity and in absorption when it is smaller. Thus "emission line" and "absorption line" describe contrast, not an intrinsic property of a transition.

When the material is in local thermodynamic equilibrium (LTE), detailed balance gives $S_\nu = B_\nu(T)$, the Planck function. LTE requires collisional processes to establish local level populations, but does not require the radiation field to be a blackbody. In a stratified, optically thick atmosphere with a slowly varying source function, the Eddington–Barbier approximation gives $I_\nu(0,\mu) \simeq S_\nu(\tau_\nu = \mu)$, where $\mu = \cos\theta$. We therefore see different physical depths at different frequencies: high-opacity line cores form closer to the surface than adjacent continuum.

2.2.1 Absorption, stimulated emission, and spontaneous emission

Consider lower and upper states l and u, populations n_l and n_u, and statistical weights g_l and g_u. Einstein coefficients describe absorption (B_{lu}), stimulated emission (B_{ul}), and spontaneous emission (A_{ul}). Their thermodynamic relations are

$$g_l B_{lu} = g_u B_{ul}, \qquad A_{ul} = \frac{2h\nu_0^3}{c^2}B_{ul}. \tag{2.8}$$

For complete redistribution with the same normalized line profile for absorption and emission,

$$j_\nu^{\mathrm{line}} = \frac{h\nu_0}{4\pi} n_u A_{ul}\phi(\nu), \tag{2.9}$$

$$\alpha_\nu^{\mathrm{line}} = \frac{h\nu_0}{4\pi}\left(n_l B_{lu} - n_u B_{ul}\right)\phi(\nu). \tag{2.10}$$

The negative stimulated-emission term matters at radio wavelengths and in population inversions. It should not be silently discarded merely because spontaneous emission dominates a particular optical application.

The absorption oscillator strength f_{lu} is related to the frequency-integrated cross section by

$$\int \sigma_\nu \, \mathrm{d}\nu = \frac{\pi e^2}{m_e c} f_{lu}, \qquad \sigma_\nu = \frac{\pi e^2}{m_e c} f_{lu} \phi(\nu). \tag{2.11}$$

Oscillator strengths are dimensionless atomic data. Their uncertainty can dominate an abundance measurement even when the spectrum has extremely high signal-to-noise.

2.3 Population equations: LTE and statistical equilibrium

In LTE, excitation follows the Boltzmann relation,

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\!\left[-\frac{E_u - E_l}{kT}\right], \tag{2.12}$$

and adjacent ionization stages follow the Saha equation,

$$\frac{n_{i+1} n_e}{n_i} = \frac{2}{\lambda_e^3} \frac{U_{i+1}(T)}{U_i(T)} \exp\!\left(-\frac{\chi_i}{kT}\right), \qquad \lambda_e = \frac{h}{\sqrt{2\pi m_e kT}}. \tag{2.13}$$

Here U_i is a partition function and χ_i the ionization potential. Nebulae, diffuse clouds, and the circumgalactic medium are generally not in LTE. Their level populations instead satisfy statistical equilibrium,

$$\sum_{j \neq i} n_j P_{ji} = n_i \sum_{j \neq i} P_{ij}, \qquad \sum_i n_i = n_{\mathrm{species}}, \tag{2.14}$$

where each P_{ij} may contain radiative, electron-impact, neutral-impact, recombination, charge-exchange, or chemical rates.

For an optically thin two-level ion excited by electron collisions, neglecting stimulated terms,

$$\frac{n_u}{n_l} = \frac{n_e q_{lu}}{A_{ul} + n_e q_{ul}}, \qquad n_{\mathrm{crit}} = \frac{A_{ul}}{q_{ul}}. \tag{2.15}$$

At $n_e \ll n_{\mathrm{crit}}$, each excitation is usually followed by a photon, so the volume emissivity scales approximately as $n_e n_l$. At $n_e \gg n_{\mathrm{crit}}$, collisions de-excite the upper state and the line no longer grows quadratically. For a multilevel atom the denominator must include every downward collider and radiative branch,

$$n_{\mathrm{crit},u} = \frac{\sum_{l<u} A_{ul}}{\sum_{j \neq u} \sum_c n_c q_{uj}^{(c)} / n_e}, \tag{2.16}$$

or, more transparently, one quotes a collider-specific critical density $n_{\mathrm{crit},c} = \sum A_{ul} / \sum q_{uj}^{(c)}$. Tabulated critical densities are therefore temperature dependent and are not universal constants of a line.

Caveat: critical density is not a hard boundary. Line emissivity changes smoothly around n_{crit}. Radiative trapping lowers the effective spontaneous escape rate, while continuum pumping raises levels without collisions. A measured density-sensitive ratio should be interpreted with a full multilevel model when optical depth, fluorescence, or several colliders matter.

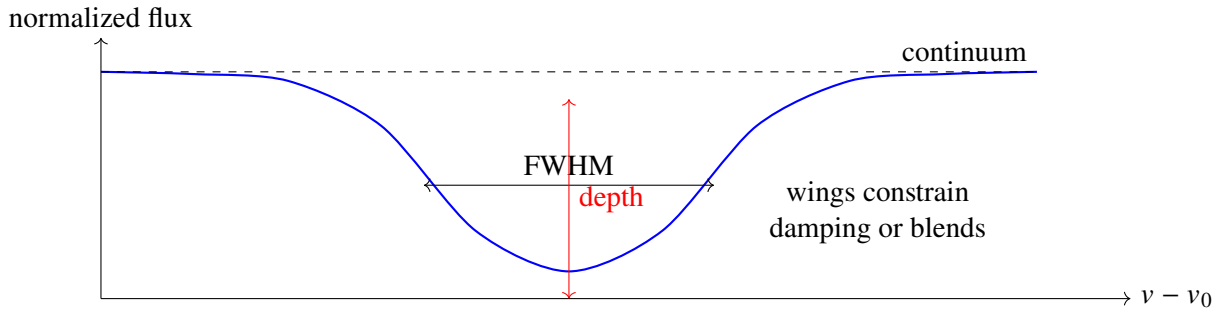


Figure 2.1: Original schematic for a normalized absorption profile. A production version can overlay intrinsic, instrument-convolved, and sampled profiles. The equivalent width is the area between the continuum and profile, not simply depth times FWHM.

2.4 Why lines have width and shape

The natural profile of an isolated damped transition is Lorentzian. Thermal and microturbulent Doppler shifts produce a Gaussian core. Their convolution is the Voigt profile,

$$\phi(\nu) = \frac{H(a,u)}{\sqrt{\pi}\Delta\nu_D}, \qquad u = \frac{\nu - \nu_0}{\Delta\nu_D}, \qquad a = \frac{\Gamma}{4\pi\Delta\nu_D}, \tag{2.17}$$

where Γ is the total damping rate and

$$\Delta\nu_D = \frac{\nu_0 b}{c}, \qquad b^2 = \frac{2kT}{m} + b_{\mathrm{nt}}^2. \tag{2.18}$$

The Doppler parameter b is related to a Gaussian velocity dispersion by $b = \sqrt{2}\sigma_v$ and to full width at half maximum by $\mathrm{FWHM} = 2\sqrt{\ln 2}\, b \simeq 1.665 b$. Mixing these conventions can bias temperatures by a factor of two.

If two ions share the same temperature and nonthermal broadening, their measured widths obey

$$b_1^2 - b_2^2 = 2kT\left(\frac{1}{m_1} - \frac{1}{m_2}\right). \tag{2.19}$$

The method is powerful only when the profiles are demonstrably co-spatial and the instrumental line-spread function has been removed. Unresolved component structure usually mimics nonthermal broadening.

Macroscopic velocity gradients produce profiles that cannot be represented by a single Voigt function. Rotation can create a double-horned integrated profile; accelerating outflows can form P-Cygni profiles; self-absorption can split an otherwise single emission peak. The measured spectrum is finally convolved with the instrument line-spread function L and integrated over detector pixels:

$$f_i = \int_{\Delta\lambda_i} \left[f_{\lambda,\mathrm{true}} * L \right](\lambda)\, R_i(\lambda)\, \mathrm{d}\lambda, \tag{2.20}$$

where R_i includes pixel response. Deconvolving noisy data is unstable; forward convolution of a physical model is normally preferable.

2.5 Optical depth, equivalent width, and the curve of growth

For a column density N_l in the lower state, a Doppler-broadened absorption line has central optical depth

$$\tau_0 = \frac{\sqrt{\pi}e^2}{m_e c}\frac{f_{lu}\lambda_0 N_l}{b} = 1.497\times10^{-15}\,\frac{N_l}{\mathrm{cm^{-2}}}\,\frac{f_{lu}(\lambda_0/\text{Å})}{b/(\mathrm{km\,s^{-1}})}. \tag{2.21}$$

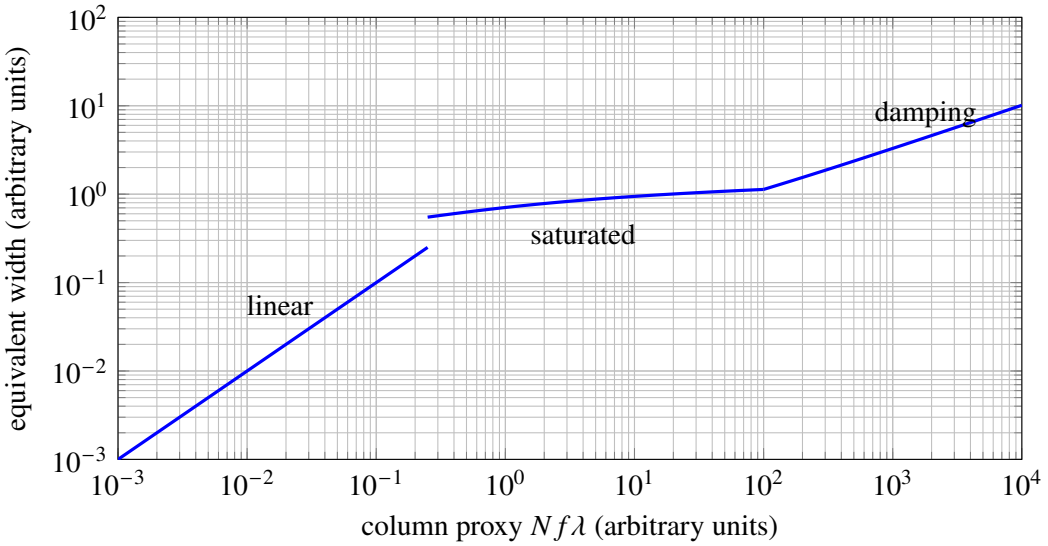


Figure 2.2: Pedagogical curve-of-growth concept generated entirely with PGFPlots. The plotted interpolation is schematic, not a substitute for evaluating a Voigt profile. A final atlas could add families of curves for several Doppler parameters.

The rest-frame equivalent width of an absorption line is

$$W_0 = \int \left(1 - \frac{F_\lambda}{F_c}\right) \mathrm{d}\lambda_{\mathrm{rest}}, \qquad W_{\mathrm{obs}} = (1+z) W_0. \tag{2.22}$$

For optically thin absorption,

$$W_0 = \frac{\pi e^2}{m_e c^2} N_l f_{lu} \lambda_0^2, \tag{2.23}$$

independent of b. As the core saturates, increasing N_l mostly broadens the black portion of the line and W grows only roughly as $b\sqrt{\ln \tau_0}$. At still larger columns, Lorentzian damping wings cause $W \propto N_l^{1/2}$. These are the linear, saturated, and damping regimes of the curve of growth.

2.5.1 Worked example: is a Lyα absorber optically thin?

Take H I Lyα with $f = 0.4164$ and $\lambda_0 = 1215.67$ Å. For $N_{\mathrm{H\,I}} = 10^{14}\,\mathrm{cm^{-2}}$ and $b = 20\,\mathrm{km\,s^{-1}}$, equation (2.21) gives

$$\tau_0 \simeq 1.497 \times 10^{-15} \frac{10^{14} \times 0.4164 \times 1215.67}{20} \simeq 3.8. \tag{2.24}$$

The line is already saturated at its center, even though its neutral-hydrogen column is modest by interstellar standards. Applying the optically thin equivalent-width formula would underestimate $N_{\mathrm{H\,I}}$. Higher Lyman-series transitions, a curve-of-growth analysis, or a forward Voigt-profile fit are needed. A formally excellent single-component fit can still be wrong if narrow components are hidden by the line-spread function.

2.6 Emission measure, surface brightness, and luminosity

For optically thin recombination radiation, the line emissivity integrated over solid angle is often written

$$\epsilon_{ul} = h\nu_{ul} n_e n_{i+1} \alpha_{ul}^{\mathrm{eff}}(T), \qquad L_{ul} = \int_V \epsilon_{ul}\, \mathrm{d}V. \tag{2.25}$$

The effective recombination coefficient includes cascades that feed the upper state. The surface brightness scales with emission measure,

$$\mathrm{EM} = \int n_e n_p \, \mathrm{d}\ell, \qquad I_{ul} = \frac{h\nu_{ul}}{4\pi} \int n_e n_p \alpha_{ul}^{\mathrm{eff}}(T) \, \mathrm{d}\ell. \tag{2.26}$$

Because EM weights density squared, dense clumps dominate emission. Absorption columns weight density linearly. Combining emission and absorption is therefore a way to constrain clumping, but only if the two measurements sample the same gas.

Collisionally excited forbidden lines have low-density emissivity

$$\epsilon_{ul} \simeq n_e n(X^i) q_{lu}(T) h\nu_{ul} \frac{A_{ul}}{\sum_k A_{uk}}, \tag{2.27}$$

with excitation coefficient

$$q_{lu}(T) = \frac{8.629 \times 10^{-6}}{g_l T^{1/2}} \Upsilon_{lu}(T) \exp\!\left(-\frac{\Delta E}{kT}\right) \quad \mathrm{cm^3\ s^{-1}}. \tag{2.28}$$

The effective collision strength Υ varies more gently with temperature than the exponential. This strong temperature sensitivity makes auroral-to-nebular ratios valuable thermometers and also makes inferred abundances sensitive to unresolved temperature structure.

2.6.1 Worked example: a density-sensitive doublet

Suppose two lines arise from metastable levels with different A/q values but close excitation energies. Their ratio can be approximated schematically by

$$R(n_e) = R_0 \frac{1 + n_e/n_{\mathrm{crit},2}}{1 + n_e/n_{\mathrm{crit},1}}. \tag{2.29}$$

At densities below both critical densities, $R \to R_0$ and the diagnostic loses leverage. At densities above both, it approaches another asymptote and again loses leverage. The ratio is informative primarily between the two critical densities. An observer should report an upper or lower limit when a measured ratio lies near an asymptote, rather than quote a spuriously precise density.

2.7 Resonant scattering, escape, and redistribution

Resonance lines such as Lyα, C IV, Mg II, and Na I D may scatter many times before escape. Scattering conserves photon number only in the idealized absence of dust absorption, collisional de-excitation, conversion to other lines, or escape outside the aperture. A simple escape-probability approximation replaces A_{ul} by $\beta(\tau)A_{ul}$, where for a static slab one rough choice is

$$\beta(\tau) \simeq \frac{1 - e^{-\tau}}{\tau}. \tag{2.30}$$

This captures the limiting trends but not detailed frequency diffusion or geometry. At large optical depth, photons escape through frequency excursions into the wings; bulk velocity and dust geometry can then shape the emergent profile more strongly than the intrinsic production spectrum.

Interpretive caution. The centroid of a resonant emission line need not equal the systemic velocity, and its integrated flux need not measure only the production rate. A spatially extended halo may be composed largely of photons created in the central region and scattered into the line of sight. Conversely, a narrow spectroscopic aperture may classify scattered photons as “destroyed” even when they merely emerge elsewhere.

2.8 A hierarchy of modeling assumptions

Line interpretation should state its rung on the following hierarchy:

1. **Optically thin emissivity or absorption.** Photons interact at most once; geometry enters only through an integral of density or density squared.
2. **One-zone statistical equilibrium.** Level populations respond to a specified temperature, density, radiation field, and abundance pattern.
3. **Photoionization or shock structure.** Thermal balance, ionization, chemistry, and attenuation vary through the source.
4. **Radiative transfer in prescribed geometry.** Optical depth, scattering, and velocity fields determine escape.
5. **Radiation hydrodynamics or magnetohydrodynamics.** Radiation changes the gas whose structure in turn changes the radiation.

Increasing sophistication does not automatically improve inference. A complex model with unconstrained geometry can be less informative than a transparent line-ratio estimate. The minimum adequate model is the simplest rung that includes all processes capable of shifting the desired quantity by more than the error budget.

2.9 Checklist for a trustworthy line interpretation

Before assigning a physical meaning to a feature, ask:

- Is the wavelength in vacuum or air, and is the velocity convention stated?
- Is the line resolved relative to the full, possibly non-Gaussian, line-spread function?
- Is the continuum model flexible enough, but not so flexible that it absorbs broad wings?
- Can a second transition reveal saturation, blending, fluorescence, or density sensitivity?
- Are LTE, optical thinness, equilibrium, and a single temperature justified?
- Does the aperture include all scattered or diffuse emission relevant to the physical question?
- Which atomic data set supplies wavelengths, A values, collision strengths, and oscillator strengths, and are their uncertainties propagated?
- Does the quoted uncertainty include calibration, model discrepancy, and component degeneracy in addition to photon noise?

The rest of the atlas repeatedly returns to these questions. A transition becomes a diagnostic only after its population mechanism, transfer regime, and measurement operator have been specified.

Chapter 3

Practical Spectroscopy and Measurement

The most elegant plasma diagnostic is only as reliable as the chain that connects a photon to a number in a table. Raw exposures contain detector signatures, sky and thermal backgrounds, atmospheric absorption, cosmic rays, geometric distortion, and an instrumental line-spread function. Reduction removes or models these effects but also introduces correlations and choices. The final spectrum is not simply the sky spectrum with noise added; it is the output of an estimator.

This chapter presents a practical workflow that applies across long-slit, fiber, echelle, and integral-field spectroscopy. Instrument-specific pipelines remain essential, but the underlying bookkeeping is common. Useful extended discussions include Gray (2005), Howell (2006), and the optimal extraction method of Horne (1986).

3.1 Design the observation around the inference

An observing proposal should begin with the physical comparison that will be made, not with a list of attractive transitions. For each required line, record:

- its observed wavelength range after redshift and expected velocity offsets;
- the continuum or background against which it will be measured;
- expected flux, surface brightness, width, and spatial extent;
- nearby blends, sky lines, telluric bands, detector gaps, and order edges;
- the angular and spectral resolution required to distinguish the alternatives;
- whether the key observable is integrated flux, equivalent width, profile shape, centroid, doublet ratio, or spatial morphology.

The resolving power is

$$R \equiv \frac{\lambda}{\Delta\lambda}, \qquad \Delta v \simeq \frac{c}{R}, \tag{3.1}$$

where $\Delta\lambda$ and Δv must refer to a stated width convention, often the FWHM of the line-spread function (LSF). Pixel sampling and resolution are not the same. Two or more pixels per FWHM avoid gross undersampling, but additional sampling does not recover information absent from the optical LSF.

For an unresolved line measured in background-limited data, signal-to-noise improves approximately as the square root of exposure time. Real observations can depart from this scaling because of read noise, residual flat-field structure, time-variable sky, correlated resampling noise, and systematic background subtraction. Exposure-time calculators should be tested at the surface brightness and extraction aperture of the science target rather than only for a point-source continuum.

3.1.1 Worked example: can a spectrograph separate two components?

Suppose two narrow components differ by $80\,\mathrm{km\,s^{-1}}$. A spectrograph with $R = 3000$ has an instrumental FWHM near $c/R \simeq 100\,\mathrm{km\,s^{-1}}$. The components may broaden or skew the feature, but will not be cleanly resolved. At $R = 10{,}000$, the nominal FWHM is $30\,\mathrm{km\,s^{-1}}$, giving meaningful separation if the signal-to-noise is adequate. This estimate is not a guarantee: the actual LSF can have wavelength-dependent wings, and two components with unequal flux are harder to identify than equal components.

3.2 Calibration frames and detector-level modeling

A simplified raw detector model is

$$D_{ij} = g^{-1}\left[S_{ij} + B_{ij} + K_{ij} + d_{ij}t\right] + b_{ij} + \epsilon_{ij}, \tag{3.2}$$

where S is source photocharge, B sky or thermal background, K scattered light, g gain, b electronic bias, d dark current, and ϵ stochastic read and photon noise. Nonlinearity, persistence, interpixel capacitance, charge transfer, and bad pixels may require additional terms.

Bias or reference-pixel correction removes the electronic zero level. Dark frames estimate thermally generated signal and sometimes hot-pixel behavior. Flat fields measure pixel response and, depending on the instrument, illumination or blaze structure. Arc lamps, etalons, sky lines, or laser combs constrain the mapping from detector position to wavelength. Calibration frames are measurements with their own noise and temporal mismatch; they should not be treated as exact arrays.

For CCD-like data in electrons, a basic variance model is

$$V_{ij} = S_{ij} + B_{ij} + K_{ij} + d_{ij}t + n_{\mathrm{read}}\sigma_{\mathrm{read}}^2, \tag{3.3}$$

with appropriate gain factors if working in analog-to-digital units. Operations such as flat-field division, interpolation, and sky modeling must propagate variance. Cosmic-ray rejection should mark altered samples; replacing a cosmic ray with a local interpolation and assigning ordinary variance falsely creates information.

3.3 Tracing, wavelength calibration, and rectification

Spectral traces curve across detectors and may vary with exposure, fiber, slit position, or echelle order. A trace model should be smooth enough to follow optics without following pixel noise. Wavelength calibration fits known features with a two-dimensional function $\lambda(x, y)$ or an instrument model. Precision should be evaluated using held-out or independent lines, not merely the residuals of lines used in the fit.

Air and vacuum wavelengths must never be mixed silently. Air's refractive index shifts optical wavelengths enough to matter for velocity work. The reference frame must also be stated: topocentric, barycentric, heliocentric, local standard of rest, or another convention. A sign error in the observatory velocity correction can exceed the physical signal being sought.

Several velocity definitions coexist. Relative to a rest frequency ν_0 or wavelength λ_0,

$$v_{\mathrm{radio}} = c\left(1 - \frac{\nu}{\nu_0}\right), \tag{3.4}$$

$$v_{\mathrm{optical}} = c\left(\frac{\lambda - \lambda_0}{\lambda_0}\right), \tag{3.5}$$

$$\beta_{\mathrm{rel}} = \frac{(\lambda/\lambda_0)^2 - 1}{(\lambda/\lambda_0)^2 + 1}. \tag{3.6}$$

The first two are approximations that differ at non-negligible redshift. A catalog column named “velocity” is incomplete without a frame and convention.

Rectification places spectra on a convenient grid but correlates neighboring pixels. Repeated interpolation broadens the effective LSF and compounds covariance. A good pipeline maps samples to the final grid once where practical, or retains native pixel coordinates for forward modeling.

3.4 Sky, thermal background, and telluric transmission

Night-sky emission is structured in wavelength and time. Near-infrared OH lines can be much brighter than the target; small wavelength or LSF mismatches then leave positive-negative residuals. Sky subtraction is a model of the sky observed through the same instrument, not a scalar offset. Nodding, dedicated sky fibers, off-source slit regions, or principal-component models each encode assumptions about spatial and temporal variation.

Atmospheric transmission multiplies the extraterrestrial spectrum. A telluric standard or synthetic atmospheric model can estimate this transmission, but division near saturated bands amplifies noise and cannot reconstruct photons that were not transmitted. Regions of very low transmission should usually be masked or fitted jointly with a telluric model. The atmospheric model and science model should be convolved to the same LSF before comparison.

Thermal emission dominates at sufficiently long wavelengths and varies with telescope and atmosphere temperature. Background subtraction can be the largest term in the error budget for faint extended emission. A local background region may remove real source emission if the object fills the field, while a remote sky exposure may not match the instantaneous background.

3.5 One-dimensional extraction

For detector row i and spatial sample j, let D_{ij} be background-subtracted counts, P_{ij} a normalized spatial profile, and V_{ij} variance. The inverse-variance optimal estimator is

$$\widehat{f_i} = \frac{\sum_j P_{ij} D_{ij}/V_{ij}}{\sum_j P_{ij}^2/V_{ij}}, \qquad \operatorname{Var}(\widehat{f_i}) = \left(\sum_j \frac{P_{ij}^2}{V_{ij}} \right)^{-1} \tag{3.7}$$

under the model assumptions (Horne, 1986). Compared with an unweighted aperture, optimal extraction downweights noisy spatial pixels. It can be biased if the profile is wrong, the object is extended, differential atmospheric refraction moves the centroid, or emission-line morphology differs from continuum morphology.

Extraction aperture is part of the observable. A narrow aperture can maximize point-source signal-to-noise but exclude diffuse line halos. Comparing line fluxes from different seeing, slit widths, or instruments requires an aperture or throughput model. For spatially resolved targets it is often preferable to preserve the data cube and define physical apertures during analysis.

3.6 Flux calibration and response

A spectrophotometric standard constrains the sensitivity function that converts count rate to physical flux density. The response contains telescope area, slit or fiber throughput, optics, detector efficiency, atmospheric extinction, and sometimes an aperture correction. A smooth response model should not fit away real standard-star absorption lines or imprint them on the target.

Absolute calibration and relative line-ratio calibration are different requirements. Lines close in wavelength can have an accurate ratio even when absolute throughput is uncertain. Widely separated lines require accurate

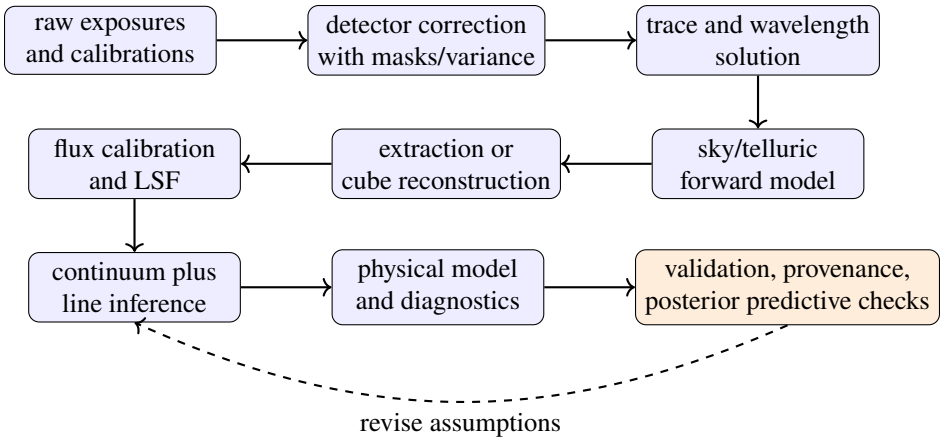


Figure 3.1: Original reduction-to-inference graph. Masks, variance or covariance, LSF, and provenance flow with the spectrum. Validation is a feedback loop rather than a final cosmetic step.

response shape and often an attenuation correction. Simultaneous photometry can rescale spectra affected by clouds or slit loss, but a single broadband point cannot repair wavelength-dependent losses.

For a source with flux density f_λ and response $R(\lambda)$, detected electrons scale as

$$N_e \propto tA \int \frac{\lambda}{hc} f_\lambda(\lambda) R(\lambda)\, \mathrm{d}\lambda. \tag{3.8}$$

The factor $\lambda/(hc)$ converts radiant energy to photon number. Confusing energy-weighted and photon-weighted response definitions produces color terms.

3.7 A transparent analysis graph

Figure 3.1 emphasizes that every product should retain auxiliary information: data-quality masks, propagated uncertainty, wavelength and spatial coordinates, spectral resolution, exposure metadata, and links to calibration versions. A flux array without these is not analysis-ready.

3.8 Continuum modeling

Line measurements are conditional on a continuum model. Possible components include stellar populations and absorption, a nonthermal power law, nebular continuum, dust emission, Fe complexes, scattered light, and residual response structure. Local polynomials are effective over narrow intervals but can absorb broad wings. Physical templates extrapolate more meaningfully but carry population and abundance mismatch.

The continuum and line should be fitted jointly whenever their uncertainties are coupled. Subtracting a best-fit continuum and then treating it as exact understates line-flux uncertainty. For Balmer emission, underlying stellar absorption can be comparable to the nebular equivalent width, especially in weakly star-forming regions. A flexible stellar-continuum model is then not optional.

Continuum placement can be stress-tested by changing windows, polynomial order, or template library and examining the shift in line parameters. These shifts are a model-systematic term, not a reason to select whichever configuration yields the smallest formal error.

3.9 Forward line-profile fitting

A common model for a local spectral interval is

$$m(\lambda) = C(\lambda; \boldsymbol{c}) + \left[\sum_{k=1}^{K} F_k \, p_k(\lambda; v_k, \sigma_k) \right] * L(\lambda), \tag{3.9}$$

sampled and integrated over the same pixels as the data. Here each p_k is normalized to unit area, so F_k is integrated flux. Multiple transitions may share velocity and dispersion; branching ratios may be fixed only when atomic physics, opacity, and blending justify doing so.

With residual vector $\boldsymbol{r} = \boldsymbol{d} - \boldsymbol{m}$ and covariance $\boldsymbol{C}$, the Gaussian log likelihood is

$$-2 \ln \mathcal{L} = \boldsymbol{r}^{\mathsf{T}} \boldsymbol{C}^{-1} \boldsymbol{r} + \ln |\boldsymbol{C}| + \text{constant}. \tag{3.10}$$

Ignoring off-diagonal covariance after resampling can underestimate uncertainties and make broad low-amplitude residuals appear significant. When the full covariance is too costly, an empirical correction from blank-sky apertures or block bootstrap is better than pretending pixels are independent.

For a Gaussian line in wavelength,

$$p(\lambda) = \frac{1}{\sqrt{2\pi}\sigma_\lambda} \exp\left[-\frac{(\lambda - \lambda_c)^2}{2\sigma_\lambda^2} \right], \quad \text{FWHM} = 2\sqrt{2 \ln 2}\, \sigma_\lambda. \tag{3.11}$$

If both source and LSF are Gaussian, $\sigma_{\text{obs}}^2 = \sigma_{\text{int}}^2 + \sigma_{\text{inst}}^2$. Subtraction in quadrature is invalid for non-Gaussian LSFs, unresolved lines near the boundary $\sigma_{\text{int}} = 0$, or blends. Forward convolution handles these cases more honestly.

3.9.1 Model complexity and components

Adding Gaussian components will almost always improve least-squares residuals. A component should be retained because it is reproducible, physically interpretable, or required by predictive checks, not merely because an optimizer found it. Compare models using held-out pixels, information criteria with their assumptions stated, or Bayesian evidence when priors are defensible. Inspect whether inferred components persist across independent transitions and spatial bins.

Parameters become non-identifiable when components are closer than the resolution or one has low contrast. In that regime, the data may constrain total flux and centroid but not the individual widths and velocities. Publishing a covariance or posterior sample makes this distinction visible.

3.10 Non-parametric profile measurements

Non-parametric velocities can summarize asymmetric profiles without assigning each feature to a Gaussian. From the continuum-subtracted cumulative flux

$$\mathcal{F}(v) = \frac{\int_{v_{\min}}^{v} F(v') \, \mathrm{d}v'}{\int_{v_{\min}}^{v_{\max}} F(v') \, \mathrm{d}v'}, \tag{3.12}$$

define v_p by $\mathcal{F}(v_p) = p/100$. Measures such as $W_{80} = v_{90} - v_{10}$ and v_{50} are common. They depend on the integration window, continuum, noise, and LSF. Negative pixels and absorption complicate the cumulative distribution, so the exact algorithm must be documented.

The flux-weighted centroid and second moment are

$$\bar{v} = \frac{\int v F(v)\,\mathrm{d}v}{\int F(v)\,\mathrm{d}v}, \qquad \sigma_v^2 = \frac{\int (v-\bar{v})^2 F(v)\,\mathrm{d}v}{\int F(v)\,\mathrm{d}v}. \tag{3.13}$$

Moments are sensitive to weak wings and window choices. Monte Carlo propagation should perturb the continuum and correlated noise, not only individual flux pixels.

3.11 Equivalent width, flux, and luminosity

Rest-frame equivalent width separates a line from the local continuum scale. For emission, some communities use positive and others negative signs; the convention must be explicit. The line flux from a fitted profile is preferable to summing a fixed window when blends or masked pixels matter. Direct integration remains a valuable model check.

Uncertainty in integrated flux for weights w_i is

$$\mathrm{Var}(F) = \boldsymbol{w}^{\mathsf{T}} \boldsymbol{C} \boldsymbol{w}, \tag{3.14}$$

not merely $\sum_i w_i^2 \sigma_i^2$ unless the covariance is diagonal. Luminosity then requires a cosmology and luminosity distance. For gravitationally lensed sources, quote whether luminosity is apparent or corrected by a magnification model.

3.12 Detection significance and upper limits

A line "detected at 5σ" is incomplete unless the search procedure is stated. If wavelength and width were scanned over many possibilities, the false-alarm rate is larger than the single-trial Gaussian tail. A matched-filter amplitude for a known template $\boldsymbol{p}$ is

$$\widehat{A} = \frac{\boldsymbol{p}^{\mathsf{T}} \boldsymbol{C}^{-1} \boldsymbol{d}}{\boldsymbol{p}^{\mathsf{T}} \boldsymbol{C}^{-1} \boldsymbol{p}}, \qquad \sigma_A^2 = \left(\boldsymbol{p}^{\mathsf{T}} \boldsymbol{C}^{-1} \boldsymbol{p} \right)^{-1}. \tag{3.15}$$

The template includes the assumed intrinsic width and LSF. Searching a template bank requires calibrating the maximum statistic using blank spectra, permutations, or simulations that reproduce the entire reduction.

An upper limit is conditional on line width, centroid window, profile, and confidence construction. For a nondetection of an unresolved line, a statement such as "3σ flux limit for a Gaussian with FWHM $100\,\mathrm{km\,s^{-1}}$ at the systemic redshift" is reusable. A bare number is not. Negative fitted fluxes should generally remain in ensemble analyses; clipping them to zero biases stacked and population estimates.

3.12.1 Worked example: width-dependent flux limit

Assume independent spectral samples of width $\delta v = 20\,\mathrm{km\,s^{-1}}$ with flux-density uncertainty s. A top-hat line of width $\Delta v = 100\,\mathrm{km\,s^{-1}}$ spans $N = 5$ samples, so the integrated-noise scale is $s\,\delta v \sqrt{N}$. If instead one assumes $\Delta v = 400\,\mathrm{km\,s^{-1}}$, $N = 20$ and the flux limit doubles. Correlated noise changes the square-root scaling. This is why line-width assumptions belong in upper-limit tables.

3.13 Integral-field spectroscopy

An integral-field unit samples two spatial dimensions and one spectral dimension, but the reconstructed cube has a three-dimensional response: a spatial point-spread function (PSF), spectral LSF, pixel footprint, and

covariance. Resampling detector pixels into spaxels correlates both neighboring wavelengths and neighboring spatial samples.

Common spatial binning methods target a continuum signal-to-noise, yet emission-line signal-to-noise may follow a different morphology. Binning based on the line being tested can also introduce selection bias. A defensible strategy defines bins using an independent quantity, fits all spaxels hierarchically, or demonstrates through injection tests that adaptive selection does not fabricate gradients.

Beam smearing mixes velocities within the PSF and can mimic high dispersion where a velocity gradient is steep. The observed cube should be compared to a source model convolved spatially and spectrally. Subtracting a scalar instrumental width from each spaxel cannot correct spatial beam smearing.

3.14 Validation by injection and repetition

End-to-end injection tests place synthetic sources into raw or minimally processed data, run the normal pipeline, and compare recovered parameters with truth. They can measure completeness, flux bias, centroid bias near sky lines, component recoverability, and the effect of covariance. Injections should span morphology, line width, equivalent width, and background conditions rather than a single ideal case.

Other high-value checks include:

- fitting independent exposures and detector subsets separately;
- comparing integrated flux with direct aperture sums;
- verifying fixed atomic branching ratios where opacity is negligible;
- checking skylines or calibration lamps for LSF and wavelength residuals;
- measuring blank apertures with the same extraction and fitting procedure;
- examining posterior predictive residuals in detector and physical coordinates;
- repeating measurements under plausible continuum and mask choices.

3.15 A minimum reproducible line record

Every published line measurement should be accompanied, directly or in machine-readable metadata, by:

1. transition identifier, rest wavelength, air/vacuum convention, and atomic-data source;
2. observed frame and velocity convention;
3. aperture, PSF, slit losses, and extraction method;
4. continuum model, fit window, masks, and blend constraints;
5. LSF model and spectral sampling;
6. flux or equivalent width with statistical and systematic uncertainties;
7. covariance or a documented approximation to it;
8. attenuation, lensing, and aperture corrections;
9. nondetection assumptions and search trials;

10. software version, configuration, calibration version, and provenance links.

This record turns a table of numbers into scientific evidence that another researcher can audit, combine, or reinterpret.

Part II

Gas in and around Galaxies

Chapter 4

Line Formation and the Logic of Spectral Diagnostics

A spectrum contains many transitions but comparatively few independent pieces of physical information. Lines from a common ion share an abundance and ionization fraction; multiplets share upper or lower states; every transition is filtered by the same foreground dust and the same instrument. The aim of diagnostic spectroscopy is therefore not to attach one physical parameter to each line. It is to identify line combinations in which unwanted factors cancel and the desired dependence remains. This chapter develops that logic for recombination lines, collisionally excited lines, fine-structure transitions, resonance absorption, and molecules. Standard references include Osterbrock and Ferland (2006), Dopita and Sutherland (2003), Draine (2011), and Tielens (2005).

4.1 Ionization balance is a rate problem

The ionic abundance of an element is controlled by competing processes rather than by a single "formation temperature." In photoionized gas, a schematic balance between stages X^i and X^{i+1} is

$$n(X^i)\Gamma_i + n_e n(X^i) C_i(T) = n_e n(X^{i+1})\alpha_{i+1}(T) + n(X^{i+1}) n_{\mathrm{H}^0} k_{\mathrm{ce}}(T), \tag{4.1}$$

where Γ_i is a photoionization rate, C_i a collisional ionization coefficient, α_{i+1} the total radiative plus dielectronic recombination coefficient, and k_{ce} represents charge exchange. The photoionization rate depends on the radiation field throughout the ionizing continuum,

$$\Gamma_i = 4\pi \int_{\nu_i}^{\infty} \frac{J_\nu}{h\nu} \sigma_i(\nu)\, \mathrm{d}\nu. \tag{4.2}$$

An ionization threshold therefore does not specify the characteristic photon energy; the product $J_\nu \sigma_i / \nu$ weights the actual photons doing the work.

The dimensionless ionization parameter is the number density of hydrogen-ionizing photons relative to hydrogen atoms,

$$U \equiv \frac{\Phi_{\mathrm{H}}}{n_{\mathrm{H}} c} = \frac{Q_{\mathrm{H}}}{4\pi r^2 n_{\mathrm{H}} c} \tag{4.3}$$

for an isotropic source emitting Q_{H} ionizing photons per second at distance r. A plane-parallel model often uses U at its illuminated face; a spherical nebula has a distribution of U. Metal-line ratios can constrain U only jointly with spectral hardness, abundance pattern, pressure, and geometry.

In low-density, collisionally ionized equilibrium (CIE), ionization fractions are primarily functions of temperature because both ionization and recombination rates scale with n_e. The familiar temperature at which an ion's CIE fraction peaks is not a unique temperature of origin. The same ion may exist in photoionized gas at a much lower kinetic temperature, in a rapidly cooling nonequilibrium plasma, or at a conductive or turbulent interface. Thus O VI is evidence for highly ionized oxygen, not by itself proof of gas at one particular temperature (Gnat and Sternberg, 2007; Tumlinson et al., 2017b).

4.1.1 Equilibrium timescales

Equilibrium is plausible only if microscopic rates are fast compared with changes in the gas. A useful recombination timescale is

$$t_{\rm rec} \simeq \frac{1}{\alpha(T) n_e}. \tag{4.4}$$

At sufficiently low density, a radiation field or temperature can change faster than ions recombine. Cooling gas can then remain overionized relative to CIE, while gas recently exposed to a source can remain underionized. The diagnostic question is not whether an equilibrium table exists, but whether $t_{\rm rec}$ and the ionization timescale are short relative to the cooling, dynamical, and source-variability timescales.

4.2 Hydrogen and helium recombination spectra

An ionizing photon absorbed by hydrogen creates a photoelectron. Recombination and cascade subsequently distribute photons among continua and lines. In the standard Case A idealization, every recombination line photon escapes. In Case B, Lyman-series photons are assumed optically thick and are converted through repeated absorption and cascade, while most Balmer and longer-wavelength photons escape. Neither case is a universal property of an H II region; they are limiting transfer prescriptions (Osterbrock and Ferland, 2006).

For an ionization-bounded nebula in steady state,

$$Q_{\rm H,abs} = \int_V n_e n_p \alpha_B(T)\, {\rm d}V, \tag{4.5}$$

and a recombination-line luminosity is

$$L_{ul} = h\nu_{ul} \frac{\alpha_{ul}^{\rm eff}(T)}{\alpha_B(T)} Q_{\rm H,abs}. \tag{4.6}$$

This relation motivates Hα and other recombination lines as star-formation tracers. It fails in predictable directions when ionizing photons escape, are absorbed by dust, arise from an active nucleus or shocks, or when the assumed stellar population does not represent the actual age and initial mass function (Kennicutt and Evans, 2012).

4.2.1 Balmer decrements and attenuation

If the intrinsic ratio $R_{\rm int} = F({\rm H}\alpha)/F({\rm H}\beta)$ is known from recombination theory, a foreground-screen color excess follows from

$$E(B-V) = \frac{2.5}{k({\rm H}\beta) - k({\rm H}\alpha)} \log_{10}\!\left(\frac{R_{\rm obs}}{R_{\rm int}}\right), \tag{4.7}$$

where $A_\lambda = k(\lambda) E(B-V)$. This compact equation hides three model choices: the intrinsic ratio depends weakly on temperature and density; the attenuation curve must suit the source; and a mixed dust–emitter geometry is not a foreground screen. Stellar Balmer absorption, flux-calibration slope, and differential aperture losses can all bias $R_{\rm obs}$. Infrared Paschen or Brackett lines extend the wavelength lever arm and are less extinguished, but cross-instrument calibration can replace one systematic with another (Fitzpatrick, 1999).

4.2.2 Worked example: screen attenuation from Hα/Hβ

Assume $R_{\rm obs} = 4.0$, $R_{\rm int} = 2.86$, and an adopted curve with $k(\mathrm{H}\beta) = 3.61$ and $k(\mathrm{H}\alpha) = 2.53$. Then

$$E(B-V) = \frac{2.5}{1.08}\log_{10}(4.0/2.86) \simeq 0.34 \text{ mag}. \tag{4.8}$$

The attenuation correction to Hα is $10^{0.4k(\mathrm{H}\alpha)E(B-V)} \simeq 2.2$. Reporting only the corrected luminosity would conceal the dominant assumptions. A reproducible measurement should state the attenuation law, intrinsic decrement, stellar-continuum treatment, and whether the correction refers to nebular or stellar reddening.

4.3 Collisionally excited lines

Forbidden and semiforbidden lines are not "unlikely" in the low-density universe. Their small spontaneous rates allow metastable levels to radiate because the time between particle collisions is long. An excitation from l to u samples the high-energy tail of the electron distribution through the exponential factor in the rate coefficient. A ratio of an auroral line from a high excitation energy to a nebular line from a lower level can therefore be a sensitive electron thermometer.

For [O III], a commonly used conceptual ratio is

$$R_{\rm OIII} = \frac{I(4959) + I(5007)}{I(4363)}. \tag{4.9}$$

At densities below the relevant critical densities, $R_{\rm OIII}$ declines as temperature rises because excitation of the auroral upper level becomes less Boltzmann-suppressed. At higher density, differential collisional de-excitation also changes the ratio. A five-level atom calculation, not a single analytic inversion, is the normal inference tool.

Doublets such as [S II] $\lambda\lambda 6716, 6731$ or [O II] $\lambda\lambda 3726, 3729$ compare levels close in energy but with different collisional de-excitation rates. Their ratios primarily constrain electron density, but become insensitive at both low- and high-density limits. Because a real aperture contains a density distribution, the inferred "density" is an emissivity-weighted effective value.

4.4 Abundances: ionic measurements and ionization corrections

An ionic abundance from an optically thin line ratio can be written schematically as

$$\frac{n(X^i)}{n(\mathrm{H}^+)} = \frac{I_\lambda(X^i)}{I(\mathrm{H}\beta)}\frac{\epsilon_{\mathrm{H}\beta}(T, n_e)}{\epsilon_\lambda(X^i; T, n_e)}. \tag{4.10}$$

The elemental abundance requires an ionization correction factor (ICF),

$$\frac{X}{\mathrm{H}} = \mathrm{ICF}(X) \sum_{i\in\text{observed}} \frac{X^i}{\mathrm{H}^+}. \tag{4.11}$$

An ICF is a model statement about unobserved ions, not an observational constant. It depends on the ionizing spectrum, geometry, and sometimes depletion into dust.

"Direct" electron-temperature abundances are direct only in the limited sense that a temperature-sensitive line is measured. If temperature fluctuates spatially, the auroral line weights hotter zones disproportionately because of its exponential emissivity. Recombination lines respond more weakly to temperature but are faint. Strong-line metallicity calibrations trade weak-line sensitivity for dependence on a model or empirical training set. Different calibrations may differ systematically; the calibration name and abundance scale are part of the result, not metadata that can be omitted (Kewley and Ellison, 2008; Kewley et al., 2019).

Table 4.1: Diagnostic design patterns. Entries describe the main sensitivity, not an exclusive interpretation. Atomic data and valid density ranges must accompany any numerical calibration.

Pattern	Desired cancellation	Principal caveats
Same-upper-level branching ratio	Population, abundance, and often temperature	Differential extinction, blends, opacity, calibration; useful as an internal check
Auroral/nebular ratio of one ion	Ionic abundance and nearly all geometry	Density dependence, recombination contribution, temperature gradients
Fine-structure doublet	Abundance and spectral hardness	Low/high-density asymptotes, more than one collider
Adjacent ion stages of one element	Elemental abundance	Ionization parameter entangled with spectral shape and geometry
Hydrogen recombination ratio	Emission measure and elemental abundance	Transfer case, dust geometry, stellar absorption
Resonance-doublet apparent optical depth	Abundance scale and continuum	Unresolved saturation, partial covering, scattered light
Isotopologue ratio	Much excitation physics if levels are matched	Optical depth, isotope-selective chemistry, abundance-ratio variations

4.4.1 Ionization-source diagrams

The Baldwin–Phillips–Terlevich family of diagrams compares a high-excitation line to a Balmer line and a low-ionization line to another Balmer line (Baldwin et al., 1981b). The closeness in wavelength of each ratio reduces sensitivity to reddening and flux-calibration slope. Star-forming sequences, AGN, and shocks occupy different regions because their ionizing spectra and partially ionized zones differ (Kewley et al., 2001a; Kauffmann et al., 2003a).

Classification boundaries are not laws of nature. Integrated spectra can mix H II regions, diffuse ionized gas, shocks, and an AGN. High-redshift abundance patterns, pressures, and stellar spectra may shift the star-forming locus. Spatially resolved data are particularly vulnerable to low equivalent-width diffuse emission and to point-spread-function mixing. Diagnostic diagrams should be treated as projections of a continuous mixture problem.

4.5 Fine-structure lines and infrared diagnostics

Fine-structure transitions connect closely spaced levels within a ground electronic term. Their excitation energies correspond to tens to thousands of kelvin, making them bright coolants of gas too cool to radiate optical collisionally excited lines. Important examples include [C II] 158 μm, [O I] 63 and 145 μm, [O III] 52 and 88 μm, [N II] 122 and 205 μm, and [Ne II] 12.8 μm. Dust extinction is greatly reduced in the infrared, though dust continuum opacity may still matter in compact, high-column sources.

Several colliders can excite the same line. For [C II], electrons, H, and H_2 have different rate coefficients and dominate in different phases. An observed [C II] luminosity can combine ionized gas, atomic photodissociation layers, and CO-faint molecular envelopes. [N II], which requires ionized gas, can help estimate the ionized contribution if abundance and density are constrained. A line frequently used as a star-formation proxy is therefore not emitted by a single “star-forming phase.”

The luminosity of a two-level fine-structure line per ion approaches a maximum in the thermalized, optically thin limit. Consequently line luminosity gives a strict minimum mass of the emitting ion only after adopting temperature, collider density, and upper-level fraction. Optical depth and subcritical excitation

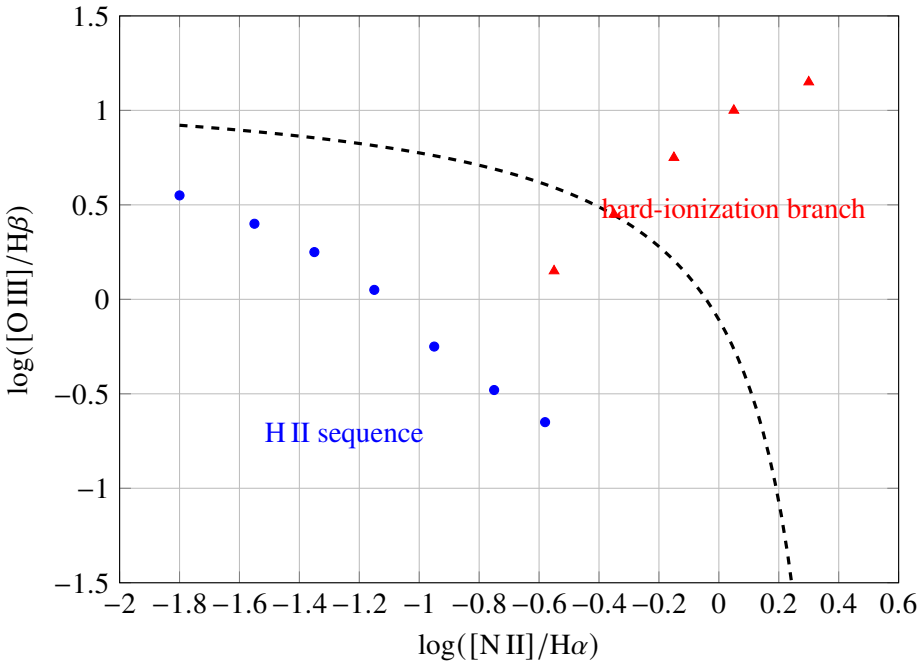


Figure 4.1: Original PGFPlots concept for an emission-line diagnostic plane. The points are synthetic and the dashed curve is illustrative; the production caption must identify any adopted empirical or theoretical boundary and its domain of validity.

increase the required mass.

4.6 Molecular rotational and vibrational lines

For a linear rigid rotor with rotational constant B, idealized level energies are

$$E_J = hBJ(J+1), \qquad g_J = 2J+1, \tag{4.12}$$

and allowed electric-dipole rotational transitions have $\Delta J = \pm 1$ when the molecule has a permanent dipole moment. CO has low-lying transitions that are easy to excite in cold molecular gas. H_2 has no permanent dipole; its quadrupole rotational lines require warmer material, so cold H_2 is usually inferred through CO, dust, gamma rays, or absorption rather than direct emission.

If lines are optically thin, share one excitation temperature, and beam filling is known, a rotational diagram uses

$$\ln\left(\frac{N_u}{g_u}\right) = \ln\left(\frac{N_{\mathrm{tot}}}{Q(T_{\mathrm{ex}})}\right) - \frac{E_u}{kT_{\mathrm{ex}}}. \tag{4.13}$$

The inverse slope is $-T_{\mathrm{ex}}$ and the intercept encodes total column through the partition function Q. Curvature can indicate multiple temperatures, non-LTE populations, opacity, or inconsistent emitting areas. A straight line does not prove LTE; limited dynamic range can hide departures.

For unresolved sources, brightness-temperature line luminosity is often denoted L' in units of K km s^{-1} pc^2. Molecular mass is then

$$M_{\mathrm{mol}} = \alpha_{\mathrm{CO}} L'_{\mathrm{CO}(1-0)}, \tag{4.14}$$

where the inclusion of helium must be stated. The conversion factor varies with metallicity, shielding, surface density, temperature, and dynamics; a Milky-Way disk value should not be applied automatically to low-metallicity dwarfs or compact starbursts (Bolatto et al., 2013b).

4.6.1 Worked example: a two-line excitation temperature

Suppose optically thin measurements yield upper-level columns N_1 and N_2 for energies $E_1 < E_2$. Equation (4.13) gives

$$T_{\rm ex} = \frac{E_2 - E_1}{k} \left[\ln\!\left(\frac{N_1 g_2}{N_2 g_1} \right) \right]^{-1}. \tag{4.15}$$

Let $(E_2 - E_1)/k = 11.1$ K and $N_1 g_2/(N_2 g_1) = 2.0$. Then $T_{\rm ex} = 16.0$ K. This is an excitation temperature between two states, not necessarily the kinetic temperature. If the density is subcritical, radiative trapping or the cosmic microwave background can set $T_{\rm ex}$. At redshift z, the background temperature $T_{\rm CMB} = 2.725(1+z)$ K both excites the molecule and reduces observed contrast.

4.7 Absorption-line column densities

Normalized absorption directly measures apparent optical depth,

$$\tau_a(v) = \ln\!\left[\frac{F_c(v)}{F(v)} \right], \tag{4.16}$$

and the apparent column-density profile is

$$N_a(v) = \frac{m_e c}{\pi e^2} \frac{\tau_a(v)}{f \lambda_0}. \tag{4.17}$$

In practical units,

$$N_a(v) = 3.768 \times 10^{14} \frac{\tau_a(v)}{f(\lambda_0/\text{Å})} \quad \text{cm}^{-2}\,(\text{km s}^{-1})^{-1}. \tag{4.18}$$

Integrating over velocity yields an apparent ionic column. Agreement between $N_a(v)$ profiles for two transitions of the same ion with different $f\lambda$ supports adequate resolution. A smaller apparent column in the stronger line flags unresolved saturation (Savage and Sembach, 1991; Fox et al., 2005b).

Partial covering changes the residual intensity to

$$\frac{F(v)}{F_c} = 1 - C_f(v) + C_f(v) e^{-\tau(v)}. \tag{4.19}$$

A saturated line can therefore have a nonzero core. Scattered light, unrelated continuum components, and instrumental background can produce the same symptom. Doublet fitting may constrain C_f and τ, but only within the chosen geometry.

4.8 Masers, inversions, and negative opacity

If $n_u/g_u > n_l/g_l$, the stimulated-emission term exceeds true absorption and $\alpha_\nu < 0$. Equation (2.7) then describes exponential amplification rather than attenuation. Astrophysical masers require a pump that creates the inversion, a velocity-coherent path, and a seed radiation field. Unsaturated maser intensity grows approximately exponentially with path length; saturation couples the radiation back to the level populations and limits growth. Narrowness, high brightness temperature, variability, and polarization can reveal masing, but isotropic luminosity is misleading because the emission may be strongly beamed.

4.9 A diagnostic is a model comparison

Every line ratio can be written abstractly as a forward model

$$\boldsymbol{F}_{\mathrm{pred}} = \mathcal{M}(T, n, \mathbf{Z}, U, J_\nu, \boldsymbol{v}, \mathcal{G}, \boldsymbol{a}), \tag{4.20}$$

where $\mathcal{G}$ denotes geometry and $\boldsymbol{a}$ atomic or molecular data. Inference asks which parameter combinations are consistent with measured fluxes and their covariance. A sound atlas therefore records for each diagnostic:

1. the transitions and adopted atomic data;
2. the dominant production and destruction mechanisms;
3. the density, temperature, abundance, and ionization ranges over which the calibration was constructed;
4. assumptions about equilibrium, opacity, geometry, and radiation field;
5. known degeneracies and at least one independent cross-check;
6. whether limits, nondetections, and calibration covariance were included.

The value of a line lies not in a single canonical interpretation, but in knowing precisely which physical alternatives it can distinguish.

Chapter 5

The Multiphase Interstellar Medium, PDRs, and Molecular Gas

The interstellar medium (ISM) is a coupled system of gas, dust, magnetic fields, cosmic rays, and radiation. Its named "phases" are useful landmarks in density and temperature, not sealed reservoirs. Material moves among them through cooling, heating, compression, rarefaction, ionization, recombination, molecule formation, feedback, and galactic dynamics. Spectral lines reveal this circulation because each transition selects a different combination of abundance, excitation, optical depth, and background.

The discussion here follows the physical framework developed in Spitzer (1978), McKee and Ostriker (1977), Draine (2011), and Tielens (2005), with phase and small-scale qualifications discussed by Cox (2005) and Stanimirović and Zweibel (2018). Photodissociation-region physics is reviewed by Hollenbach and Tielens (1999), and molecular conversion factors by Bolatto et al. (2013b).

5.1 A phase is an equilibrium branch, not merely a temperature bin

For an ideal gas with total particle density $n_{\rm tot}$,

$$P_{\rm th} = n_{\rm tot} k T. \tag{5.1}$$

Observers often quote $P_{\rm th}/k$ in units of $\mathrm{K\,cm^{-3}}$. Thermal pressure is only one part of the momentum budget:

$$P_{\rm tot} = P_{\rm th} + P_{\rm turb} + P_B + P_{\rm cr} + P_{\rm rad}, \qquad P_B = \frac{B^2}{8\pi}. \tag{5.2}$$

These terms need not be locally isotropic or in equipartition. Magnetic tension, for example, is not captured by treating P_B as an ordinary scalar pressure.

The thermal energy equation can be written schematically as

$$\frac{3}{2} n_{\rm tot} k \frac{\mathrm{d}T}{\mathrm{d}t} = \Gamma - n_e n_{\rm H} \Lambda(T, \boldsymbol{x}) - P_{\rm th} \boldsymbol{\nabla}\cdot\boldsymbol{v} + \boldsymbol{\nabla}\cdot(\kappa \boldsymbol{\nabla} T), \tag{5.3}$$

where Γ includes heating per volume, Λ summarizes radiative cooling, and the final terms represent compression or expansion and conduction. In diffuse neutral gas, photoelectrons ejected from small grains and polycyclic aromatic hydrocarbons often provide important heating. Cosmic-ray ionization, X-rays, turbulent dissipation, and chemical heating can dominate in shielded or extreme environments. Cooling proceeds through species and transitions whose importance changes with temperature, ionization, density, metallicity, and optical depth.

Thermal equilibrium satisfies $\Gamma = n_e n_{\rm H} \Lambda$. If heating scales roughly as n while cooling scales roughly as n^2, the equilibrium curve in pressure-density space can have two stable branches separated by an unstable branch. This motivates cold and warm neutral media that coexist over a range of pressure (Wolfire et al., 2003). Turbulence continuously drives gas away from exact equilibrium, so observed distributions are broad rather than delta functions.

For isobaric perturbations, the Field criterion may be expressed using the net cooling per mass or volume with care about convention. A practical statement is that an equilibrium is unstable when a small temperature decrease at fixed pressure makes net cooling stronger, amplifying the perturbation. Quoting "thermally unstable" gas from temperature alone assumes a heating and cooling model.

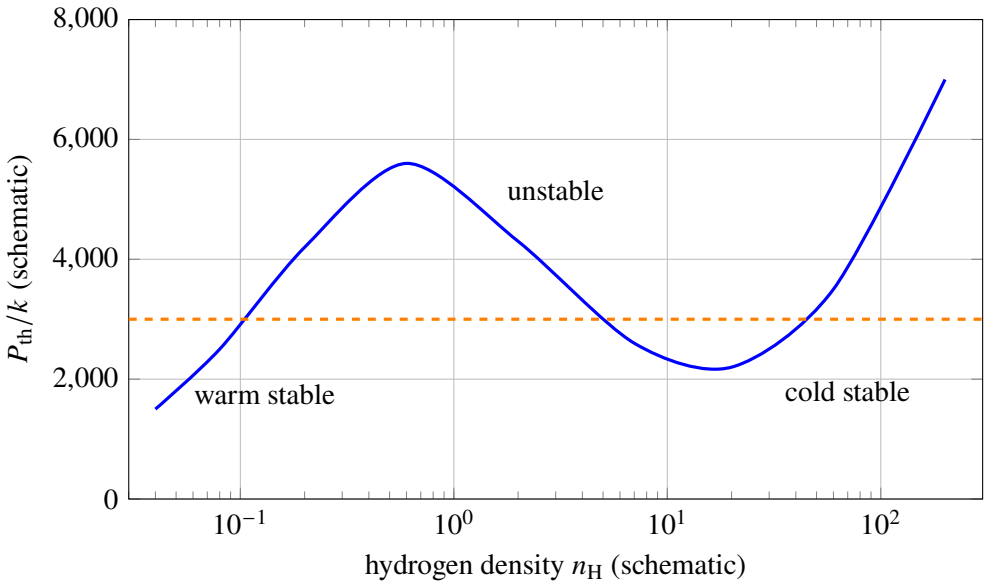


Figure 5.1: Original schematic of a two-phase thermal-equilibrium curve. The shape is pedagogical rather than a numerical cooling calculation. A horizontal pressure can intersect warm and cold stable branches, with an intermediate unstable solution. Radiation field, metallicity, dust, cosmic rays, and shielding shift the curve.

5.2 Working inventory of ISM regimes

Table 5.1 gives orientation values. They are deliberately shown as broad ranges; a phase label should never replace a measured density or temperature.

The hot phase has large volume filling potential but small mass density; molecular structures carry substantial mass in a small volume. Filling factors are scale and environment dependent. A pencil-beam absorption spectrum samples column-weighted material, while diffuse emission weights emissivity and telescope beam. Apparent disagreement can therefore reflect different selection functions rather than inconsistent physics.

5.2.1 Neutral atomic gas and the 21-cm line

The H I hyperfine transition at 1420.4058 MHz connects the triplet and singlet ground-state configurations. Its excitation is described by a spin temperature T_s. For a homogeneous layer in the Rayleigh–Jeans limit,

$$T_B(v) = T_{\mathrm{bg}} e^{-\tau(v)} + T_s \left[1 - e^{-\tau(v)}\right]. \tag{5.4}$$

Against a bright continuum source, absorption constrains $\tau(v)$; emission and absorption together constrain T_s. In the optically thin limit,

$$N_{\mathrm{HI}} = 1.823 \times 10^{18} \int T_B(v)\, \mathrm{d}v \quad \mathrm{cm}^{-2}, \tag{5.5}$$

with T_B in kelvin and v in $\mathrm{km\,s^{-1}}$. Cold opaque components cause the optically thin estimate to miss column. Beam mismatch between emission and pencil-beam absorption adds another systematic.

Thermal H I width sets an upper limit on temperature because turbulence also broadens the profile. Decomposing emission into Gaussians does not uniquely assign physical clouds; regularization and component-count assumptions can dominate the phase fractions.

Table 5.1: Orientation map for interstellar regimes. Values vary with galactic environment and are not classification boundaries.

Regime	Typical T	Indicative n_{H}	Representative probes
Cold neutral medium	tens to a few hundred K	~ 1–$10^2\ \mathrm{cm}^{-3}$	H I 21-cm absorption/emission, C I fine structure, [C II], Na I, K I
Warm neutral medium	several 10^3–10^4 K	~ 0.1–$1\ \mathrm{cm}^{-3}$	H I 21 cm, UV low-ion absorption, broad H I absorption
Warm ionized medium	roughly 10^4 K	$\sim 10^{-2}$–$1\ \mathrm{cm}^{-3}$	Hα, radio free–free, pulsar dispersion, [N II], [S II]
Classical H II region	roughly 10^4 K	~ 10–$10^4\ \mathrm{cm}^{-3}$	H and He recombination lines, [O II], [O III], [N II], [S II]
Hot ionized medium	10^5–10^7 K	$\sim 10^{-4}$–$10^{-2}\ \mathrm{cm}^{-3}$	soft X-ray lines and continuum, O VI at interfaces, highly ionized absorption
Diffuse molecular gas	~ 20–100 K	$\sim 10^2$–$10^3\ \mathrm{cm}^{-3}$	CO, CH, OH, HCO^+ absorption, dust, gamma rays
Dense molecular structures	~ 10–100 K	$\gtrsim 10^4\ \mathrm{cm}^{-3}$	NH_3, N_2H^+, HCN, HCO^+, CS, dust continuum

5.2.2 Warm ionized and hot gas

Diffuse warm ionized gas is traced by recombination emission, free–free emission, Faraday rotation, and pulsar dispersion measure

$$\mathrm{DM} = \int n_e \, \mathrm{d}\ell. \tag{5.6}$$

Combining DM with an emission measure $\int n_e^2 \mathrm{d}\ell$ can constrain a characteristic density and filling factor, provided the paths match. Elevated low-ionization line ratios in diffuse gas reflect its ionization and heating state and can contaminate classifications based on integrated galaxy spectra.

Hot gas emits bremsstrahlung and metal lines in the extreme ultraviolet and X-ray. Its inferred density depends on an emission measure and volume, hence on geometry and filling factor. Soft X-ray absorption by foreground gas complicates the coolest hot components. Intermediate-temperature ions at interfaces may be far from ionization equilibrium, making single-temperature CIE fits an approximation.

5.3 Photodissociation regions

A photodissociation region (PDR) is predominantly neutral gas whose heating and chemistry are controlled substantially by far-ultraviolet photons below the hydrogen ionization edge. PDRs occur on molecular-cloud surfaces, within clumpy clouds, around H II regions, in diffuse atomic gas, and on galaxy-wide scales (Hollenbach and Tielens, 1999). The term describes a regulatory regime, not a simple geometric shell.

Two useful control parameters are gas density and incident far-UV field. The latter is often normalized as G_0 to a conventional interstellar band-integrated field, or as χ under a different normalization. These conventions differ; the unit and reference field must be stated. Ratios such as G_0/n_{H} influence grain charging, photoelectric heating efficiency, and the location of chemical transitions.

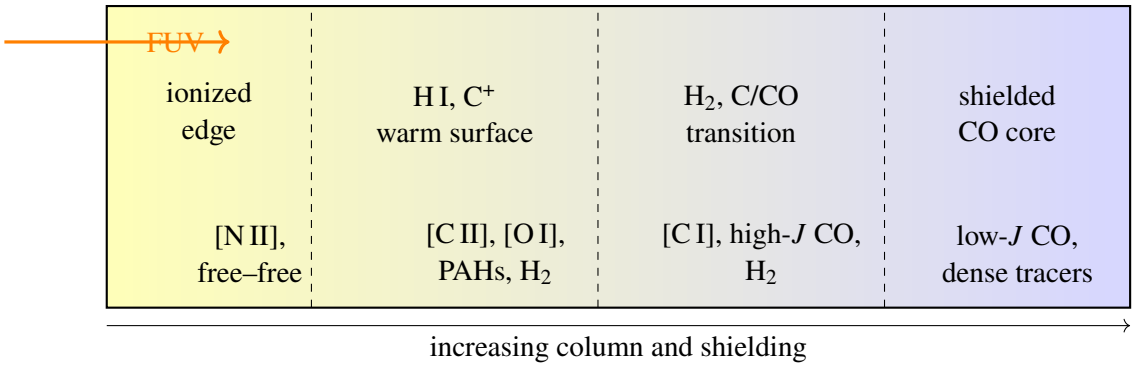


Figure 5.2: Original plane-parallel PDR concept. Real clouds are porous and illuminated from multiple directions, so zones overlap in projection. Transition depths depend on density, metallicity, dust, radiation spectrum, cosmic rays, and dynamics.

5.3.1 Hydrogen formation and shielding

In ordinary dusty gas, H_2 forms efficiently on grain surfaces. A schematic formation rate per volume is

$$\mathcal{R}_{\rm form} = R_{\rm d}\, n_{\rm H}\, n({\rm H}), \tag{5.7}$$

where $R_{\rm d}$ depends on grain abundance, temperature, and sticking physics. The unshielded photodissociation rate is reduced by dust attenuation and H_2 self-shielding:

$$\mathcal{R}_{\rm diss} = D_0 G_0\, f_{\rm sh}(N_{\rm H_2}) e^{-\sigma_d N_{\rm H}} n({\rm H_2}). \tag{5.8}$$

The H/H_2 transition occurs where formation balances the attenuated destruction. H_2 can self-shield through its line spectrum at columns where CO remains photodissociated. This produces “CO-faint” or “CO-dark” molecular gas, especially at low metallicity where dust shielding per hydrogen nucleus is smaller.

5.3.2 Worked example: an H_2 formation time

Adopt $R_d = 3 \times 10^{-17}\,{\rm cm^3\,s^{-1}}$ as an illustrative Milky-Way value and $n_{\rm H} = 100\,{\rm cm^{-3}}$. The characteristic formation time from mostly atomic gas is

$$t_{\rm form} \sim (R_d n_{\rm H})^{-1} \simeq 1.1 \times 10^7\ {\rm yr}. \tag{5.9}$$

At ten times higher density the time is roughly ten times shorter, absent changes in grain properties. Comparing this time with a turbulent crossing or cloud assembly time tests chemical equilibrium. The calculation does not locate the H/H_2 front, because photodissociation and shielding were omitted.

5.3.3 Carbon chemistry and principal coolants

At an illuminated surface, carbon is often C^+ while hydrogen may already be neutral. Increasing shielding permits neutral carbon and eventually CO to become abundant. This cartoon is chemically useful but not a strict series of planar layers: turbulent mixing, cosmic rays, clumping, and nonequilibrium reactions can place C^+, C, and CO together.

[C II] 158 μm and [O I] 63 μm are major coolants over portions of PDR parameter space. Their intensities relative to far-infrared dust emission and to one another are commonly compared with PDR grids to infer density and radiation field. Degeneracies are common, [C II] has ionized-gas contributions, [O I] can be

optically thick and self-absorbed, and multiple cloud faces occur within a beam. A fitted "G_0" is therefore an effective model parameter unless geometry is constrained.

The photoelectric heating efficiency falls when grains become more positively charged because electrons are harder to eject. This helps explain why line-to-FIR ratios can decline in intense radiation fields, but high gas density, dust-bounded ionized regions, line opacity, and continuum optical depth can contribute to observed "line deficits."

5.4 Molecular clouds are dynamical structures

Molecular clouds are organized hierarchically into complexes, filaments, clumps, cores, and velocity-coherent structures. Boundaries defined by CO intensity, extinction, or a dendrogram need not select the same mass. A cloud catalog is the output of a segmentation algorithm and resolution scale as well as an astrophysical population.

For a structure of mass M, effective radius R, and one-dimensional velocity dispersion σ, the commonly quoted virial parameter is

$$\alpha_{\rm vir} = \frac{5\sigma^2 R}{GM} \tag{5.10}$$

for an idealized uniform sphere without surface pressure or magnetic terms. Values near two are often associated with a marginally bound object under particular density profiles. Treating equation (5.10) as an exact binding criterion ignores geometry, density gradients, external pressure, magnetic support, mass uncertainty, and time dependence (McKee and Ostriker, 2007).

The turbulent Mach number is

$$\mathcal{M} = \frac{\sigma_{\rm 3D}}{c_s}, \qquad c_s = \sqrt{\frac{kT}{\mu m_{\rm H}}}. \tag{5.11}$$

Molecular gas is commonly supersonic on cloud scales. Shocks create density structure and dissipate energy; turbulence can both support a region globally and compress subregions locally. Line width includes opacity broadening, multiple components, systematic gradients, and unresolved feedback in addition to turbulence.

5.4.1 CO as a mass tracer

For the CO $J = 1 \rightarrow 0$ line,

$$N({\rm H_2}) = X_{\rm CO} W_{\rm CO}, \qquad M_{\rm mol} = \alpha_{\rm CO} L'_{\rm CO}. \tag{5.12}$$

$X_{\rm CO}$ is a column conversion and $\alpha_{\rm CO}$ a luminosity-to-mass conversion. Authors must state whether helium is included. CO(1–0) is usually optically thick in individual clouds, yet integrated intensity can correlate with mass because velocity structure exposes different surfaces and cloud-scale linewidth and size correlate with mass. This emergent calibration is environmental rather than fundamental (Bolatto et al., 2013b).

At low metallicity, the H_2-bearing envelope outside the CO-rich region becomes a larger fraction of the cloud, increasing $X_{\rm CO}$. In warm, high-dispersion galaxy centers or starbursts, CO luminosity per unit mass can rise, lowering $\alpha_{\rm CO}$. Dust-based gas masses offer an independent route but require dust temperature, opacity, dust-to-gas ratio, and subtraction of atomic gas.

5.4.2 Worked example: CO luminosity to molecular mass

Let a galaxy have $L'_{\rm CO(1-0)} = 2.0 \times 10^9\ {\rm K\,km\,s^{-1}\,pc^2}$. Adopting $\alpha_{\rm CO} = 4.35\, M_\odot ({\rm K\,km\,s^{-1}\,pc^2})^{-1}$ including helium gives

$$M_{\rm mol} = 8.7 \times 10^9\, M_\odot. \tag{5.13}$$

If the environment instead justified $\alpha_{\rm CO} = 0.8$ in the same units, the mass would be $1.6 \times 10^9\,M_\odot$. The factor-of-five spread is a model uncertainty, not a rounding error. A publication should state the adopted conversion, environmental rationale, transition ratio if a higher-J line was observed, and helium convention.

5.5 Dense-gas tracers and effective excitation

Molecules with large dipole moments, such as HCN or HCO^+, have large spontaneous rates and correspondingly high optically thin critical densities. Calling one line a "dense-gas tracer" does not mean it arises only above its tabulated critical density. Radiative trapping reduces effective critical density, multilevel cascades alter populations, infrared pumping can matter, electron collisions can contribute, and an extended density distribution emits over a broad range.

A more observationally relevant quantity is an effective excitation density: the density required to produce a specified line brightness for an adopted column, temperature, and radiation field. It is model dependent by design. Comparing several transitions and isotopologues provides leverage on opacity and excitation. Even then, abundance variations from freeze-out, photochemistry, shocks, or cosmic rays may dominate ratios.

NH_3 inversion transitions provide a useful thermometer because ratios of metastable states respond to kinetic temperature under common conditions. N_2H^+ often survives where CO freezes onto grains, while CO desorption can destroy it. SiO can be enhanced when shocks process grains. These are chemical tendencies, not one-to-one labels; the local reaction network and timescale determine the result.

5.6 Dust, extinction, and continuum mass

Optically thin thermal dust emission is commonly modeled as

$$S_{\nu,\rm obs} = \frac{(1+z)\,\kappa_{\nu,\rm rest} M_d B_{\nu,\rm rest}(T_d)}{D_L^2}, \tag{5.14}$$

under one widespread convention for opacity per dust mass. Factors of 4π differ among emissivity definitions, so dimensional consistency with the adopted κ is essential. A single modified blackbody compresses a temperature distribution into an effective T_d; short-wavelength points weight warm dust and can bias mass low. At high optical depth, emission approaches $B_\nu(T)$ and no longer scales linearly with mass.

Extinction measures a pencil-beam column weighted by grain cross section, whereas emission measures dust luminosity and temperature. Comparing them tests geometry and opacity models. Converting dust mass to total gas mass requires a dust-to-gas ratio that varies with metallicity and chemical evolution.

5.7 Shocks, interfaces, and phase exchange

A shock converts bulk kinetic energy into heat, compression, magnetic energy, cosmic rays, and radiation. For a strong adiabatic shock in a monatomic gas, the immediate post-shock temperature is approximately

$$T_s = \frac{3\mu m_{\rm H}}{16k} v_s^2 \simeq 1.4 \times 10^5\ {\rm K} \left(\frac{\mu}{0.61}\right) \left(\frac{v_s}{100\,{\rm km\,s^{-1}}}\right)^2. \tag{5.15}$$

Radiative shocks develop a cooling and recombination zone that emits ions spanning a large temperature range. Their line ratios depend on shock speed, preshock density, magnetic parameter, abundance, precursor radiation, and completeness of the cooling zone (Dopita and Sutherland, 2003).

Conductive boundaries, turbulent mixing layers, and cloud ablation also populate intermediate ions. A sightline through many interfaces can produce substantial columns without a large volume of gas at the ion's CIE peak temperature. Kinematic alignment, multiple ions, and cooling-time arguments are needed to distinguish interfaces from a volume-filling phase.

5.8 Combining tracers without forcing them into one phase

A robust multiphase analysis uses a generative mixture rather than assigning each line to exactly one component. For line ℓ,

$$F_\ell = \sum_p F_{\ell,p}(T_p, n_p, Z_p, U_p, N_p, \ldots), \tag{5.16}$$

where phases p may include ionized gas, atomic PDR surfaces, CO-faint H_2, and shielded molecular gas. The data constrain only combinations for which sufficiently distinct tracers exist. Priors on pressure balance, abundance continuity, or geometry should be explicit and tested rather than assumed silently.

The following cross-checks are especially valuable:

- compare H I emission with absorption to test opacity and spin temperature;
- use [N II] or other ionized-only lines to estimate the ionized contribution to [C II];
- compare CO-based gas with dust-based gas to expose CO-faint material or conversion-factor changes;
- use isotopologues and multiple J levels to test CO opacity and excitation;
- compare density-squared emission with density-linear absorption or dispersion measures to constrain clumping;
- align velocity components before combining ionic ratios across transitions.

The ISM's complexity is not a failure of the phase framework. It is the central observable: line combinations map how matter cycles among thermal, chemical, and dynamical states.

Chapter 6

Star Formation Across Scales and the Lines That Trace It

Star formation connects parsec-scale self-gravitating structures to the integrated light of galaxies. Spectral lines enter at every stage: molecular transitions map the reservoir, dense-gas species select compact structures, infall and outflow shape profiles, recombination lines count ionizing photons, and metal lines diagnose the H II regions around massive stars. No single transition measures an instantaneous, scale-independent "star-formation rate." Each tracer averages over a population, timescale, extinction geometry, and aperture.

This chapter builds a hierarchy from collapse physics to population calibrations. Broad theoretical and observational syntheses are provided by McKee and Ostriker (2007) and Kennicutt and Evans (2012); dense-core physics is reviewed by Bergin and Tafalla (2007).

6.1 Gravity, pressure, and characteristic scales

For isothermal gas of density ρ and sound speed c_s, the Jeans length and a commonly used Jeans mass convention are

$$\lambda_J = c_s \left(\frac{\pi}{G\rho}\right)^{1/2}, \qquad M_J = \frac{\pi^{5/2}}{6}\frac{c_s^3}{G^{3/2}\rho^{1/2}}. \tag{6.1}$$

Perturbations larger than the Jeans length can grow in the idealized infinite, uniform, nonmagnetic medium. Real clouds are finite, turbulent, magnetized, and externally pressured; Jeans quantities remain useful local scales rather than exact fragmentation predictions.

The free-fall time of a uniform sphere is

$$t_{\rm ff} = \left(\frac{3\pi}{32G\rho}\right)^{1/2}. \tag{6.2}$$

Star formation is inefficient per mean-density free-fall time. A common descriptive parameter is

$$\epsilon_{\rm ff} = \frac{\dot{M}_\star t_{\rm ff}}{M_{\rm gas}}. \tag{6.3}$$

Its inferred value depends on how a cloud boundary, gas mass, density, stellar population, and averaging time are defined. A region that is assembling or dispersing violates the steady-state interpretation of $\dot{M}_\star$.

Thermal pressure is rarely the dominant cloud-scale support. A useful effective one-dimensional dispersion may combine thermal and nonthermal motions,

$$\sigma_{\rm eff}^2 = c_s^2 + \sigma_{\rm nt}^2, \tag{6.4}$$

but calling every nonthermal line width "support" is dangerous. Converging flows, rotation, collapse, opacity broadening, and unresolved components all contribute. Support is a statement about the momentum equation and spatial correlations, not about linewidth alone.

6.1.1 Worked example: free-fall time of a dense clump

Take molecular gas with number density $n_{\mathrm{H_2}} = 10^4\,\mathrm{cm^{-3}}$ and mean mass per H_2 molecule including helium $\mu_{\mathrm{H_2}} m_{\mathrm{H}} \simeq 2.8 m_{\mathrm{H}}$. Then $\rho \simeq 4.7 \times 10^{-20}\,\mathrm{g\,cm^{-3}}$ and

$$t_{\mathrm{ff}} \simeq 3.1 \times 10^5\ \mathrm{yr}. \tag{6.5}$$

If the clump has $M = 10^4 M_\odot$ and $\epsilon_{\mathrm{ff}} = 0.01$, its time-averaged star-formation rate would be $\dot{M}_\star \simeq 3.2 \times 10^{-4} M_\odot\,\mathrm{yr^{-1}}$. This estimate is conditional on the mean density and on a steady efficiency; substructure has shorter local free-fall times.

6.2 Magnetic fields and mass-to-flux balance

Magnetic fields affect compression, angular-momentum transport, anisotropic motion, and the coupling of protostellar outflows to gas. A cloud's mass-to-flux ratio is often compared with a critical value,

$$\mu_\Phi \equiv \frac{(M/\Phi_B)}{(M/\Phi_B)_{\mathrm{crit}}}. \tag{6.6}$$

Subcritical structures ($\mu_\Phi < 1$) cannot collapse across ordered field lines in ideal magnetohydrodynamics, while supercritical structures can. The numerical critical coefficient depends on geometry. Zeeman splitting measures a line-of-sight field component in selected gas, dust polarization traces projected orientation with grain-alignment assumptions, and Faraday rotation weights $n_e B_\parallel$. Combining these into a three-dimensional field demands a geometric model.

Nonideal effects—ambipolar diffusion, Ohmic dissipation, and the Hall effect—allow field and neutral matter to drift on small scales. Turbulent reconnection and field topology further complicate a single diffusion-time picture. Magnetic criticality should therefore be inferred as a distribution with projection uncertainty, not a binary label for one projected map.

6.3 From clouds to filaments, clumps, and cores

Cloud structure is scale free over substantial ranges, and names such as clump and core partly reflect resolution and tracer choice. A dense continuum peak may contain several velocity components; a velocity-coherent feature may have little contrast in column density. Catalog completeness depends on spatial filtering and segmentation.

Dense cores are often categorized as starless, prestellar, or protostellar. A prestellar designation implies gravitational boundedness, which inherits uncertainty from mass, temperature, radius, linewidth, density profile, magnetic field, and external pressure. A source without an infrared counterpart is not automatically starless: deeply embedded low-luminosity objects can evade a survey.

Chemical differentiation provides an evolutionary clock only after physical conditions are modeled. CO freeze-out at cold high density can enhance N_2H^+ relative to CO; protostellar heating returns volatile species to the gas; shocks release grain material and enhance molecules such as SiO. Reaction and adsorption timescales vary with density, temperature, ionization, and grain surface area. A chemical "age" is model dependent.

6.4 Infall and accretion signatures

An optically thick line with an excitation temperature rising inward can show a brighter blue peak than red peak when the envelope is collapsing and a thinner line defines the systemic velocity. The foreground redshifted layer preferentially absorbs red emission. This classic blue asymmetry is suggestive, not unique:

multiple clouds, rotation, outflow, and spatially varying excitation can mimic it. Mapping and comparison of optically thick and thin isotopologues are essential.

Mass infall is sometimes estimated as

$$\dot{M}_{\rm inf} = 4\pi r^2 \rho(r) v_{\rm inf}(r). \tag{6.7}$$

Every factor is model dependent. A fitted profile constrains projected velocity through a radiative-transfer model; it does not directly measure a spherical mass flux. Filamentary accretion, disks, and episodic flow can make the spherical estimate an orientation-level summary only.

Accretion luminosity onto a protostar of mass $M_\star$ and radius $R_\star$ is approximately

$$L_{\rm acc} \simeq f_{\rm acc} \frac{G M_\star \dot{M}_{\rm acc}}{R_\star}, \tag{6.8}$$

where $f_{\rm acc}$ accounts for energy advected, radiated elsewhere, or carried by outflows. Hydrogen lines such as Brγ may correlate empirically with accretion luminosity in calibrated samples, but geometry, extinction, winds, and stellar mass limit extrapolation.

6.5 Protostellar jets and molecular outflows

Magnetocentrifugal launching extracts angular momentum and produces fast jets or winds. Their interaction entrains molecular gas into broad line wings. CO maps can estimate outflow mass, momentum, and kinetic energy after corrections for excitation, opacity, missing low-velocity gas, abundance, spatial filtering, and inclination. For channel mass $M(v)$ relative to systemic velocity,

$$P_{\rm out} = \sum_v M(v) |v - v_{\rm sys}|, \tag{6.9}$$

$$E_{\rm out} = \frac{1}{2} \sum_v M(v) (v - v_{\rm sys})^2. \tag{6.10}$$

Energy is especially sensitive to faint high-velocity wings, while mass is sensitive to confusion near the cloud velocity. A dynamical time $t_{\rm dyn} = R_{\rm proj}/v_{\rm los}$ combines two projection-dependent quantities and is not necessarily the true age. Reporting uncorrected observables alongside model-corrected rates helps future reinterpretation.

Shock-excited H_2, [Fe II], SiO, and high-J CO trace different zones of an outflow. [Fe II] can indicate fast, partially ionized shocks and grain processing; near-infrared H_2 lines select warm molecular material; low-J CO carries much of the cool entrained mass. Line luminosity alone cannot inventory the outflow without a temperature distribution.

6.6 H II regions as ionizing-photon calorimeters

For uniform hydrogen density $n_{\rm H}$ around a source emitting $Q_{\rm H}$ ionizing photons per second, the ideal Strömgren radius is

$$R_S = \left(\frac{3 Q_{\rm H}}{4\pi \alpha_B n_e n_p} \right)^{1/3}. \tag{6.11}$$

After rapid ionization, overpressure drives expansion. An idealized uniform-medium solution is

$$R(t) = R_S \left(1 + \frac{7 c_i t}{4 R_S} \right)^{4/7}, \tag{6.12}$$

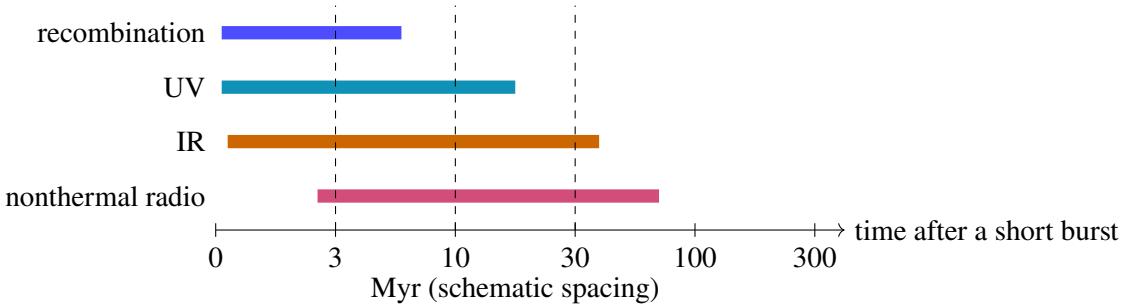


Figure 6.1: Original qualitative response-window concept. Bars indicate relative sensitivity after a short stellar burst, not universal top-hat lifetimes. Actual kernels depend on metallicity, IMF, binaries, dust, and observing band.

where c_i is ionized-gas sound speed. Density gradients, winds, radiation pressure, leakage, gravity, magnetic fields, and cloud porosity change the evolution.

The observed recombination luminosity counts ionizations absorbed by gas. If a fraction $f_{\rm esc}$ escapes and $f_{\rm dust}$ is absorbed by dust before ionizing hydrogen, then schematically

$$Q_{\rm H,gas} = (1 - f_{\rm esc} - f_{\rm dust}) Q_{\rm H,total}. \tag{6.13}$$

Age and IMF determine $Q_{\rm H,total}$ per unit star-formation rate. Binary evolution and stellar rotation can extend hard ionizing emission beyond the lifetime predicted by a simple single-star population.

6.6.1 Worked example: Strömgren scale

For $Q_{\rm H} = 10^{49}\,{\rm s}^{-1}$, $n_e \simeq n_p = 100\,{\rm cm}^{-3}$, and $\alpha_B = 2.6 \times 10^{-13}\,{\rm cm}^3\,{\rm s}^{-1}$,

$$R_S \simeq 3.1\ {\rm pc}. \tag{6.14}$$

Reducing density by a factor of ten increases R_S by $10^{2/3} \simeq 4.6$. The calculation assumes uniform fully ionized hydrogen and no dust or leakage; a clumpy nebula with the same mean density need not have the same recombination rate.

6.7 Star-formation-rate tracers as temporal filters

A luminosity-based calibration can be written

$$\dot{M}_\star = C_X L_X, \tag{6.15}$$

but C_X encodes an IMF, metallicity, stellar-evolution model, star-formation history, dust treatment, and luminosity definition. More fundamentally, the luminosity is a convolution of the past star-formation history with a response kernel,

$$L_X(t) = \int_0^\infty \dot{M}_\star(t - \tau)\, \ell_X(\tau, Z, {\rm IMF})\, {\rm d}\tau. \tag{6.16}$$

Ionizing-recombination tracers weight the youngest massive stars; ultraviolet continuum persists longer; supernova-related radio emission is delayed; infrared emission depends on dust heating and can include older stars. Differences among tracers may reveal burstiness rather than calibration failure.

Table 6.1: Common star-formation tracers and dominant qualifications. Timescales are population dependent and should be computed for the adopted stellar model.

Tracer	What supplies the luminosity	Important qualifications
H recombination lines	Ionizations absorbed by gas	Dust attenuation, stellar absorption, photon leakage and dust absorption, AGN/shocks, IMF sampling
Thermal radio free–free	Ionized gas powered by ionizing photons	Nonthermal separation, electron temperature, photon leakage; low extinction
UV continuum	Photospheres of young and intermediate-age stars	Dust attenuation, star-formation history, metallicity, older-population contamination
Total or monochromatic IR	Dust heated by absorbed radiation	Cirrus from older stars, unobscured escape, dust temperature and spectral shape, AGN
Hybrid Hα+IR or UV+IR	Direct plus dust-reprocessed components	Aperture matching and empirical coefficient depend on population and spatial scale
Nonthermal radio	Cosmic-ray electrons accelerated after stellar death	Time delay, escape and cooling, magnetic field, AGN
[C II] and other cooling lines	Gas cooling linked indirectly to FUV heating	Multiphase origin, metallicity, line deficits, opacity, AGN/shocks

6.7.1 Stochastic populations and small apertures

Standard calibrations assume that the stellar IMF and evolutionary phases are adequately sampled. In a small region or low-SFR galaxy, the number of ionizing stars may be small. Hα luminosity then fluctuates strongly at fixed longer-timescale SFR and can be zero between massive-star births. This is a probability-distribution problem, not necessarily evidence that the IMF has a different upper cutoff.

Spatial drift also matters. Molecular gas, embedded infrared sources, exposed H II regions, and young clusters are separated in both time and position. On sufficiently small apertures, gas and SFR tracers sample different evolutionary stages and the depletion time scatters dramatically. Galaxy-scale averages combine many regions and recover more stable relations.

6.8 Gas–star formation scaling relations

Surface densities are often related by

$$\Sigma_{\mathrm{SFR}} = A \left(\frac{\Sigma_{\mathrm{gas}}}{\Sigma_0} \right)^N \tag{6.17}$$

or expressed through a depletion time

$$t_{\mathrm{dep}} = \frac{M_{\mathrm{gas}}}{\dot{M}_\star}. \tag{6.18}$$

The slope N depends on gas phase, dynamic range, spatial scale, regression method, conversion factors, nondetections, and shared distance or inclination terms. A correlation does not identify the rate-limiting physics by itself.

Relations between HCN-like luminosity and star formation are sometimes interpreted as evidence that a fixed-density gas component forms stars with constant efficiency. However, line luminosity depends on excitation, opacity, abundance, and the entire density probability distribution. A linear relation can emerge from several models. Multiple transitions, resolved density constraints, and independent masses are needed to distinguish them.

6.9 Spatially resolved evolutionary sequencing

An idealized region passes from gas concentration to embedded sources, exposed H II region, and feedback-cleared stellar association. Observed maps contain many regions at different phases. Cross-correlating gas peaks and young-star peaks as a function of aperture can constrain relative phase durations and separation scales, but only under assumptions about region independence, steady formation rate, detectability, and tracer lifetimes.

Line ratios add physical context to the sequence:

- dense molecular tracers and deuterated species probe cold pre-stellar gas;
- warm H_2, high-J CO, and complex molecules identify heating and shocks;
- radio recombination and free–free emission reveal deeply embedded massive stars;
- optical recombination and forbidden lines map exposed ionized gas;
- broad molecular, optical, and X-ray features trace feedback and dispersal.

Selection by one stage biases inferred lifetimes of another. A complete sequence requires a parent sample defined independently of the tracers being timed.

6.10 Inference checklist for a star-formation measurement

Before turning a line luminosity into a rate or efficiency, record:

1. the aperture and whether gas and stellar tracers are resolution matched;
2. the assumed distance, attenuation, IMF, metallicity, and stellar models;
3. the response timescale and whether steady star formation is plausible;
4. contamination from AGN, shocks, diffuse ionized gas, evolved stars, or cirrus;
5. photon escape and absorption by dust inside ionized regions;
6. molecular excitation, abundance, opacity, and conversion-factor assumptions;
7. stochastic population effects and treatment of nondetections;
8. whether a quoted efficiency uses a mean-density, dense-gas, orbital, or other characteristic time.

A star-formation tracer is best viewed as a calibrated temporal and spatial filter. Its limitations become opportunities when several filters are modeled together.

Chapter 7

Stellar Feedback, Superbubbles, and Galactic Outflows

Massive stars reshape their surroundings before, during, and after their deaths. Photoionization heats gas; radiation transfers momentum; winds and jets inflate hot cavities; supernovae generate shocks, metals, cosmic rays, and turbulence. These channels overlap in space and time. Their collective action can disperse molecular clouds, maintain a multiphase disk, launch fountains and winds, and seed the circumgalactic medium.

"Feedback" is not a synonym for broad lines. It is a causal claim that a source has transferred energy, momentum, mass, or newly synthesized material to gas. Establishing that claim requires a budget, geometry, and timescale. Reviews spanning local and galactic scales include McKee and Ostriker (2007), Naab and Ostriker (2017), Veilleux et al. (2005b), and Veilleux et al. (2020).

7.1 Bookkeeping: rates, integrated yields, and coupling

For a feedback channel with injection rate $\dot{E}_{\rm inj}$, only a fraction ϵ_E may appear as kinetic plus thermal energy of the observed phase:

$$\dot{E}_{\rm phase} = \epsilon_E \dot{E}_{\rm inj} - \dot{E}_{\rm rad} - \dot{E}_{\rm transfer} - \dot{E}_{\rm work}. \tag{7.1}$$

Similarly, momentum and mass move between phases. A low energy measured in cool gas does not demonstrate weak feedback if most energy resides in a faint hot wind. Conversely, comparing a cool phase's momentum with a source's instantaneous rate can mislead if the flow accumulated over a long episode.

The useful quantities have distinct dimensions:

$$\dot{M}_{\rm out} \quad [M\,T^{-1}], \tag{7.2}$$

$$\dot{p}_{\rm out} = \dot{M}_{\rm out} v \quad [M\,L\,T^{-2}], \tag{7.3}$$

$$\dot{E}_{\rm kin} = \tfrac{1}{2}\dot{M}_{\rm out} v^2 \quad [M\,L^2\,T^{-3}]. \tag{7.4}$$

Papers sometimes call $p = Mv$ a "momentum rate" or call E/t "energy." Dimensional labels and equations prevent such ambiguities.

Define mass loading relative to star formation as

$$\eta_M = \frac{\dot{M}_{\rm out}}{\dot{M}_\star}, \tag{7.5}$$

with analogous metal or energy loading factors. Numerator and denominator should refer to compatible times and spatial boundaries. An outflow at 10 kpc records a past launch episode, whereas Hα may weight the most recent few Myr.

7.2 Photoionization heating and expansion

Each hydrogen photoionization leaves the excess photon energy above 13.6 eV as photoelectron energy, which heats the gas. Thermal balance near 10^4 K raises the pressure relative to cold molecular surroundings, driving

an ionization front and shock. The Strömgren and ideal expansion solutions in Chapter 6 provide baseline scales.

Photoionization can both suppress and promote local collapse. It erodes and accelerates cloud surfaces, lowers the mass available to form stars, and opens low-density escape channels. Its shock can also compress pre-existing dense structures. Demonstrating "triggered" star formation requires showing that stars would not have formed on the same timescale without the feedback, a counterfactual rarely established by spatial coincidence alone.

Optical line ratios identify ionization fronts and partially ionized zones. Enhanced [S II]/Hα or [O I]/Hα can arise in shocks, diffuse ionized gas, or hard radiation. Kinematics and multiple diagnostic planes are required. Free–free and recombination emission trace ionized gas even when optical emission is obscured.

7.3 Radiation pressure

Absorption of luminosity L supplies a direct momentum rate L/c. A schematic radiative force is

$$\dot{p}_{\rm rad} \simeq \left(1 - e^{-\tau_{\rm UV}} + f_{\rm trap}\tau_{\rm IR}\right) \frac{L}{c}. \tag{7.6}$$

The first term accounts for direct absorption; the second is an idealized infrared multiple-interaction contribution. The factor $f_{\rm trap}$ recognizes that porosity, radiation–matter instabilities, and geometry can reduce coupling below a homogeneous optical-depth estimate. It should be constrained, not assumed to be one.

Resonance-line photons also transfer momentum through repeated scattering, but dust destruction, frequency diffusion, and escape through velocity or spatial channels limit the number of effective interactions. The emergent line profile is a probe of the same transfer process and cannot be interpreted as an independent intrinsic emission spectrum.

Comparing an observed momentum flux with L/c is useful, but a ratio above unity does not uniquely prove radiation trapping. Energy-conserving hot gas, time variability, geometry, radius estimates, and phase conversion can all produce an apparent "momentum boost."

7.4 Stellar winds and wind-blown bubbles

A stellar wind with mass-loss rate $\dot{M}_w$ and terminal speed v_w carries

$$\dot{E}_w = \frac{1}{2}\dot{M}_w v_w^2, \qquad \dot{p}_w = \dot{M}_w v_w. \tag{7.7}$$

The freely expanding wind terminates at a reverse shock, creating a hot interior that pushes a swept-up shell. For constant mechanical luminosity L_w in a uniform ambient density ρ_0, the ideal energy-conserving similarity solution scales as

$$R_b(t) = C\left(\frac{L_w}{\rho_0}\right)^{1/5} t^{3/5}, \qquad v_b = \frac{3R_b}{5t}, \tag{7.8}$$

where C depends on the exact solution convention (Weaver et al., 1977). Radiative cooling, conduction, turbulent mixing, leakage, density gradients, and episodic input alter both normalization and exponent.

Bubble interiors can be faint because emissivity scales roughly as density squared. Soft X-ray luminosity often probes dense interfaces or mixed material rather than the volume-filling wind. Optical shells trace cooled swept-up gas, and expansion may be seen as split profiles or coherent velocity ellipses. A ring on the sky without kinematic expansion is not sufficient evidence for a bubble.

7.4.1 Worked example: distinguishing energy- and momentum-driven scalings

An energy-driven bubble in uniform density has $R \propto t^{3/5}$, while a shell driven by constant momentum injection scales approximately as $R \propto t^{1/2}$. Suppose a bubble has $R = 100$ pc and $v = 30\,\mathrm{km\,s^{-1}}$. The similarity ages are

$$t_E = \frac{3R}{5v} \simeq 2.0\ \mathrm{Myr}, \qquad t_p = \frac{R}{2v} \simeq 1.6\ \mathrm{Myr}. \tag{7.9}$$

The difference is smaller than typical uncertainties from projection and acceleration through a density gradient. Radius and speed alone rarely establish the driving regime; hot-gas pressure, shell mass, energy losses, and source history add leverage.

7.5 Supernova remnants and terminal momentum

A supernova injects ejecta mass, metals, and energy of order 10^{51} erg for a canonical energetic event. Early free expansion gives way to a Sedov–Taylor stage when swept-up mass dominates. For explosion energy E in uniform density ρ_0,

$$R_{\mathrm{ST}}(t) = \xi \left(\frac{E t^2}{\rho_0}\right)^{1/5}, \qquad v_{\mathrm{ST}} = \frac{2R_{\mathrm{ST}}}{5t}. \tag{7.10}$$

When shocked gas cools efficiently, a dense radiative shell forms. Energy declines, but radial momentum continues to grow modestly before approaching a terminal value. Numerical calculations for representative conditions find a weak density dependence and terminal momentum of order a few $10^5\ M_\odot\,\mathrm{km\,s^{-1}}$ per event, with environment, clustering, and microphysics affecting the result (Kim and Ostriker, 2015).

An unresolved simulation that deposits thermal energy into too much mass can radiate it immediately and miss the physical momentum-generating stage. Feedback implementations therefore require adequate resolution or a calibrated terminal momentum prescription. Injecting both full energy and terminal momentum without accounting for the resolved stage can double count.

Observed remnants span nonradiative X-ray and Balmer shocks, radiative optical shells, radio synchrotron structures, and molecular interactions. Shock speeds inferred from line ratios or X-ray temperature should be checked against proper-motion and line-width measurements. Electron and ion temperatures may not equilibrate immediately behind a fast collisionless shock.

7.6 Clustering, superbubbles, and breakout

Massive stars form in groups, so winds and supernovae overlap. Explosions inside an existing low-density cavity lose less energy than isolated explosions in dense gas, while shell impacts and mixing can enhance cooling. The interval between events, cluster compactness, ambient stratification, and pre-supernova clearing determine whether energy accumulates coherently.

In a vertically stratified disk, a superbubble may accelerate as it reaches lower density, fragment, and vent hot gas into the halo. Breakout connects local feedback to a galactic fountain or wind. The dense shell may remain near the disk while hot, metal-rich gas escapes through channels, so mass, metal, and energy loading need not track each other.

7.7 Cosmic rays and magnetic coupling

Supernova shocks and other accelerators populate nonthermal particles. Cosmic rays contribute pressure, ionize shielded gas, heat through wave damping, and can drive outflows if their pressure gradient couples to

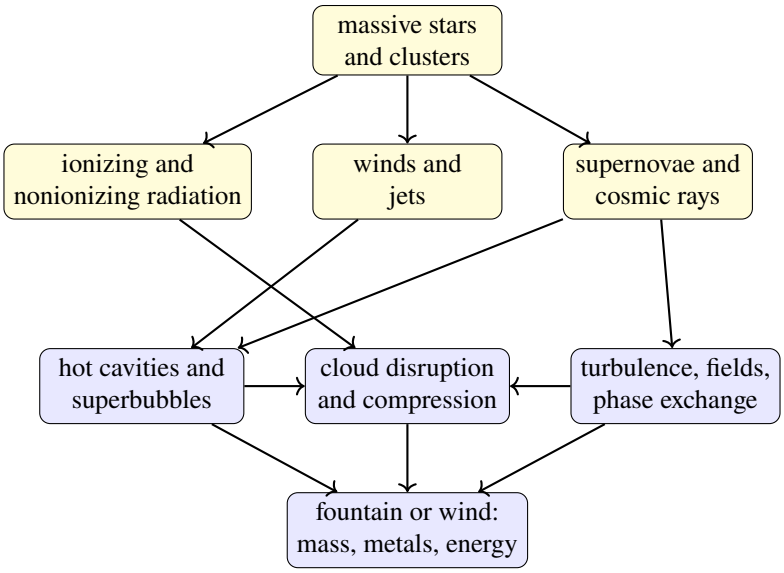


Figure 7.1: Original feedback coupling map. Arrows represent transfer, not a strict time sequence. Observed phases exchange mass and energy, so summing independently derived phase rates can double count the same material.

plasma. Transport may involve advection, diffusion, and streaming, each anisotropic in an ordered magnetic field.

Gamma rays probe hadronic interactions and energetic particles; synchrotron emission depends on electron spectra and magnetic field; ion chemistry constrains low-energy ionization. These observables sample different portions of the cosmic-ray spectrum. A single "cosmic-ray energy density" inferred from radio equipartition is not a direct measurement of the particles that dominate molecular ionization.

Cosmic-ray-driven winds can accelerate gas more gradually than a purely thermal wind and may remain effective where radiative cooling removes thermal energy. Quantitative predictions depend on transport coefficients and wave damping, which are not universal constants. Line diagnostics of ionization state, temperature, and kinematics provide important indirect constraints.

7.8 How multiphase outflows are observed

Different tracers select different phases and geometries:

Absorption detects only gas projected against a background source, yielding a covering-weighted pencil beam. Emission can map the flow but strongly weights dense clumps. Resonant emission may be scattered into locations far from photon production. Agreement of velocity alone does not prove that phases are co-spatial.

7.9 Mass, momentum, and energy of an outflow

For an ionized phase whose recombination luminosity is $L \propto n_e n_p V$, mass scales as

$$M_{\rm ion} \simeq \frac{\mu m_{\rm H} L_{\rm rec}}{h\nu \alpha^{\rm eff}(T) n_e}, \tag{7.11}$$

after extinction and contamination corrections. Density uncertainty maps directly to mass uncertainty. Density-sensitive doublets often measure bright clumps rather than the low-density material that may dominate

Table 7.1: Outflow tracers and principal selection effects.

Phase or component	Representative diagnostics	Main selection effects
Hot volume-filling wind	X-ray continuum and highly ionized lines	Emission weights dense interfaces; absorption requires a bright background; temperature and abundance degenerate with emission measure
Warm ionized	Hα, [N II], [S II], [O III], UV ions	Extinction, shocks versus photoionization, density-squared weighting, projection and beam smearing
Warm neutral	Na I D, Mg II, H I absorption and emission	Resonant scattering, stellar contamination, ionization/depletion corrections, covering fraction
Molecular	CO, OH, HCN, HCO^+, warm H_2	Conversion factor, excitation, opacity, abundance, interferometric filtering, orientation
Dusty gas	Far-IR continuum, extinction, scattered light	Dust temperature, opacity, dust-to-gas ratio, foreground versus outflow geometry
Cosmic rays	Synchrotron, gamma rays, ionization chemistry	Magnetic field, transport, loss time, unrelated AGN contribution

volume or mass.

For a thin partial shell at radius R subtending solid angle Ω, an absorption column gives

$$M_{\rm shell} \simeq \Omega R^2 \mu m_{\rm H} N_{\rm H}, \tag{7.12}$$

where $N_{\rm H}$ requires ionization, abundance, depletion, and covering corrections. The radius is often the least directly constrained quantity for unresolved absorption.

Several incompatible conventions are used for outflow rate. For a shell crossing time R/v,

$$\dot{M}_{\rm out} \simeq \frac{M_{\rm out} v}{R}. \tag{7.13}$$

A uniformly filled steady cone may introduce a factor of order a few. Rather than calling one convention correct in every geometry, authors should provide M, v, R, and the exact equation so readers can translate.

Velocity choices are likewise consequential. A centroid measures bulk shift; a maximum or percentile velocity emphasizes wings; deprojected speed depends on cone inclination. Because energy scales as v^2, wing definitions dominate kinetic-power comparisons.

7.9.1 Worked example: geometry dominates an outflow rate

Suppose a molecular outflow has $M_{\rm out} = 10^8 M_\odot$, characteristic speed $v = 300\,{\rm km\,s^{-1}}$, and radius $R = 1$ kpc. The flow-time estimate is

$$\dot{M}_{\rm out} \simeq 31 \left(\frac{M_{\rm out}}{10^8 M_\odot}\right) \left(\frac{v}{300\,{\rm km\,s^{-1}}}\right) \left(\frac{R}{1\,{\rm kpc}}\right)^{-1} M_\odot\,{\rm yr^{-1}}. \tag{7.14}$$

If the emission is actually concentrated at 0.3 kpc, the rate rises by more than a factor of three. If the molecular conversion factor shifts by a factor of four, so does the rate. A formal line-flux uncertainty of five percent is not the relevant total uncertainty.

7.10 Launch, survival, and phase creation

Cool clouds placed in a fast hot wind experience ram pressure, shocks, shear, thermal conduction, magnetic draping, and radiative cooling. A characteristic cloud-crushing time is

$$t_{\rm cc} = \frac{\chi^{1/2} r_c}{v_{\rm rel}}, \tag{7.15}$$

where $\chi = \rho_c/\rho_h$. In simple nonradiative hydrodynamics, disruption can occur before acceleration to the hot-wind speed. Radiative cooling in mixed gas, magnetic fields, conduction, self-gravity, and continuous cloud creation can extend survival or create new cool material in situ.

Therefore cool gas at large radius need not be the same parcels launched cold from the disk. It may have condensed from a mixed outflow. Metallicity, dust survival, ion ratios, cloud size distribution, and velocity alignment can test origin, but no single diagnostic is decisive.

7.11 Escape, fountains, and recycling

Comparing line-of-sight velocity with an escape speed is only a first check. The gas has a three-dimensional velocity distribution, the gravitational potential evolves, and hydrodynamic drag, cooling, and pressure forces continue after observation. A parcel below local escape speed can be accelerated; one above it can transfer energy and fall back.

Material that returns in a galactic fountain still matters: it redistributes metals and angular momentum, exchanges mass with the halo, and delays star formation. The CGM can contain first-time inflow, recycled wind, stripped satellite gas, and material circulating within the halo (Tumlinson et al., 2017b). An absorption component labeled "outflow" from its velocity sign alone may be ambiguous without galaxy orientation and systemic-velocity precision.

7.12 A rigorous feedback claim

A strong feedback analysis proceeds through five tests:

1. **Identification:** show that the spectral component is distinct from rotation, tidal structure, beam smearing, and unrelated foreground gas.

2. **Excitation:** determine whether photoionization, shocks, mixing, or an AGN supplies the observed lines; allow mixtures.

3. **Geometry:** state the aperture, radius, solid angle, inclination, covering fraction, and whether absorption and emission sample the same structure.

4. **Budget:** derive mass, momentum, and energy with conversion-factor, density, ionization, extinction, and phase corrections made explicit.

5. **Causality and fate:** compare source history with flow time and assess whether gas is dispersed, recycled, or escaping rather than equating detection with quenching.

The ideal output is not one "feedback efficiency" with a tiny statistical error. It is a phase-resolved budget with transparent conditional ranges. Such a budget can be compared meaningfully with simulations, which must apply the same apertures, line emissivities, and selection functions to their synthetic observations.

Chapter 8

The Circumgalactic Medium, the Intergalactic Medium, and the Cosmic Web

8.1 A working map of diffuse baryons

The gaseous Universe is not naturally divided into boxes. Gas crosses the interstellar medium (ISM), circumgalactic medium (CGM), intragroup medium, and intergalactic medium (IGM); the same parcel may be accreted, form stars, be ejected, cool, mix, and return. The names are nevertheless useful if they are treated as operational definitions rather than immutable phases. In this chapter, the *CGM* is gas outside the bright stellar body and cold star-forming disk but inside the region dynamically associated with a galaxy's dark-matter halo. The *IGM* is gas outside individual halos. The *cosmic web* is the evolving network of sheets, filaments, nodes, and voids created by gravitational instability. These definitions deliberately leave a transition zone: around groups, interacting systems, and splashback regions, assignment to one galaxy can be physically ambiguous (Tumlinson et al., 2017a; Faucher-Giguère and Oh, 2023).

A convenient halo scale is defined by

$$M_\Delta = \frac{4\pi}{3}\,\Delta\,\rho_{\rm ref}(z)\,R_\Delta^3, \tag{8.1}$$

where the reference density may be the critical or mean matter density. Authors who quote R_{200c}, R_{200m}, or a "virial radius" are not necessarily using the same boundary. The conversion can be tens of percent and should never be hidden when normalizing impact parameters. Associated velocity and temperature scales are

$$V_\Delta = \left(\frac{GM_\Delta}{R_\Delta}\right)^{1/2}, \tag{8.2}$$

$$T_{\rm vir} \simeq \frac{\mu m_{\rm p} V_\Delta^2}{2k_{\rm B}}, \tag{8.3}$$

with mean molecular weight μ. For an ionized, Milky-Way-mass halo, $T_{\rm vir}$ is of order 10^6 K. This number is a guide, not a claim that the halo is isothermal.

The most important conceptual correction to a one-phase cartoon is that temperature labels do not define unique structures. "Cool" gas near 10^4 K may occur in infalling filaments, stripped satellite wakes, radiatively cooled wind material, turbulent mixing layers, or self-shielded clouds. "Warm" gas near 10^5–10^6 K may be cooling through a short-lived temperature interval rather than occupying a stable volume. "Hot" gas can be supported approximately hydrostatically, driven by winds, or shock heated during accretion. Observed velocity coincidence between two ions does not prove that they are co-spatial. A spectrograph collapses a three-dimensional, time-dependent medium into optical depth versus one velocity coordinate.

8.2 Why the CGM is central to galaxy evolution

A galaxy's stellar mass is only the integrated residue of a larger baryon cycle. Cosmological inflow supplies gas; cooling and angular-momentum transport deliver some of it to the ISM; stars and black holes return

mass, metals, momentum, cosmic rays, and radiation; satellites donate gas by winds and stripping; and some material escapes or is exchanged between halos. Consequently, the CGM is simultaneously a fuel reservoir, a feedback calorimeter, and a record of environmental processing (Tumlinson et al., 2017a; Faucher-Giguère and Oh, 2023; Hafen et al., 2019).

For a control surface at radius r, a formal instantaneous mass flux is

$$\dot{M}(r) = \int \rho(\boldsymbol{x})\,[\boldsymbol{v}(\boldsymbol{x}) - \boldsymbol{v}_{\rm frame}] \cdot \hat{\boldsymbol{r}}\, r^2\, \mathrm{d}\Omega. \tag{8.4}$$

This equation is straightforward in a simulation and extremely difficult to infer from a single sightline. An absorber supplies a column density and a line-of-sight velocity, but generally not its distance along the sightline, solid angle, transverse velocity, or time history. Thus labels such as "inflow" and "outflow" should be probabilistic unless geometry supplies additional evidence. Major-axis/minor-axis trends, metallicity, comparison with the escape speed, resolved emission, or multiple nearby sightlines can strengthen the interpretation, but none is individually decisive.

The cool-CGM mass inferred from absorption illustrates the inference chain. For a hydrogen column profile $N_{\rm H}(R)$ and azimuthal covering fraction $f_{\rm c}(R)$,

$$M_{\rm cool}(< R_{\rm max}) = 2\pi\mu_{\rm H} m_{\rm p} \int_0^{R_{\rm max}} N_{\rm H}(R)\, f_{\rm c}(R)\, R\, \mathrm{d}R. \tag{8.5}$$

The directly constrained quantity is often $N_{\rm H\,I}$, not $N_{\rm H}$. Writing $f_{\rm H\,I} = N_{\rm H\,I}/N_{\rm H}$ gives $N_{\rm H} = N_{\rm H\,I}/f_{\rm H\,I}$. The neutral fraction can be tiny and depends on density, spectrum and amplitude of the ionizing field, temperature, shielding, and non-equilibrium history. Large masses inferred for cool halo gas are therefore important but model conditioned (Werk et al., 2014; Prochaska et al., 2017). A robust paper reports which part of Equation 8.5 is measured and which part is assumed.

8.3 Thermal competition and multiphase structure

Two timescales provide a useful first organization. An approximate radiative cooling time is

$$t_{\rm cool} \simeq \frac{\frac{3}{2} n_{\rm tot} k_{\rm B} T}{n_{\rm e} n_{\rm H} \Lambda(T, Z, J_\nu)}, \tag{8.6}$$

where Λ includes radiative losses and, if desired, photoheating. A local free-fall estimate is

$$t_{\rm ff}(r) = \left[\frac{2r}{g(r)}\right]^{1/2}. \tag{8.7}$$

When shock-heated gas cools slowly compared with its dynamical time, an extended hot atmosphere can persist. When cooling is rapid, cold streams, condensation, and radiative interfaces become important. The ratio $t_{\rm cool}/t_{\rm ff}$ is a diagnostic, not a universal one-number criterion; rotation, turbulence, uplift, conduction, magnetic topology, cosmic-ray pressure, and ongoing forcing all alter the outcome (Faucher-Giguère and Oh, 2023).

Pressure balance is similarly informative but not guaranteed:

$$P_{\rm tot} = P_{\rm th} + P_{\rm turb} + P_{\rm B} + P_{\rm CR} + \cdots. \tag{8.8}$$

A cool component can have $n_{\rm cool} T_{\rm cool} \neq n_{\rm hot} T_{\rm hot}$ while remaining confined by non-thermal pressure, ram pressure, gravity, or time dependence. An apparent pressure discrepancy between photoionization models

and a coronal model may therefore diagnose missing physics, but it may also expose an incorrect ionizing background or an assumption that non-coincident velocity components form one cloud.

Interfaces are disproportionately visible. Conductive boundaries, radiative shocks, and turbulent mixing can generate ions such as C IV, N V, O VI, and Ne VIII while containing little of the total mass. In collisional ionization equilibrium, each ion fraction has a characteristic temperature range, but gas that cools faster than it recombines retains an ionization memory. Photoionization can also produce high ions at much lower kinetic temperature if the density is sufficiently small. Ion names are therefore not thermometers by themselves (Gnat and Sternberg, 2007; Sutherland and Dopita, 1993).

8.4 A tracer ladder for circumgalactic gas

8.4.1 Neutral hydrogen

H I is usually the entry point because the Lyman series spans an enormous range of column density. Lyα is sensitive but saturates readily. Higher-order Lyman lines have smaller oscillator strengths and recover information hidden in saturated Lyα. At sufficiently high $N_{\mathrm{H\,I}}$, the continuum optical depth at the Lyman edge,

$$\tau_{912} = N_{\mathrm{H\,I}}\sigma_{912}, \tag{8.9}$$

constrains the neutral column even when line cores are black. At still larger columns, Lorentzian damping wings determine $N_{\mathrm{H\,I}}$. The lesson is to design coverage of a *ladder*, not a single line.

Absorber names encode approximate column regimes, but boundaries vary with author and purpose. The Lyα forest generally denotes mostly photoionized systems below the classical Lyman-limit threshold. Partial Lyman-limit systems and Lyman-limit systems occupy the transition to optical thickness near 1 Ryd. Sub-damped and damped Lyα systems contain progressively larger neutral columns; the conventional DLA boundary is $N_{\mathrm{H\,I}} = 2 \times 10^{20}$ cm^{-2} (Wolfe et al., 2005). These are observational classes, not one-to-one maps onto IGM, CGM, and ISM.

8.4.2 Low and intermediate ions

Mg II, C II, Si II, Fe II, and O I preferentially trace cool or low-ionization material. Their diagnostic value differs:

- O I is coupled to H I by charge exchange over useful conditions, making O/H comparatively direct in neutral gas.
- Si II and Fe II have several transitions spanning oscillator strengths, valuable for diagnosing saturation.
- Refractory Fe and Si can be depleted onto dust, so gas-phase abundance ratios need not equal nucleosynthetic ratios.
- Mg II is observable from the ground after modest redshifting and has built large statistical samples, but equivalent width mixes column, velocity spread, covering fraction, and saturation.

Intermediate ions such as Si III and C III are often strong because they can be abundant in photoionized gas, but their strongest transitions saturate easily. A precise equivalent width is not necessarily a precise column density.

8.4.3 High ions and hot gas

C IV, N V, O VI, and Ne VIII probe harder ionization conditions and often reveal interfaces or warm-hot gas. O VI is especially important because its 1031.93 and 1037.62 Å doublet is accessible to far-UV spectroscopy at low redshift and to optical spectroscopy at sufficiently high redshift. Strong, extended O VI around star-forming galaxies established a major metal reservoir (Tumlinson et al., 2011). Yet converting $N_{\mathrm{O\,VI}}$ into oxygen or total gas mass requires both the ion fraction $f_{\mathrm{O\,VI}}$ and metallicity:

$$N_{\mathrm{H}} = \frac{N_{\mathrm{O\,VI}}}{f_{\mathrm{O\,VI}}\,(\mathrm{O/H})_{\odot}\,10^{[\mathrm{O/H}]}}. \tag{8.10}$$

A factor-of-ten uncertainty in metallicity becomes a factor-of-ten uncertainty in mass.

At $T \gtrsim 10^6$ K, abundant ions emit and absorb soft X-rays. Oxygen O VII and O VIII are central WHIM and halo tracers; the Fe-L complex diagnoses hotter, metal-rich plasma. Emission scales approximately as $\int n_{\mathrm{e}} n_{\mathrm{H}} \Lambda_E \, \mathrm{d}\ell$ and is weighted toward dense clumps. Absorption scales as $\int n_i \, \mathrm{d}\ell$ and is more sensitive to diffuse columns but requires an exceptionally bright background source and exquisite control of instrumental structure. The two methods do not sample the same density weighting.

8.5 The IGM as a fluctuating medium

8.5.1 Photoionized forest

The classical cloud picture has largely been replaced by a continuous-density view. Much Lyα forest absorption arises in mildly overdense, photoionized sheets and filaments whose peculiar velocities, temperature, thermal broadening, and Hubble expansion shape the spectrum (Rauch, 1998; McQuinn, 2016). For highly ionized hydrogen in equilibrium,

$$n_{\mathrm{H\,I}} \Gamma_{\mathrm{H\,I}} \simeq n_{\mathrm{e}} n_{\mathrm{p}} \alpha_{\mathrm{A}}(T), \tag{8.11}$$

so the neutral fraction is extremely sensitive to the photoionization rate $\Gamma_{\mathrm{H\,I}}$ and moderately sensitive to temperature through the recombination coefficient. The transmitted flux is

$$F(v) = C(v) \exp[-\tau(v)] + N(v), \tag{8.12}$$

with continuum C, absorption field τ, and noise N. In forest cosmology, continuum fitting is not merely cosmetic: fitting away long-wavelength modes distorts the measured correlation function and must be represented in the forward model.

A useful analytic argument sets an absorber's characteristic scale near a local Jeans or filtering scale. Combining photoionization equilibrium with that scale produces a relation between $N_{\mathrm{H\,I}}$ and gas overdensity (Schaye, 2001). The relation explains why the column-density distribution can encode the matter distribution, but its coefficients depend on temperature, ionizing background, gas fraction, and dynamical assumptions. It should be used as a calibrated scaling, not an exact cloud ruler.

8.5.2 Mean opacity, thermal history, and reionization

Even a small neutral fraction produces substantial Lyα opacity because the transition is strong. The Gunn–Peterson argument therefore constrains diffuse neutral hydrogen far below the fraction needed for a predominantly neutral Universe (Gunn and Peterson, 1965). A dark trough at high redshift does not by itself measure a unique neutral fraction: once Lyα saturates, small changes in transmission can correspond to a broad range of neutral fractions. Higher Lyman lines, damping-wing shape, proximity-zone structure, and ensembles of sightlines are needed.

The forest also remembers heat input. Photoheating during hydrogen and helium reionization establishes a temperature–density relation often written

$$T(\Delta) = T_0 \Delta^{\gamma - 1}, \tag{8.13}$$

for a restricted low-density regime, where $\Delta = \rho_{\rm b}/\bar{\rho}_{\rm b}$. Thermal broadening, pressure smoothing, and peculiar velocities affect line widths and small-scale flux power differently. Inferring T_0 and γ therefore requires synthetic spectra with the instrumental line-spread function and the same continuum and masking operations as the data.

8.5.3 The warm-hot IGM

As large-scale structure grows, accretion and merger shocks heat some diffuse gas to 10^5–10^7 K, the conventionally defined warm-hot intergalactic medium (WHIM) (Cen and Ostriker, 1999; Davé et al., 2001). No single tracer is complete:

Broad Lyα absorbers are metallicity independent in principle, but blends, continuum errors, and non-thermal broadening can mimic large Doppler parameters.

O VI and Ne VIII are powerful in the UV, but their baryon conversion depends on metallicity and ionization history.

O VII and O VIII access hotter gas in X-rays, but line equivalent widths are tiny and claimed detections demand independent confirmation, careful trials accounting, and stable calibration.

Thermal Sunyaev–Zel'dovich and dispersion measures respond to electrons rather than a metal abundance, but projection and halo separation are nontrivial.

Historically, "missing baryons" has meant baryons permitted by early-Universe constraints but absent from low-redshift inventories, not non-baryonic dark matter. Census estimates evolve with surveys and ionization corrections (Bregman, 2007; Shull et al., 2012; Péroux and Howk, 2020). Individual X-ray WHIM claims, including influential O VII results, should be presented with their stated significance and subsequent replication status rather than as closure of the census (Nicastro et al., 2018).

8.6 Metals as history, not merely abundance

Metals in diffuse gas prove prior processing by stars, but translating them into a transport history is difficult. The logarithmic abundance is

$$[\mathrm{X/H}] = \log_{10}\left(\frac{N_{\rm X}}{N_{\rm H}}\right) - \log_{10}\left(\frac{\mathrm{X}}{\mathrm{H}}\right)_{\odot}, \tag{8.14}$$

after ionization and depletion corrections. The adopted solar reference must be stated (Asplund et al., 2021). A metallicity posterior inherits uncertainty from $N_{\rm H\,I}$, the radiation field, gas density, relative abundance pattern, dust depletion, phase assignment, and atomic data.

Low metallicity can support an accretion interpretation, while enhanced α/Fe or N/O can encode enrichment timescales. However, metal-poor gas may be old IGM material mixed only weakly with winds; metal-rich gas may arise from outflow, recycling, stripping, or enrichment by another galaxy. Metallicity alone does not provide the sign of radial velocity. Observed metallicity distributions are also selection functions: metal-selected samples and H I-selected samples answer different questions. Early evidence for a bimodal low-redshift Lyman-limit metallicity distribution was scientifically influential (Lehner et al., 2013); later analyses should be compared at matched H I range, redshift, sensitivity, and priors rather than reduced to a slogan.

8.7 From pencil beams to maps

Absorption is extraordinarily sensitive because detectability does not suffer the same density-squared penalty as emission, but it sparsely samples the sky. Emission maps morphology but is surface-brightness limited and vulnerable to sky subtraction, scattered light, and resonant transfer. The strongest programs combine:

1. a background quasar or galaxy grid for column density and kinematics;
2. deep galaxy redshifts to establish environment and impact parameters;
3. integral-field emission mapping for morphology and velocity;
4. X-ray, millimeter, or radio measurements for phases missed in the UV;
5. synthetic observations of simulations passed through the same selection, resolution, and noise.

At high redshift, extended Lyα emission can illuminate cool gas around galaxies and quasars. Lyα is resonant: a photon can emerge far from where it was produced and at a frequency altered by scattering through a moving medium. Morphology and line centroid therefore cannot be read as a direct emissivity or velocity map without radiative-transfer modeling. Non-resonant recombination lines and metal lines provide critical controls.

Dense background-source fields enable tomography. Each sightline remains a one-dimensional convolution of density, ionization, peculiar velocity, and thermal state, but regularized reconstruction can reveal coherent filaments and voids. On much larger scales, surveys treat transmitted forest flux as a biased tracer and measure its auto-correlation and cross-correlation with quasars. DESI has demonstrated the statistical power of hundreds of thousands of forest spectra for baryon acoustic oscillations (DESI Collaboration, 2025). This regime is conceptually continuous with single-line analysis: both require a forward model for the continuum, absorption, instrument, masking, and selection.

8.8 A disciplined interpretation protocol

For any diffuse-gas claim, write the chain in the following order.

1. **Observed:** normalized flux, surface brightness, equivalent width, profile shape, angular separation, and their covariance.
2. **Instrumental:** spectral and spatial line-spread functions, wavelength zero point, sensitivity, background model, detector artifacts, aperture, and completeness.
3. **Atomic inference:** transition identity, oscillator strength, column density, excitation, saturation assessment, and blend model.
4. **Plasma inference:** ionization mechanism, density, temperature, radiation field, abundance pattern, depletion, and phase association.
5. **Geometric inference:** distance, path length, covering fraction, opening angle, inclination, and association with a galaxy or filament.
6. **Derived physics:** total mass, flux, cooling rate, metal mass, energy, and comparison with binding or feedback budgets.

Uncertainty normally grows down this list. Reporting a precise equivalent width with a loosely constrained total mass is not contradictory; it is an honest consequence of the inference hierarchy.

8.9 Common category errors

- **Ion equals phase.** O VI can arise from collisionally ionized, photoionized, non-equilibrium cooling, or mixed gas.
- **Velocity equals radial motion.** A blueshift is not automatically an outflow if systemic redshift, inclination, rotation, and environment are uncertain.
- **Equivalent width equals column.** Saturated equivalent width can increase through velocity spread with little sensitivity to column.
- **Emission and absorption sample the same gas.** Their density weighting and selection functions differ.
- **A halo boundary is observed.** $R_{\rm vir}$ is convention and model; gas dynamics need not change abruptly there.
- **Metallicity is origin.** Enrichment constrains history but rarely identifies a unique transport channel.
- **A simulation cell is a cloud.** Numerical resolution, mixing, and subgrid feedback can alter synthetic absorber structure; convergence of galaxy properties does not guarantee convergence of CGM interfaces.

8.10 Research frontiers

Several measurements would transform the field. Closely spaced sightlines can determine transverse coherence and cloud sizes without assuming a path length. Joint UV and X-ray spectroscopy can connect cool, interface, and virial-temperature gas. Deep, absolutely calibrated emission mapping can measure faint cooling and fluorescence. Time-domain absorption against moving or variable background sources can probe small-scale structure. Finally, hierarchical analyses that fit populations rather than one absorber at a time can propagate non-detections, selection functions, ionizing-field uncertainty, and halo-to-halo scatter.

The central scientific target is not a census of ion labels. It is a quantitative account of how mass, metals, momentum, and energy move between galaxies and the cosmic web. Spectral lines are the measuring apparatus, and their greatest power comes from combining transitions whose degeneracies are different.

Chapter 9

Absorption-Line Spectroscopy: From Pixels to Physical Gas

9.1 What an absorption spectrum actually measures

An absorption spectrum is a comparison between received flux and an estimate of the flux that would have been received without the absorber. In its most useful forward form,

$$F_{\rm pix}(\lambda) = \left\{ C(\lambda) \left[(1 - C_{\rm f}) + C_{\rm f} \exp\left(- \sum_k \tau_k(\lambda) \right) \right] \right\} \otimes \mathcal{L}(\lambda) + B(\lambda) + \epsilon(\lambda), \tag{9.1}$$

where C is the source continuum and broad emission spectrum, $C_{\rm f}$ is the covering fraction, $\mathcal{L}$ is the instrumental line-spread function (LSF), B is any residual background or scattered light, and ϵ is noise. Each physical velocity component and each blended transition contributes an optical-depth profile τ_k. Equation 9.1 is a reminder that a fitted Voigt component is not directly observed. The detector records a sampled convolution of source, absorber, instrument, and background.

For an intervening quasar absorber the covering fraction is often taken as unity, but this should not become reflex. Partial coverage is common in intrinsic AGN absorption, stellar winds, compact continuum sources occulted by small clouds, and spatially mixed emission/absorption. A nonzero flux floor in a strong doublet can arise from partial coverage, unresolved velocity structure, scattered light, an erroneous background, or unrelated emission. The competing explanations must be fitted at the flux level.

The line identification itself is a hypothesis. A persuasive identification uses multiple transitions at one redshift, expected strength ordering, absence of plausible alternatives, and consistency with instrumental coverage. Atomic wavelengths and oscillator strengths must come from a versioned source such as Morton (2003); precision claims should not mix air and vacuum wavelengths or oscillator-strength compilations silently.

9.2 Coordinates, frames, and velocity conventions

The first redshift estimate is

$$1 + z = \frac{\lambda_{\rm obs}}{\lambda_0} = \frac{\nu_0}{\nu_{\rm obs}}. \tag{9.2}$$

For velocity structure within a system, define the convention explicitly. A commonly used local coordinate about reference redshift $z_{\rm ref}$ is

$$v = c\,\frac{z - z_{\rm ref}}{1 + z_{\rm ref}}. \tag{9.3}$$

At large offsets, use the exact relativistic relation or work directly in redshift. Wavelength calibration must also specify frame: observatory, heliocentric, barycentric, Local Standard of Rest, or a galaxy systemic frame. Optical, radio, and relativistic velocity definitions differ at non-negligible redshift. Combining UV, optical, and radio spectra without harmonizing definitions can manufacture offsets comparable to the line widths being interpreted.

The adopted systemic redshift often dominates an outflow velocity's uncertainty, especially if it comes from a broad or asymmetric line. Narrow stellar absorption, molecular lines, or spatially integrated nebular emission may each define a different practical zero point.

9.3 Continuum placement is part of the inference

Dividing by a single best continuum and forgetting its uncertainty produces overconfident line measurements. The preferred approach is to model local continuum parameters jointly with the absorption. For an isolated line, a low-order polynomial or spline over windows demonstrably free of absorption may suffice. For a quasar Lyα forest, broad emission complexes and correlated forest absorption require a global template, principal components, Gaussian process, or forward model. For galaxy spectra, stellar photospheric features and the instrumental response may be comparable to the interstellar line.

Continuum choices should be tested by:

1. changing the fit-window boundaries;
2. increasing or decreasing continuum flexibility;
3. injecting synthetic lines into unnormalized data;
4. propagating continuum samples into equivalent width and column posteriors;
5. checking blank regions for biased residuals and underestimated covariance.

For broad, shallow features, continuum uncertainty can dominate photon noise. A claimed broad Lyα absorber or AGN trough is not secure if it disappears under plausible continuum models.

9.4 Equivalent width: robust summary, incomplete physics

For continuum-normalized flux $F_n(\lambda)$, the observed-frame equivalent width is

$$W_{\lambda,\mathrm{obs}} = \int_{\lambda_1}^{\lambda_2} [1 - F_n(\lambda)]\ \mathrm{d}\lambda, \qquad W_{\lambda,\mathrm{r}} = \frac{W_{\lambda,\mathrm{obs}}}{1+z}. \tag{9.4}$$

Equivalent width is conserved by a normalized LSF when all absorbed flux is included and the continuum is correct. This makes it useful at modest resolution. It does not make it a unique measure of column density.

In the optically thin regime, the rest equivalent width satisfies

$$N\ [\mathrm{cm}^{-2}] = 1.13 \times 10^{20}\ \frac{W_{\lambda,\mathrm{r}}\ [\text{Å}]}{f \lambda_0^2\ [\text{Å}^2]}, \tag{9.5}$$

where f is the oscillator strength. Once the line saturates, equivalent width becomes sensitive mainly to Doppler width and component spread. Eventually damping wings restore a strong column dependence. These are the linear, saturated, and damping portions of the curve of growth (Spitzer, 1978; Draine, 2011).

For a pixel sum with independent errors, a simple statistical variance is

$$\sigma^2(W_\lambda) = \sum_i \left(\Delta\lambda_i\, \sigma_{F_n,i}\right)^2, \tag{9.6}$$

but reduced spectra often have correlated noise from resampling, flat fields, and background subtraction. Continuum uncertainty, integration-window uncertainty, fixed-pattern noise, and blends must be added or marginalized. An empirical distribution from line-free windows with the same width is often more realistic than a diagonal-noise formula.

9.5 Optical depth and column density

9.5.1 The apparent-optical-depth method

For full coverage and a reliable continuum, define

$$\tau_{\rm a}(v) = \ln\left[\frac{1}{F_n(v)}\right]. \tag{9.7}$$

The apparent column density per unit velocity is

$$N_{\rm a}(v) = \frac{m_{\rm e}c}{\pi e^2}\frac{\tau_{\rm a}(v)}{f\lambda_0} = 3.768\times10^{14}\frac{\tau_{\rm a}(v)}{f\lambda_0[\text{Å}]}\frac{\text{cm}^{-2}}{\text{km s}^{-1}}. \tag{9.8}$$

Integrating over velocity gives $N_{\rm a}$. The method preserves kinematic information and enables a direct saturation test: two transitions of the same ion should yield the same $N_{\rm a}(v)$ when resolved and unsaturated (Savage and Sembach, 1991).

If the stronger transition yields systematically smaller $N_{\rm a}(v)$ than the weaker one, unresolved saturation is likely. Equality is necessary but not always sufficient: both transitions can be severely saturated and unresolved. Negative fluxes, low signal-to-noise, and logarithmic transformation introduce biases; simulations tailored to the data help quantify them (Fox et al., 2005a). Report lower limits rather than forcing a precise column from black cores.

9.5.2 Voigt profiles

The intrinsic absorption cross section combines Doppler and natural broadening. For component j,

$$\tau_j(v) = N_j\frac{\pi e^2}{m_{\rm e}c}f\,\phi_{\rm V}(v;v_j,b_j,\Gamma), \tag{9.9}$$

where the normalized Voigt profile $\phi_{\rm V}$ is the convolution of a Gaussian and Lorentzian, b_j is the Doppler parameter, and Γ is the radiative damping constant. The model flux must be convolved with the wavelength-dependent LSF before comparison with pixels.

Component decomposition is rarely unique. A high-column narrow component can resemble a lower-column broader component after convolution. Additional weak components can improve the likelihood without corresponding to distinct physical clouds. Useful safeguards are:

- fit all usable transitions of an ion simultaneously;
- tie redshifts only where a shared kinematic phase is physically justified;
- allow thermal widths to vary by ion mass rather than forcing one b;
- compare models using predictive checks, not reduced χ^2 alone;
- report integrated columns and velocity percentiles, which are often more stable than the number of fitted components.

9.5.3 Thermal and non-thermal broadening

For a Gaussian component,

$$b_i^2 = b_{\rm nt}^2 + \frac{2k_{\rm B}T}{m_i}, \tag{9.10}$$

where $b_{\rm nt}$ represents unresolved turbulent, bulk, and structural broadening. Two ions believed to be co-spatial can in principle solve for T and $b_{\rm nt}$:

$$T = \frac{m_1 m_2}{2k_{\rm B}(m_2 - m_1)} \left(b_1^2 - b_2^2\right). \tag{9.11}$$

This inference fails if the ions are not co-spatial, if one profile contains unresolved subcomponents, or if their fitted component boundaries differ. Hydrogen plus a heavy ion gives the largest mass leverage. A broad H I line without aligned metal absorption can constrain temperature only after blends and continuum systematics are controlled.

9.6 Saturation, doublets, and covering fraction

The line-center optical depth for a Doppler component can be written

$$\tau_0 \simeq 1.497 \times 10^{-15} \frac{N[\mathrm{cm}^{-2}]\, f\, \lambda_0[\text{Å}]}{b[\mathrm{km\,s}^{-1}]}. \tag{9.12}$$

This quick calculation should be made during observing design. A high signal-to-noise spectrum does not recover information erased by saturation; one needs weaker transitions, higher resolution, damping wings, or a continuum edge.

For a resolved doublet with optical-depth ratio R and a velocity-dependent covering fraction,

$$I_{\rm weak}(v) = 1 - C_{\rm f}(v) + C_{\rm f}(v) e^{-\tau(v)}, \tag{9.13}$$

$$I_{\rm strong}(v) = 1 - C_{\rm f}(v) + C_{\rm f}(v) e^{-R\tau(v)}. \tag{9.14}$$

Solving these equations can separate optical depth from coverage under a homogeneous-screen assumption. Real sources may have a distribution of optical depth across the emitting surface, and broad-line and continuum regions can have different coverage. Consequently, the algebraic solution is a compact model, not a geometric image.

Doublet ratios are also distorted by blending. The O VI 1037 line, for example, can overlap unrelated interstellar or intergalactic absorption; C IV and Si IV doublets can overlap themselves in sufficiently broad AGN winds. Global multiplet fitting is safer than measuring each trough independently.

9.7 The curve of growth as a diagnostic

The curve of growth predicts $W_\lambda(N, b)$ for one component. Fitting several transitions with different $f\lambda$ can constrain N and b even when profiles are not resolved. The method is valuable for sparse data but compresses away velocity information. Multi-component structure produces an effective b whose physical meaning is limited.

A practical curve-of-growth analysis should display the data and posterior in W/λ versus $Nf\lambda$ space, list the transitions excluded for blending, and test whether a single (N, b) fits both weak and strong lines. If only saturated transitions are available, the posterior often has a long column-density tail. Quoting symmetric errors in that situation is misleading. The output may properly be a lower bound.

9.8 Blends and line confusion

Line confusion increases toward short rest wavelength, high source redshift, and low resolution. A candidate feature should be tested against a line list at every known absorber redshift, Galactic ISM and molecular features, geocoronal emission, detector artifacts, and telluric absorption for ground-based data. The most reliable workflow is simultaneous:

1. construct a master list of expected transitions for all systems;
2. map every transition into observed wavelength;
3. fit overlapping optical depths in one flux model;
4. require shared parameters only within physically motivated groups;
5. inspect residuals in detector coordinates as well as wavelength.

Airglow can be time dependent; separating orbital night may reduce it at the cost of exposure. Telluric absorption is multiplicative and depends on airmass, humidity, and time. Sky emission is additive and has a spatial profile. Treating all three with a generic mask discards information and can leave biased wings. A forward model or physically chosen exclusion is preferable.

9.9 Instrumental resolution and sampling

Resolving power $R = \lambda/\Delta\lambda$ corresponds approximately to

$$\Delta v_{\mathrm{FWHM}} \simeq \frac{c}{R}. \tag{9.15}$$

The equation is a planning shorthand. Real LSFs can be non-Gaussian, wavelength dependent, and altered by source extent, slit illumination, focus, or detector location. A model using a Gaussian with the nominal R can bias narrow-line columns and component widths. Use the tabulated or empirical LSF for the actual setting.

Nyquist sampling is not the same as spectral resolution. Rebinning can improve visual signal-to-noise per bin but does not create information and generates covariance. Keep native or minimally resampled pixels for the likelihood; rebin for presentation only, or carry the covariance matrix. Coaddition is also a model: exposures can differ in wavelength zero point, LSF, sensitivity, fixed-pattern structure, and background. Fitting exposures jointly is often safer than coadding first.

9.10 From ionic columns to a physical phase

9.10.1 Photoionization models

Define the ionization parameter

$$U = \frac{\Phi_{\mathrm{H}}}{n_{\mathrm{H}} c} = \frac{n_\gamma(E > 13.6\,\mathrm{eV})}{n_{\mathrm{H}}}. \tag{9.16}$$

Given an incident spectrum, elemental abundances, geometry, stopping column, and equilibrium assumptions, a code such as Cloudy predicts ionic columns (Ferland et al., 2017). Ratios of successive stages of one element can constrain U while reducing abundance sensitivity. Combining H I with metal ions then constrains metallicity. In practice, multiphase structure often prevents one U from fitting every ion.

The radiation field is part of the model, not a universal constant. Metagalactic UV-background prescriptions differ in amplitude and hardness (Haardt and Madau, 2012); nearby galaxies or an AGN can dominate locally.

Because U constrains a photon-density to gas-density ratio, changing the field changes the inferred $n_{\rm H}$ and path length

$$L = \frac{N_{\rm H}}{n_{\rm H}}. \tag{9.17}$$

Path lengths that exceed the halo or imply implausible pressure are powerful model checks.

A reproducible photoionization analysis reports the code version, atomic database, incident spectrum and normalization, solar abundance reference, relative-abundance priors, dust/depletion treatment, geometry, density law, stopping rule, and whether upper limits enter the likelihood. Publishing the model grid or scripts is more useful than publishing a corner plot alone.

9.10.2 Collisional and non-equilibrium models

In collisional ionization equilibrium (CIE), ion fractions depend mainly on temperature for a specified abundance set. This makes ion ratios appear to be thermometers. The assumptions are restrictive: the gas must have had time to ionize and recombine, photoionization must be negligible, and electrons must be close to a Maxwellian distribution. Rapidly cooling gas can remain overionized, while a shock can remain underionized. Non-equilibrium cooling calculations show that the same ionic ratio may correspond to different thermal histories (Gnat and Sternberg, 2007).

The cooling time and recombination time should be compared:

$$t_{\rm rec}(X^i) = \frac{1}{\alpha_i(T) n_{\rm e}}. \tag{9.18}$$

If $t_{\rm cool} \lesssim t_{\rm rec}$, equilibrium is doubtful. Turbulent mixing and conduction create distributions of temperature rather than one T. In that case, integrated interface columns and kinematic alignment may be more meaningful than a single-temperature fit.

9.11 Abundances, depletion, and ionization corrections

An elemental abundance inferred from ions is

$$[{\rm X/H}] = \log\left(\frac{N_{X^i}}{N_{\rm H\,I}}\right) + \log\left(\frac{f_{\rm H\,I}}{f_{X^i}}\right) - \log({\rm X/H})_\odot. \tag{9.19}$$

Every term needs a convention. The solar scale changes at the 0.01–0.1 dex level for important elements (Asplund et al., 2021). Dust removes refractory elements from the gas. Nucleosynthetic patterns can depart from solar, especially at low metallicity or after delayed Type Ia and AGB enrichment. Charge exchange can couple some ions to H I. Treating every difference as ionization produces a deceptively precise but wrong metallicity.

Useful abundance combinations include:

- O I/H I for predominantly neutral gas, with charge-exchange caveats;
- S II or Zn II as comparatively weakly depleted tracers where usable;
- Fe/alpha-element ratios for depletion plus enrichment history;
- N/O for secondary versus primary nitrogen, provided ionization phases match.

Upper limits are scientifically active. A non-detection of a low ion can rule out a high-density solution; a non-detection of N V can disfavor one hard spectrum. Censoring limits from a likelihood because they are not detections throws away this leverage.

9.12 Population measurements

For an absorber survey, completeness is a function $C(W, z, \lambda, b, \mathrm{S/N}, \ldots)$, not one threshold. The incidence is often written

$$\frac{\mathrm{d}\mathcal{N}}{\mathrm{d}z} = \frac{1}{\Delta z_{\mathrm{eff}}} \sum_j \frac{1}{C_j}, \qquad \Delta z_{\mathrm{eff}} = \int g(z)\,\mathrm{d}z, \tag{9.20}$$

where $g(z)$ accounts for coverage blocked by detector gaps, foreground features, proximity exclusions, and sensitivity. For cosmological comparison, one can use absorption distance

$$\mathrm{d}X = \frac{(1+z)^2 H_0}{H(z)}\,\mathrm{d}z. \tag{9.21}$$

Blind injection and recovery through the actual detection pipeline is the most direct completeness calibration.

Association with galaxies introduces another selection function: magnitude limit, angular coverage, redshift success, fiber collision, and group-finding algorithm. Calling an absorber isolated when the galaxy survey is incomplete is not a physical result. The low-redshift COS survey literature provides useful examples of documenting redshift path and line-finding methodology (Danforth et al., 2016).

9.13 A reproducible analysis recipe

1. Preserve calibrated exposures and quality arrays; do not begin from a screenshot or display-only coadd.
2. Verify wavelength frames and align exposures using narrow, well-identified features without erasing real shifts.
3. Define a transition list with versioned atomic data.
4. Fit continuum, background, absorption, and the correct LSF jointly where practical.
5. Measure equivalent widths and apparent columns as minimally model-dependent summaries.
6. Test saturation with weaker lines, doublets, higher Lyman lines, or Lyman-edge and damping information.
7. Fit physically motivated multiplets, but separate robust integrated quantities from non-unique component structure.
8. Compare photoionized, collisionally ionized, and non-equilibrium hypotheses; use posterior predictive spectra.
9. Propagate atomic, continuum, calibration, radiation-field, abundance, depletion, and geometry systematics into derived mass or energetics.
10. Release normalized and unnormalized spectra, masks, LSFs, code, model grids, posterior samples, and machine-readable measurements.

9.14 Failure modes worth remembering

Black core, precise column: impossible without weaker transitions or damping and edge information; the correct result is often a limit.

Same velocity, same phase: not guaranteed in a projected, multi-component medium.

Large b, high temperature: only an upper limit if non-thermal and unresolved broadening are possible.

Doublet ratio looks right: both members may be unresolved and saturated, or one may be blended.

One-zone model fits most ions: the excluded ion may be the clearest evidence for a second phase.

High formal significance: unmodeled continuum flexibility, line searching across many resolution elements, and calibration features change the false-positive rate.

Precise mass from one sightline: geometry and covering fraction can dominate the error even when the column is secure.

Absorption spectroscopy is powerful because it detects very little material. That same sensitivity makes it vulnerable to hidden saturation, unrelated blends, and model-dependent ionization corrections. The defensible strategy is layered: preserve the pixel-level evidence, extract model-light measurements, and add physical assumptions one at a time.

Part III

Energetic and Multiwavelength Spectroscopy

Chapter 10

Active Galactic Nuclei: Accretion, Line Regions, and Outflows

10.1 The central engine as an energy-conversion system

An active galactic nucleus (AGN) is not defined by one spectral line. It is a coupled system in which gas loses angular momentum, releases gravitational binding energy near a supermassive black hole, produces a broadband radiation field, and sometimes launches winds and relativistic jets. The observables depend on black-hole mass, accretion rate, spin, magnetic flux, viewing angle, obscuration, time variability, and host-galaxy contamination. A taxonomy based only on apparent line width or radio brightness therefore mixes intrinsic physics with orientation and selection (Urry and Padovani, 1995).

The Eddington luminosity follows from balancing electron-scattering radiation force against gravity in ionized hydrogen:

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_{\rm p} c}{\sigma_{\rm T}} \simeq 1.26 \times 10^{38} \left(\frac{M_{\rm BH}}{M_\odot}\right) \,{\rm erg\,s^{-1}}. \tag{10.1}$$

The Eddington ratio is $\lambda_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$. If the radiative efficiency is η,

$$L_{\rm bol} = \eta \dot{M}_{\rm acc} c^2. \tag{10.2}$$

Neither η nor the bolometric correction is universal. They depend on accretion state, spin, spectral energy distribution (SED), and anisotropy.

In a geometrically thin, optically thick disk, the local effective temperature approximately follows

$$\sigma T_{\rm eff}^4(R) = \frac{3 G M_{\rm BH} \dot{M}}{8\pi R^3} \left[1 - \left(\frac{R_{\rm in}}{R}\right)^{1/2}\right], \tag{10.3}$$

with relativistic corrections near the innermost stable orbit (Shakura and Sunyaev, 1973). Integrating blackbody-like annuli motivates the optical/UV big blue bump. A hot, compact corona inverse-Compton scatters seed photons into an X-ray power law. Dust outside its sublimation radius absorbs UV/optical radiation and reradiates in the infrared. These components illuminate different line-forming regions with an SED that is not directly observed in every direction.

At very low Eddington ratio, radiatively inefficient flows can be hot and geometrically thick; at high ratio, advection and radiation pressure modify the thin-disk picture. Applying a thin-disk continuum fit outside its regime can produce precise but model-specific masses and accretion rates.

10.2 A scale hierarchy

Event-horizon and inner-disk scales: gravitational radii $r_{\rm g} = G M_{\rm BH}/c^2$, probed by rapid variability, relativistic reflection, and sometimes interferometric imaging.

X-ray corona: compact region producing the hard continuum; its geometry remains uncertain and may change with accretion state.

Broad-line region (BLR): dense, rapidly moving gas producing permitted lines with widths of thousands of $\mathrm{km\,s^{-1}}$; forbidden lines are collisionally suppressed at high density.

Dusty structure: an anisotropic, clumpy distribution near the dust sublimation scale, not necessarily a smooth torus.

Narrow-line region (NLR): lower-density gas extending from tens of parsecs to kiloparsecs, producing forbidden and permitted lines with narrower profiles.

Radio jet and lobes: magnetized relativistic flow spanning sub-parsec to megaparsec scales in the most powerful sources.

Galaxy-scale outflow: ionized, neutral, atomic, and molecular gas accelerated or pressurized by radiation, winds, jets, star formation, or their combination.

This hierarchy matters because apertures mix it. A fiber spectrum may combine BLR light, NLR emission, H II regions, diffuse ionized gas, shocks, and stellar continuum. An X-ray spectrum can mix coronal continuum, disk reflection, distant reflection, warm absorption, soft excess, and host emission. Spectral decomposition is therefore a spatial model even when the object is unresolved.

10.3 Emission-line regions

10.3.1 Permitted broad lines

Hydrogen Balmer lines, Mg II, C IV, Lyα, and other permitted transitions dominate many type-1 AGN spectra. Their profiles encode a luminosity-weighted transfer function rather than a simple orbit. Rotation, inflow, outflow, anisotropic emission, obscuration, responsivity, and radiation pressure all contribute. C IV is particularly susceptible to a blueshifted wind component; treating its full width as purely virial can bias single-epoch black-hole masses.

The virial mass estimator is

$$M_{\mathrm{BH}} = f_{\mathrm{vir}} \frac{R_{\mathrm{BLR}} \Delta V^2}{G}, \tag{10.4}$$

where ΔV is a line-width measure and f_{vir} absorbs BLR geometry, inclination, and kinematics. Different use of FWHM versus line dispersion and mean versus rms spectra changes the calibration. The result is not a direct dynamical mass unless R_{BLR} and the transfer geometry are constrained.

10.3.2 Reverberation mapping

If a variable continuum drives line variability, the observed line light curve is

$$L_{\mathrm{line}}(v, t) = \int \Psi(v, \tau)\, C(t - \tau)\, \mathrm{d}\tau, \tag{10.5}$$

where $\Psi(v, \tau)$ is the velocity-delay map (Blandford and McKee, 1982; Peterson, 1993). A characteristic lag gives $R_{\mathrm{BLR}} \simeq c\tau$ in the rest frame. Velocity-resolved lags can distinguish idealized rotation, inflow, and outflow patterns, although finite cadence, seasonal gaps, red-noise variability, and line blending blur the response.

Campaign design must match three timescales: sampling cadence shorter than the expected lag structure, duration several times the largest lag, and enough continuum variability to drive a measurable response. The observed lag is dilated by $(1 + z)$. Flux calibration must be stable across epochs; aperture changes and seeing-dependent host contamination can imitate variability.

10.3.3 Narrow-line diagnostics

Classical diagnostic diagrams compare lines close in wavelength so that dust and flux-calibration effects are modest. The most familiar plane is

$$\log \frac{[\mathrm{O\,III}]\ 5007}{\mathrm{H}\beta} \quad \text{versus} \quad \log \frac{[\mathrm{N\,II}]\ 6583}{\mathrm{H}\alpha}, \tag{10.6}$$

with companion planes using [S II]/Hα and [O I]/Hα (Baldwin et al., 1981a; Kewley et al., 2001b; Kauffmann et al., 2003b; Kewley et al., 2006). Hard AGN continua extend partially ionized zones and enhance low-ionization lines, while stellar H II regions occupy a sequence controlled by metallicity and ionization parameter.

The boundaries are not laws of nature. They depend on abundance pattern, stellar populations, redshift, density, diffuse ionized gas, shocks, and aperture mixing. At high redshift, star-forming galaxies can shift relative to local training samples. A spatially resolved mixing sequence is often more informative than one integrated classification.

Density-sensitive doublets such as [S II] 6716/6731 and [O II] 3726/3729 constrain electron density only over their responsive range and for the zone emitting the ions. Temperature-sensitive auroral-to-nebular ratios such as [O III] 4363/(4959+5007) measure a strongly emissivity-weighted temperature. In an inhomogeneous NLR, a single density and temperature are summaries, not a full structure (Osterbrock and Ferland, 2006).

10.4 Obscuration and orientation

The basic unification insight is that an intrinsically similar central engine can look different when the BLR is directly visible, hidden by dusty material, or seen through a relativistically beamed jet (Urry and Padovani, 1995). Polarized broad lines, ionization cones, radio morphology, and infrared emission support this view. Modern unification is statistical rather than absolute. Covering factor, luminosity, host dust, accretion state, and time variability also matter; some changing-look AGN alter spectral type on human timescales.

Line ratios involving widely separated wavelengths require an extinction model. For a foreground-screen approximation,

$$F_{\mathrm{obs}}(\lambda) = F_{\mathrm{int}}(\lambda) 10^{-0.4 E(B-V) k(\lambda)}. \tag{10.7}$$

The Balmer decrement can constrain reddening, but intrinsic ratios vary with density, optical depth, collisional excitation, and radiative transfer. Mixed dust-emitter geometry does not obey a simple screen.

10.5 Absorption from nuclear winds

10.5.1 UV absorption classes

AGN spectra show narrow absorption lines, mini-broad troughs, and broad absorption lines (BALs). Common ions include C IV, Si IV, N V, O VI, Mg II, and metastable or excited transitions. Blueshift establishes motion toward the observer relative to the systemic frame, but distance and mass flux remain unknown without a density or geometry constraint.

Doublet residuals often reveal saturation plus partial coverage. Excited-state to ground-state ratios can constrain density; combining density with the ionization parameter gives a radial estimate

$$r = \left(\frac{Q_{\mathrm{H}}}{4\pi c\, U n_{\mathrm{H}}} \right)^{1/2}, \tag{10.8}$$

where $Q_{\rm H}$ is the ionizing photon rate. The result inherits uncertainty from the unobserved extreme-UV SED, level populations, shielding, and whether the transitions share a phase.

Variability supplies a complementary constraint. A response to continuum changes can bound recombination time and density; transverse changes in covering can instead reflect cloud motion. Distinguishing ionization change from occultation requires multiple ions and a measured ionizing continuum.

10.5.2 X-ray warm absorbers and ultra-fast outflows

Soft-X-ray spectra of many AGN contain photoionized absorption from oxygen, neon, iron, and other elements, often in multiple ionization and velocity components. At Fe K energies, blueshifted Fe XXV and Fe XXVI absorption can indicate velocities near a tenth of the speed of light in ultra-fast outflows (Tombesi et al., 2010; King and Pounds, 2015). The identification is demanding: the continuum and reflection model, detector features, blind energy search, and number of trials all affect significance. Confirmation through multiple transitions or epochs is more persuasive than one residual.

X-ray ionization is commonly described by

$$\xi = \frac{L_{\rm ion}}{n_{\rm H} r^2} \quad [\mathrm{erg\,cm\,s^{-1}}], \tag{10.9}$$

over a stated ionizing energy range. Conventions for $L_{\rm ion}$ differ, so ξ values from different codes or papers are not automatically interchangeable. A distribution of column over ionization parameter may be more realistic than a small number of discrete slabs.

10.6 Outflow mass, momentum, and power

For a thin partial shell of radius R, column $N_{\rm H}$, solid-angle fraction $\Omega/4\pi$, and characteristic speed v, a widely used estimate is

$$\dot{M}_{\rm out} \simeq \Omega\, R\, \mu m_{\rm p} N_{\rm H} v, \tag{10.10}$$

where here Ω is in steradians. Authors sometimes absorb 4π into a dimensionless covering factor; the convention must be stated. The associated momentum and kinetic-power rates are

$$\dot{p}_{\rm out} = \dot{M}_{\rm out} v, \qquad \dot{E}_{\rm kin} = \frac{1}{2}\dot{M}_{\rm out} v^2. \tag{10.11}$$

Another common estimator uses $M_{\rm out} v/R$ or $3M_{\rm out} v/R$ depending on whether the assumed flow is a shell or a volume-filling, steady cone. This factor is not bookkeeping trivia: it encodes a different history and density profile. Radius, deprojection, electron density, conversion factor, and low-velocity separation from the host can each dominate uncertainty. Tables comparing outflow energetics should homogenize these choices before claiming physical trends (Veilleux et al., 2005a; Fiore et al., 2017).

Ionized mass from a recombination line scales roughly as

$$M_{\rm ion} \propto \frac{L_{\rm recomb}}{n_{\rm e} \alpha_{\rm eff} h\nu}, \tag{10.12}$$

so adopting an electron density that is too low inflates the mass. Density doublets may preferentially weight bright dense clumps, while most mass could occupy a lower-density component. Molecular mass from CO requires an $\alpha_{\rm CO}$ conversion; neutral atomic mass from absorption requires a spin temperature and covering factor. Multi-phase outflow budgets are therefore sums of differently conditioned estimates, not directly measured totals (Bolatto et al., 2013a; Cicone et al., 2014).

10.7 Jets and radio diagnostics

Synchrotron emission identifies relativistic electrons and magnetic fields. For $S_\nu \propto \nu^\alpha$, a steep spectrum generally characterizes optically thin lobes while a flat compact core can be a superposition of self-absorbed jet components. Doppler beaming changes observed flux, timescales, and apparent speed. Polarization and Faraday rotation probe ordered fields and intervening magnetized plasma, but depolarization can be internal or external.

Jet mechanical power is often inferred from cavities in hot X-ray gas:

$$P_{\rm cav} \simeq \frac{E_{\rm cav}}{t_{\rm age}}, \qquad E_{\rm cav} \simeq \frac{\gamma}{\gamma - 1} pV. \tag{10.13}$$

For relativistic plasma, the enthalpy is commonly approximated as $4pV$. Buoyancy, sound-crossing, and refill times give different ages. Radio luminosity is an imperfect proxy because synchrotron efficiency, spectral age, and environment vary.

Jets can drive shocks, entrain cool gas, inflate bubbles, or maintain turbulence. A spatial coincidence between radio and line emission is suggestive, but velocity-resolved line ratios and pressure estimates are needed to distinguish shock excitation, compression-enhanced star formation, and AGN photoionization. The Blandford–Znajek and Blandford–Payne mechanisms provide limiting physical pictures for spin-powered and disk-launched outflows (Blandford and Znajek, 1977; Blandford and Payne, 1982).

10.8 X-ray reflection and the Fe K complex

Hard coronal photons striking optically thick material produce a reflected continuum, absorption edges, and fluorescent lines. Near-neutral iron emits Fe Kα near 6.4 keV; He-like and H-like iron produce features near 6.7 and 6.97 keV. Reflection from the inner disk is shifted and skewed by Doppler motion, gravitational redshift, and light bending (George and Fabian, 1991; Reynolds, 2014).

Spin inference uses the inner radius implied by the full relativistic reflection spectrum, not merely the red wing of one line. The fit depends on disk ionization, density, elemental abundance, inclination, coronal illumination profile, continuum curvature, absorption, and the assumption that the disk reaches the innermost stable circular orbit. Robust analyses compare alternative continua and reflection codes, fit a broad energy band, and test whether spin is constrained by the data rather than a parameter boundary.

A narrow 6.4-keV component can arise in distant Compton-thick material. The Compton shoulder, Kβ line, reflection hump, and variability help locate it. In Compton-thick AGN, the direct continuum can be strongly suppressed, making equivalent widths large and naive bolometric corrections unreliable.

10.9 Feedback across phases and timescales

Nuclear winds can begin at sub-parsec scales while molecular outflows are measured over hundreds of parsecs or kiloparsecs. Equality of their instantaneous energy fluxes is not required: the phases are observed at different epochs of an intermittent source. In an ideal momentum-driven picture, shocked wind energy cools efficiently and the large-scale momentum rate is of order L/c. In an ideal energy-conserving bubble, retained thermal energy can produce a larger momentum flux (King and Pounds, 2015). Leakage, gravity, radiative losses, clumpy coupling, and source variability blur this distinction.

Evidence for feedback should connect at least three levels:

1. a credible AGN energy or momentum source;
2. a kinematically and spatially resolved response in surrounding gas;

3. an effect on cooling, star formation, or gas retention beyond what stellar feedback and gravitational motions plausibly explain.

A broad line alone demonstrates disturbed gas, not quenching. Conversely, maintenance-mode heating may prevent cooling without an obvious fast outflow.

10.10 An AGN line-analysis checklist

1. Establish systemic redshift with a documented tracer.
2. Separate instrumental resolution from intrinsic line width.
3. Fit stellar continuum and Balmer absorption before narrow-line ratios.
4. Decompose narrow and broad components jointly across linked lines, respecting fixed atomic ratios only where justified.
5. Test reddening, aperture mixing, and spatially resolved diagnostic ratios.
6. For absorption, fit doublets with optical depth and covering fraction; search all plausible blends.
7. For mass outflow rates, publish the geometry, radius, density or conversion factor, projection convention, and velocity statistic.
8. For X-ray residuals, include continuum and reflection alternatives and the blind-search trials factor.
9. Compare phases at matched aperture and timescale where possible.
10. Distinguish measured luminosity or column from inferred mass, power, and causal feedback.

AGN spectra are richest when treated as a connected system. The UV line that measures a nuclear wind, the optical lines that map an ionization cone, the CO wings that reveal accelerated molecular gas, the radio lobes that carry mechanical energy, and the X-ray lines that probe the corona and fastest wind are not competing descriptions. They are different density, temperature, radius, and time weightings of one evolving engine.

Chapter 11

X-ray, Far-Infrared, Submillimeter, and Radio Diagnostics

11.1 One gas cycle, several observational weightings

No band inventories every phase of cosmic gas. Optical and ultraviolet spectra are highly sensitive to 10^4 K ionized gas and to absorption through low-density material, but dust, cold molecular reservoirs, and hot plasma can remain inconspicuous. X-ray lines expose highly ionized metals; far-infrared fine-structure lines carry cooling from neutral and weakly ionized gas; millimeter rotational lines make cold molecular gas observable; and centimeter data trace atomic hydrogen, free–free emission, synchrotron radiation, and magnetized plasma. These are complementary density, temperature, abundance, opacity, and geometry weightings, not interchangeable mass estimators.

The common discipline is forward modeling. A source spectrum is altered by radiative transfer and cosmological redshift, sampled through a point-spread function or synthesized beam, and redistributed among detector channels. Continuum, aperture, response, and background therefore belong in the definition of every reported line measurement.

11.2 Thermal X-ray plasma

For a low-density, optically thin collisionally excited line,

$$L_{ul} = \int n_{\mathrm{e}} n_{\mathrm{H}} A_X f_{X^i}(T, n_{\mathrm{e}} t) \epsilon_{ul}(T, n_{\mathrm{e}})\, \mathrm{d}V, \tag{11.1}$$

where A_X is elemental abundance, f_{X^i} the ionic fraction, and ϵ_{ul} the emissivity coefficient. The $n_{\mathrm{e}} n_{\mathrm{H}}$ weighting favors dense structure, so a uniform-density conversion from luminosity to mass is biased when gas is clumpy.

In collisional ionization equilibrium (CIE), ionization balances recombination and ion fractions depend mainly on temperature. A recently shocked plasma may be underionized; rapidly cooling or expanding gas may be overionized and recombining. Nonequilibrium calculations introduce an ionization age such as $n_{\mathrm{e}} t$. Photoionized plasma instead responds to an external spectral energy distribution and ionization parameter. One ion can occur in more than one regime, so a temperature assigned from a line name alone is model dependent (Draine, 2011; Sutherland and Dopita, 1993; AtomDB Team, 2025).

H-like Lyα lines, He-like triplets, the Fe L complex, and Fe K lines provide complementary constraints. For a He-like triplet,

$$G = \frac{f + i}{r}, \qquad R = \frac{f}{i}, \tag{11.2}$$

where r, i, and f denote resonance, intercombination, and forbidden components. These ratios respond to excitation mechanism, temperature, density, and ultraviolet pumping, but blends and radiative transfer must be included.

A multiphase source can be represented by a differential emission measure,

$$L_E = \int \Lambda_E(T, Z) \frac{\mathrm{d}EM}{\mathrm{d}\log T}\,\mathrm{d}\log T, \qquad EM = \int n_\mathrm{e} n_\mathrm{H}\,\mathrm{d}V. \tag{11.3}$$

Fitting a one-temperature spectrum to such gas can bias both continuum and abundance. The adopted solar abundance scale must be named (Asplund et al., 2021). Likewise, the cooling relation $\mathcal{L} = n_\mathrm{e} n_\mathrm{H} \Lambda(T, Z)$ hides dependencies on ionization state, abundance pattern, and radiation field.

11.2.1 Absorption, fluorescence, and the detector response

Photoelectric absorption may be written schematically

$$F_E^\mathrm{obs} = F_E^\mathrm{int} \exp[-N_\mathrm{H} \sigma_\mathrm{eff}(E, Z, \xi)]. \tag{11.4}$$

The fitted equivalent hydrogen column inherits assumptions about abundances, cross-sections, ionization, and covering. Fe Kα near 6.4 keV can arise in distant obscuring matter or disk reflection. Near a black hole, Doppler shifts, gravitational redshift, and light bending broaden the full reflection spectrum; an isolated Gaussian is a detection model, not a complete physical interpretation (George and Fabian, 1991; Reynolds, 2014).

Counts predicted in channel j are

$$\mu_j = t_\mathrm{exp} \int R_j(E) A_\mathrm{eff}(E) S(E)\,\mathrm{d}E + B_j, \tag{11.5}$$

so photon models must be folded through redistribution and effective-area responses. Sparse counts call for Poisson likelihoods rather than forced Gaussian errors. A free search in line energy also requires a trials correction. Weak O VII or O VIII absorption claims in the WHIM are especially sensitive to continuum choice, wavelength calibration, fixed-pattern residuals, Galactic foregrounds, and the full searched redshift path (Bregman, 2007; Nicastro et al., 2018).

Microcalorimeters resolve velocity information without dispersion. XRISM Resolve provides few-eV-class spectroscopy over much of its operating band (XRISM Science Operations Team, 2026). For approximately Gaussian components,

$$\sigma_v^2 = \frac{k_\mathrm{B} T}{m_\mathrm{ion}} + \sigma_\mathrm{turb}^2 + \sigma_\mathrm{bulk,unresolved}^2. \tag{11.6}$$

Ions of different mass can help separate thermal and nonthermal widths only if they trace the same volume.

11.3 Far-infrared fine-structure cooling

Low-lying fine-structure transitions can be excited in gas too cool for optical forbidden lines. Central examples are [C II] 158 μm, [O I] 63 and 145 μm, [N II] 122 and 205 μm, [O III] 52 and 88 μm, and [C I] 370 and 609 μm. For an optically thin two-level approximation,

$$n_\mathrm{crit} = \frac{A_{ul}}{\sum_c q_{ul}^{(c)}}. \tag{11.7}$$

A critical density is incomplete unless collider and temperature are stated. Electrons, H, and H_2 have different rate coefficients, and radiative trapping changes the effective excitation requirement.

[C II] is not phase unique: C^+ occurs in ionized gas, diffuse neutral gas, and photodissociation-region (PDR) surfaces. [N II] helps estimate the ionized contribution, while [O I] weights warmer, denser neutral

material but may be optically thick or self-absorbed. [O III] selects harder radiation. A PDR-grid solution is conditional on geometry, abundance, column, filling factor, and the partition of [C II] among phases (Tielens and Hollenbach, 1985; Hollenbach and Tielens, 1999).

The observed decline of [C II]/far-infrared luminosity in many compact luminous systems is called the [C II] deficit. High ionization parameter, grain charging, dust-bounded regions, saturation of the upper level, opacity, and an AGN-heated continuum may all contribute. The name describes a ratio; it does not select one cause.

11.4 Molecular rotational lines

Cold H_2 has no permanent dipole moment, so CO is widely used as a proxy. The velocity-integrated CO flux gives the brightness-temperature luminosity

$$L'_{\rm CO} = 3.25\times10^{7}(S_{\rm CO}\Delta v)\nu_{\rm obs}^{-2}D_{\rm L}^{2}(1+z)^{-3}\quad {\rm K\,km\,s^{-1}\,pc^{2}}, \tag{11.8}$$

for flux in Jy km s^{-1}, frequency in GHz, and distance in Mpc. Molecular mass is then

$$M_{\rm mol} = \alpha_{\rm CO}L'_{\rm CO(1-0)}. \tag{11.9}$$

The factor $\alpha_{\rm CO}$ depends on metallicity, surface density, temperature, dynamics, cloud structure, and the helium convention; it is a physical prior rather than a unit conversion (Bolatto et al., 2013a).

A high-J line additionally requires an excitation ratio $r_{J1} = L'_{\rm CO[J-(J-1)]}/L'_{\rm CO(1-0)}$. A CO spectral-line energy distribution constrains a distribution of density and temperature and rarely has a unique one-component interpretation. Dense-gas tracers such as HCN, HCO^+, and CS are also affected by opacity, trapping, chemistry, and infrared pumping (Carilli and Walter, 2013).

At high redshift, $T_{\rm CMB} = 2.725(1+z)$ K both excites low levels and provides a background. Ignoring it can hide cold gas whose excitation temperature lies only slightly above the background. Main-isotope CO is often optically thick; isotopologues constrain opacity but introduce isotope chemistry and selective photodissociation.

11.5 H I at 21 cm

The hyperfine line has rest frequency 1420.40575177 MHz. A uniform layer obeys

$$T_{\rm B}(v) = T_{\rm s}(1-e^{-\tau}) + T_{\rm bg}e^{-\tau}. \tag{11.10}$$

For optically thin foreground-subtracted emission,

$$N_{\rm HI} = 1.823\times10^{18}\int T_{\rm B}(v)\,{\rm d}v\quad {\rm cm^{-2}}, \tag{11.11}$$

with $T_{\rm B}$ in K and velocity in km s^{-1}. In absorption,

$$N_{\rm HI} = 1.823\times10^{18}\frac{T_{\rm s}}{f_{\rm c}}\int \tau_{\rm obs}(v)\,{\rm d}v. \tag{11.12}$$

Thus absorption alone constrains a combination of spin temperature and covering factor (Chengalur et al., 2013). For unresolved optically thin emission,

$$\frac{M_{\rm HI}}{M_\odot} = 2.356\times10^{5}\frac{D_{\rm L}^{2}}{1+z}\left(\frac{\int S_\nu {\rm d}v}{\rm Jy\,km\,s^{-1}}\right), \tag{11.13}$$

with $D_{\rm L}$ in Mpc. Self-opacity, beam dilution, stray radiation, and resolved-out diffuse structure violate simple conversions.

11.6 Radio continuum, magnetic fields, and interferometers

Optically thin free–free radiation is tied to the ionizing-photon rate, whereas synchrotron emission depends on cosmic-ray electrons, magnetic fields, losses, and transport. Their spectral decomposition is approximate when regions are mixed or when low-frequency absorption and curvature matter (Condon and Ransom, 2016; Kennicutt and Evans, 2012).

For a simple foreground screen,

$$\chi(\lambda) = \chi_0 + \mathrm{RM}\lambda^2, \qquad \mathrm{RM} = 0.812 \int n_\mathrm{e} B_\parallel \, \mathrm{d}l \quad \mathrm{rad\,m^{-2}}, \tag{11.14}$$

with n_e in $\mathrm{cm^{-3}}$, $B_\parallel$ in μG, and path in pc. Internal rotation, multiple components, and beam depolarization require Faraday-depth modeling rather than one slope.

An interferometer samples Fourier components:

$$V(u,v) = \int A(l,m) I(l,m) e^{-2\pi i(ul+vm)} \, \mathrm{d}l \, \mathrm{d}m. \tag{11.15}$$

The beam, weighting, flagging, and deconvolution are part of the measurement. Long baselines sharpen resolution but do not ensure surface-brightness sensitivity; absent short spacings suppress large-scale flux. A reproducible line cube records channel response, synthesized beam, primary-beam correction, continuum subtraction, mask construction, and correlated noise.

11.7 A cross-band audit

The strongest combinations break different degeneracies: Hα plus free–free emission tests extinction; [N II] partitions [C II]; CO ladders constrain excitation while dust supplies an independent mass prior; and H I emission plus absorption constrains spin temperature. Comparisons require matched aperture, beam, velocity interval, and calibration convention.

Always preserve the inference chain

$$\begin{aligned} &\text{counts or visibilities} \rightarrow \text{line flux and profile} \rightarrow \text{excitation and opacity} \\ &\rightarrow \text{column or mass} \rightarrow \text{physical claim.} \end{aligned}$$

A bright line need not trace the dominant mass, a nondetection is not a phase-independent zero, critical density is not a hard boundary, and a higher-resolution interferometric image may recover less total flux. Every arrow in the chain adds assumptions; an atlas should display them beside the diagnostic.

Chapter 12

Redshift Windows, Instrument Choice, and Survey Planning

12.1 Begin with the observable, not the facility

An observing plan should start with a physical measurement: ionic column, line-ratio temperature, gas velocity, density, luminosity, spatial gradient, or population distribution. The line, required precision, and confounding features then determine wavelength coverage, resolving power, sensitivity, aperture, and sample design. Beginning with an available instrument and only later asking what its spectrum measures invites an exposure-time calculation for the wrong observable.

Cosmological redshift maps rest and observed coordinates through

$$\lambda_{\rm obs} = (1+z)\lambda_0, \qquad \nu_{\rm obs} = \frac{\nu_0}{1+z}. \tag{12.1}$$

An instrument spanning $\lambda_{\rm min}$ to $\lambda_{\rm max}$ can observe a line over

$$z_{\rm min} = \max\left[0, \frac{\lambda_{\rm min}}{\lambda_0} - 1\right], \qquad z_{\rm max} = \frac{\lambda_{\rm max}}{\lambda_0} - 1, \tag{12.2}$$

before accounting for gaps, throughput, atmospheric absorption, detector edges, and configuration. A line lying formally inside the range can still be unusable.

12.2 Resolution, sampling, and velocity information

Resolving power and approximate velocity resolution are

$$R = \frac{\lambda}{\Delta\lambda}, \qquad \Delta v \simeq \frac{c}{R}. \tag{12.3}$$

This Δv describes an instrumental resolution element, not necessarily the centroid precision. A high signal-to-noise unresolved line can be centroided more precisely than its width, while intrinsic width requires deconvolution of the measured line-spread function (LSF). Real LSFs may have non-Gaussian wings and depend on wavelength, source extent, slit illumination, detector position, or observing epoch.

Sampling and resolution are distinct. Two or more pixels per resolution element may sample an LSF, but correlated extraction and resampling mean the pixels are not independent measurements. For doublets or kinematic components, compute separation in velocity and simulate the convolved profile at the anticipated signal-to-noise ratio. A nominal R alone does not prove that the scientific parameters are identifiable.

Extended-source spectroscopy adds spatial–spectral coupling. A slit can turn a velocity gradient or off-center illumination into an apparent line shift. In a slitless or dispersive X-ray spectrum, source extent broadens features along the dispersion direction. Integral-field spectroscopy removes some aperture ambiguity but leaves point-spread-function mixing between spaxels.

12.3 Flux, luminosity, surface brightness, and exposure time

For an integrated line flux $F_{\rm line}$,

$$L_{\rm line} = 4\pi D_{\rm L}^2 F_{\rm line}. \tag{12.4}$$

Flux density transformations require the appropriate $(1+z)$ factor:

$$F_{\nu,\rm obs}(\nu_{\rm obs}) = \frac{L_{\nu,\rm em}[(1+z)\nu_{\rm obs}]}{4\pi D_{\rm L}^2 (1+z)}. \tag{12.5}$$

Equivalent width in wavelength units scales as $W_{\rm obs} = (1+z)W_0$. Surface brightness from a resolved source suffers Tolman dimming proportional to $(1+z)^{-4}$, apart from bandpass and evolutionary effects. A high-redshift galaxy can have detectable integrated flux while its diffuse halo remains inaccessible per resolution element.

For background-dominated counting, signal-to-noise grows roughly as $t^{1/2}$, but read noise, source Poisson noise, cosmic rays, confusion, calibration floors, and overheads modify the scaling. Radio sensitivity also depends on bandwidth, integration time, number of baselines, weighting, and weather. Official exposure calculators are preferable to scaling a published observation with different source size and background.

The correct sensitivity unit follows the science. An unresolved-line limit in erg $\rm s^{-1}$ $\rm cm^{-2}$, a brightness-temperature limit per beam per channel, and a column-density limit per velocity interval cannot be substituted for one another. For a Gaussian line, the peak signal changes inversely with assumed width at fixed integrated flux; every upper limit must state its adopted width.

12.4 Atmospheric and foreground windows

Ground-based optical and infrared data encounter telluric absorption, night-sky emission, thermal background, and variable transmission. Redshift selection can place a target line on an OH skyline or inside a water band even when it lies within the nominal spectrograph range. A planning calculation should propagate the target redshift uncertainty and barycentric velocity, plot all required lines against sky and transmission spectra, and reserve continuum on both sides.

Radio-frequency interference and Galactic foreground emission are the analogous frequency-dependent hazards in radio work. At X-ray energies, detector background, solar-wind charge exchange, Galactic hot gas, and photoelectric absorption can dominate faint diffuse sources. Space observing removes the terrestrial atmosphere but not foregrounds, scattered light, detector systematics, or scheduling constraints.

12.5 A current facility map

The summary below is a planning orientation, not a substitute for the call-specific handbook. Instrument modes, offered filters, sensitivities, and status change. Values were checked against official documentation available in September 2026 and should be rechecked at proposal submission.

Table 12.1: Representative facilities and line-spectroscopy roles. Coverage and resolution are deliberately approximate.

Facility or mode	Approximate coverage or resolution	Strong use case	Planning caution
HST/COS FUV and NUV	roughly 900–3200 Å across modes; medium-resolution FUV modes commonly $R \sim 10^4$–2.4×10^4	high-throughput UV absorption in the low-redshift CGM and IGM	coverage has detector gaps and lifetime-position constraints; use the current LSF and sensitivity tables (Dixon et al., 2026)
HST/STIS	UV through optical, with low to high spectral resolution by grating	high-resolution UV/optical work, spatially resolved slits, and wavelengths outside optimal COS modes	usually lower UV throughput than COS; slit and extended-source effects matter
JWST/NIRSpec	0.6–5.3 μm; prism $R \sim 100$, gratings near $R \sim 1000$ or 2700	rest-optical diagnostics at high redshift; fixed slits, microshutters, and IFU	configuration gaps, source centering, spectral overlap, and wavelength- dependent LSF (Space Telescope Science Institute, 2026b)
JWST/MIRI	mid-infrared imaging and spectroscopy; LRS and medium-resolution IFU modes	warm molecular lines, ionic fine-structure lines, dust, and obscured nuclei	thermal/background behavior, fringing, and spatially varying IFU response (Space Telescope Science Institute, 2026a)
VLT/UVES	approximately 300–1100 nm; typical slit resolving power near 4×10^4 and higher in narrow configurations	precise optical absorption, metal systems, and kinematics	setting-dependent gaps, slit losses, atmospheric cutoff, and tellurics (European Southern Observatory, 2026)
VLT/X-shooter	simultaneous UV to near-IR, approximately 0.3–2.48 μm; moderate R	broad line inventories and uncertain-redshift transients or galaxies	resolution changes by arm and slit; telluric and inter-arm calibration
VLT/MUSE	optical integral-field spectroscopy, approximately 465–930 nm in common modes; R of a few thousand	spatially resolved emission, Lyα halos, and blind line searches	sky residuals, PSF chromaticity, and surface-brightness selection
ALMA	receiver bands collectively cover much of 35–950 GHz; offered modes are cycle specific	CO and other molecules, [C I], redshifted fine-structure lines, and dust	atmospheric windows, configuration, maximum recoverable scale, and excitation (ALMA Observatory, 2026)

Facility or mode	Approximate coverage or resolution	Strong use case	Planning caution
Karl G. Jansky VLA	centimeter coverage chiefly across 1–50 GHz plus lower-frequency systems	H I, radio recombination lines, continuum, polarization, and molecular masers	radio-frequency interference, array configuration, bandwidth smearing, and missing short spacings (National Radio Astronomy Observatory, 2026)
XRISM/Resolve	non-dispersive few-eV-class X-ray spectroscopy over its operating band	hot-plasma velocities, temperatures, abundances, and line complexes	use current in-orbit effective-area and gate-valve status, not prelaunch curves (XRISM Science Operations Team, 2026)
DESI	thousands of simultaneous fibers over a wide optical band	large quasar samples, Lyα forest statistics, galaxy redshifts	fiber aperture, target selection, spectral covariance, and survey masks (DESI Collaboration, 2016, 2025)
SDSS-V	multi-epoch optical and near-infrared spectroscopy plus local-volume mapping	Milky Way, black-hole variability, and spatially resolved nearby-galaxy surveys	subsurvey selection and data-release-specific calibrations (Kollmeier et al., 2017; Sloan Digital Sky Survey, 2026)

12.6 Worked window logic

For Lyα at 1215.67 Å, an optical spectrograph beginning at 3600 Å first accesses the line near $z = 1.96$. This arithmetic says nothing about the atmospheric cutoff, quasar brightness, forest path length, or whether higher-order Lyman lines are covered. For Hα at 6562.8 Å, the NIRSpec upper wavelength near 5.3 μm corresponds formally to $z \approx 7.1$; in practice sensitivity, configuration, line width, and neighboring [N II] determine feasibility.

For a line with rest frequency 115.271 GHz, CO(1–0) enters a 30 GHz receiver near $z = 2.84$. A detected feature at one frequency is not by itself a secure rotational identification: adjacent CO transitions at other redshifts, [C I], and fine-structure lines can be confused. Wide bandwidth, a continuum counterpart, a second line, or an external redshift collapses this ambiguity.

Velocity coverage must also include uncertainty. A systemic redshift error of 300 km s^{-1} is comparable to an entire narrow-band tuning margin in some experiments. For outflows or cluster atmospheres, expected wings and member velocities may require thousands of km s^{-1} around systemic. Baseline for continuum fitting belongs inside the requested band.

12.7 A reproducible planning workflow

1. State the primary physical parameter and the accuracy needed to change the scientific conclusion.
2. List required and corroborating lines with vacuum rest wavelengths or rest frequencies, atomic database version, and expected ratios.

3. Propagate the full redshift probability distribution into observed wavelength, not just the best redshift.
4. Overlay configurations, detector gaps, skylines, telluric bands, foregrounds, and known blends.
5. Choose R by forward-simulating the narrowest components or doublets that must be separated.
6. Predict line flux, width, spatial size, continuum, and background in the exact aperture or beam used by the sensitivity calculation.
7. Model acquisition, slit or shutter losses, dithers, nods, mosaics, calibration, overhead, and maximum recoverable scale.
8. Define the detection statistic, extraction aperture, upper-limit width, and treatment of nondetections before inspecting the result.
9. Include a validation route: a doublet partner, independent redshift, second transition, blank sky, standard star, or repeat epoch.
10. Record software, calibration reference files, ephemeris corrections, data release, exposure-calculator version, and all manual masks.

12.8 Survey selection is part of the astrophysics

A targeted program and a survey answer different questions. Target selection based on UV brightness can exclude dusty galaxies; requiring background quasars weights CGM sightlines by the quasar luminosity function and angular availability; fiber collisions and finite apertures reshape environmental samples; line-flux selection favors high equivalent width. Population statistics require a selection function $S(\boldsymbol{x})$ that enters the likelihood or completeness correction.

For an absorption survey, the usable path length depends on wavelength masks, signal-to-noise, proximity exclusions, and the column and width of a hypothetical line. An incidence estimate is therefore based on effective path, not merely number of spectra. For an emission-line survey, inject-and-recover tests should span size, width, continuum, and wavelength because completeness is multidimensional.

Large facilities do not eliminate small systematics. DESI has transformed Lyα forest statistics through hundreds of thousands of spectra, while SDSS-V adds multi-epoch and spatial dimensions; their power comes from documented calibration and selection as much as sample size (DESI Collaboration, 2025; Sloan Digital Sky Survey, 2026). A line atlas should link each diagnostic to both the physical model and the survey transfer function.

12.9 The final pre-proposal audit

Before submission, verify five boundaries: the line is in a high-throughput region for the full redshift uncertainty; the resolution answers the stated kinematic or deblending question; the sensitivity uses the correct line width and source extent; the spatial configuration recovers the relevant scales; and at least one alternative identification or systematic has a planned test. Then repeat the check against the current official call. A proposal built around a static instrument table can become obsolete, but a plan built around explicit observables and a reproducible forward model remains portable.

Chapter 13

Inference, Modelling, and Reproducible Line Workflows

A measured line flux is the endpoint of a reduction model and the input to a physical model. Treating it as a context-free number loses calibration covariance, nondetections, selection, response information, and the assumptions used to separate line from continuum. A reproducible workflow instead carries a generative model from latent source spectrum to detector samples, exposes the approximation used at each compression step, and stores enough provenance to repeat the result.

13.1 Begin with the data-generating process

Let $\boldsymbol{s}(\boldsymbol{\theta})$ be a rest-frame source spectrum for physical parameters $\boldsymbol{\theta}$, A an aperture or spatial mixing operator, R the spectral response, and $B\boldsymbol{\phi}$ a background model. A general expectation in detector space is

$$\boldsymbol{\mu}(\boldsymbol{\theta}, \boldsymbol{\phi}) = R\, A\, \boldsymbol{s}(\boldsymbol{\theta}) + B\boldsymbol{\phi}. \tag{13.1}$$

For photon counts n_i, the elementary likelihood is

$$\ln \mathcal{L}_{\mathrm{P}} = \sum_i \left[n_i \ln \mu_i - \mu_i - \ln \Gamma(n_i + 1) \right]. \tag{13.2}$$

Background counts measured in another region should be modeled jointly with their area and exposure scaling, not subtracted and assigned an ad hoc Gaussian error. This is the logic behind count-space statistics used in X-ray spectroscopy (Cash, 1979; Arnaud, 1996).

For approximately Gaussian extracted spectra $\boldsymbol{y}$ with covariance C, the corresponding log likelihood is

$$\ln \mathcal{L}_{\mathrm{G}} = -\frac{1}{2} \left[(\boldsymbol{y} - \boldsymbol{m})^{\mathsf{T}} C^{-1} (\boldsymbol{y} - \boldsymbol{m}) + \ln |C| + N \ln(2\pi) \right]. \tag{13.3}$$

Resampling, sky subtraction, continuum fitting, common calibration, PSF matching, and shared denominators create covariance. Ignoring off-diagonal terms can narrow intervals and exaggerate the information content of a line profile. If full C is too large, preserve a sparse covariance, a correlation kernel, or the linear operator that generated the spectrum. Merely inflating diagonal errors does not reproduce parameter-dependent correlations.

Equation 13.1 is also a reminder not to "deconvolve first" unless deconvolution is itself part of the likelihood. A model profile should normally be convolved with the wavelength-dependent line-spread function and integrated over detector pixels. For an integral-field cube, the spatial PSF, slice geometry, differential atmospheric refraction, and source velocity field can couple spatial and spectral structure. Beam smearing can then mimic a broad component or flatten a velocity gradient. Forward-modeling the native samples avoids asking a noisy inverse operation to create information.

13.1.1 Fit the continuum and line together when they interact

A local line model might include a polynomial or physically motivated continuum, one or more response-convolved profiles, and nuisance terms for sky residuals. Continuum uncertainty is especially important for

weak equivalent widths and broad lines. Estimating a continuum, subtracting it, and then fitting the line as if the continuum were known discards covariance. Simultaneous fitting or posterior propagation preserves it.

Line parameters need constraints that represent physics without manufacturing detections. A doublet may share velocity and width and have a ratio bounded by atomic physics; blended species need not share kinematics. Positive-only flux priors change the null distribution at zero. Consequently, the improvement in χ^2 from adding a non-negative line does not necessarily follow the usual asymptotic likelihood-ratio law. Calibrate searches with posterior predictive checks or simulations that reproduce the continuum fit and look-elsewhere scan (Protassov et al., 2002).

13.1.2 Identifiability, priors, and parameterization

A parameter is identifiable only to the extent that distinct values predict distinguishable data after nuisance parameters and response are considered. For an optically thin single line, column density, emitting area, abundance, and ion fraction may appear only as a product. No sampler can separate them without additional information. Report identifiable combinations, add well-motivated complementary data, or acknowledge prior dominance.

Priors should be specified in the parameterization where the scientific belief is intended. Uniform priors in flux, log flux, equivalent width, and line amplitude divided by noise are not equivalent. A bounded "uninformative" prior can set an implicit physical scale through its endpoints. Transforming from positive width σ to $\log \sigma$ changes the density by a Jacobian. Prior predictive simulation - drawing parameters from the prior and generating synthetic spectra - is an efficient way to expose absurd widths, fluxes, doublet ratios, or continua before seeing the data.

Parameterization also affects computation. Centered hierarchical models can mix poorly when individual lines are weak, whereas a non-centered form can be more stable. Highly covariant amplitude and width may benefit from parameters closer to integrated flux and log width. Convergence diagnostics should be reported for the actual posterior draws: multiple chains, effective sample size, rank-based diagnostics, trace inspection, and sensitivity to initialization. A maximum-likelihood covariance matrix is a local curvature approximation and can miss boundaries, tails, multiple modes, and discrete alternatives.

13.1.3 Model comparison without a significance shortcut

Deciding between one and two kinematic components is a scientific model- comparison problem, not simply a request for the model with lower residual sum of squares. Information criteria approximate predictive tradeoffs under specific regularity assumptions; marginal likelihoods depend on the full prior volume; cross-validation evaluates predictive performance for a declared unit of held-out data. Mixture components and lines at a boundary violate common asymptotic assumptions. The preferred tool therefore depends on the claim.

When the question is whether a broad outflow component is required, simulate spectra from the best adequate narrow-component model, repeat the same search over centroid and width, and compare the chosen detection statistic with the simulated distribution. When the question is prediction for new spectra, leave out independent objects or exposures rather than correlated pixels. When physical alternatives are all approximations, present model-averaged or model-stratified results instead of converting a small score difference into certainty about ontology.

Do not confuse a fit with a measurement

A converged optimizer establishes neither identifiability nor calibration. Inspect alternative starting points, parameter correlations, residual structure, prior sensitivity, and simulation-based recovery. Quote the observable and its likelihood or posterior, not only the best-fit parameter vector.

13.2 Ratios, calibration terms, and uncertainty budgets

For a ratio $r = F_1/F_2$, first-order propagation gives

$$\mathrm{Var}(\ln r) = \frac{\mathrm{Var}(F_1)}{F_1^2} + \frac{\mathrm{Var}(F_2)}{F_2^2} - 2\frac{\mathrm{Cov}(F_1, F_2)}{F_1 F_2}. \tag{13.4}$$

The covariance term is often beneficial when both lines share flux calibration, aperture correction, or continuum placement. It can also be negative after deblending. Treating catalogue columns as independent can therefore either overstate or understate ratio uncertainty.

A useful hierarchical calibration model for observation j is

$$F_{ij}^{\mathrm{obs}} = g_j\, a_{ij}\, F_i^{\mathrm{true}} + \epsilon_{ij}, \qquad \ln g_j \sim \mathcal{N}(0, \sigma_{g,j}^2), \tag{13.5}$$

where g_j is a shared multiplicative calibration and a_{ij} represents line-dependent attenuation or aperture effects. Marginalizing these nuisance terms propagates correlations naturally. Keep statistical repeatability, absolute calibration, model discrepancy, atomic data, distance, lensing, attenuation, and aperture terms separate in the budget; they average down in different ways.

Avoid constructing a diagnostic analysis entirely from logarithmic ratios when one or more denominator lines are weak. The ratio likelihood becomes skewed and heavy-tailed, and a catalogue cut on both numerator and denominator creates a complicated truncated sample. A safer population model predicts the vector of line fluxes, applies attenuation and calibration, and compares that vector with all measured flux estimators and their covariance. Ratios can still be plotted as summaries. The model, however, retains the information in the overall normalization and admits negative noise realizations without undefined logarithms.

Table 13.1: A minimum uncertainty ledger for a derived line quantity.

Class	Example	Recommended representation
Counting/noise	source, sky, read noise	detector-space likelihood or covariance
Reduction	flat field, sky basis, extraction	nuisance parameters or reduction ensembles
Calibration	throughput, beam, standard star	shared multiplicative/additive terms
Model	continuum, profiles, phase mixture	competing models and posterior predictive checks
Physics data	collision strengths, abundances	versioned grid and sensitivity reruns
Population	distance, lensing, selection	hierarchical latent variables and selection function

13.3 Nondetections and censored information

An upper limit is not a flux measurement with value zero. If an analysis retains a Gaussian estimator $\hat{F}$ and uncertainty σ_F, use its full likelihood even when $\hat{F}/\sigma_F$ fails a reporting threshold. If only the event $\hat{F} < F_{\mathrm{lim}}$ is available, its likelihood under a model predicting $F(\boldsymbol{\theta})$ is

$$p(\hat{F} < F_{\mathrm{lim}} \mid \boldsymbol{\theta}) = \Phi\left(\frac{F_{\mathrm{lim}} - F(\boldsymbol{\theta})}{\sigma_F}\right), \tag{13.6}$$

where Φ is the normal cumulative distribution. This censored likelihood is less informative than the original estimator. Publishing signed fluxes and uncertainties, alongside quality flags, generally supports better population inference than publishing limits alone.

Classical survival-analysis methods are valuable when their censoring assumptions hold (Feigelson and Nelson, 1985). Astronomical censoring is often informative: the limit varies with wavelength, background, source size, and target selection, all of which can correlate with the physical quantity. Then the measurement and selection process must be modeled jointly. Stacking is not a universal escape. A mean stack estimates a weighted population functional set by the normalization, weights, and missing data; it does not recover the median object or the ratio of mean latent luminosities without assumptions.

13.4 Selection effects and population likelihoods

Suppose objects have latent parameters $\boldsymbol{\theta}$ drawn from $p(\boldsymbol{\theta} \mid \boldsymbol{\Lambda})$ and enter the sample with probability $P_{\rm det}(\boldsymbol{\theta})$. Conditioning on a detected sample introduces the efficiency

$$\alpha(\boldsymbol{\Lambda}) = \int P_{\rm det}(\boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \boldsymbol{\Lambda})\, d\boldsymbol{\theta}. \tag{13.7}$$

For fixed detected sample size, a schematic population likelihood includes one factor of α^{-1} per object,

$$\mathcal{L}(\boldsymbol{\Lambda}) \propto \prod_{k=1}^{N} \frac{\int p(d_k \mid \boldsymbol{\theta}_k) p(\boldsymbol{\theta}_k \mid \boldsymbol{\Lambda})\, d\boldsymbol{\theta}_k}{\alpha(\boldsymbol{\Lambda})}. \tag{13.8}$$

The exact form changes if the total event rate is modeled, but the principle is stable: selection normalizes the population model (Mandel et al., 2019).

Injection and recovery is the most direct way to estimate a complicated $P_{\rm det}$. Inject sources spanning flux, width, wavelength, continuum, size, and environment into realistic data; run the exact detection and quality pipeline; and model recovery probability in that multi-dimensional space. Injecting only isolated Gaussian lines into white noise misses sky residuals, blends, continuum errors, and human vetting. Avoid applying a one-dimensional "90 percent flux completeness" curve to a population whose width or size distribution evolves.

13.5 Physical models: useful approximations and model discrepancy

13.5.1 Photoionized nebulae and slabs

Photoionization codes solve ionization, thermal balance, and radiative transfer through a specified geometry and incident spectrum. Cloudy is widely used for nebulae, AGN, and diffuse gas (Ferland et al., 2017). A grid must record code version, atomic-data set, incident spectral energy distribution, gas density or pressure law, abundances and depletion, dust prescription, stopping criterion, geometry, covering factor, and whether emission is integrated or emergent from a specified face. "Metallicity" alone is not a complete model coordinate.

Interpolation across a grid should be tested against direct runs, especially near ionization fronts or thermal instabilities. Grid priors are not innocuous: a uniform prior in U differs from one uniform in $\log U$, and a rectangular grid can assign weight to chemically inconsistent abundance combinations. Model discrepancy should be included when observed line ratios cannot be matched within measurement errors; otherwise the posterior may become extremely precise about the least-wrong point in an inadequate model.

Large physical grids are often replaced by interpolators or machine-learning emulators. Validate the emulator on withheld grid points using errors in the space actually fit - often log luminosity or line ratio - and propagate emulator uncertainty. Dense validation near sharp ionization fronts and grid edges is more valuable than a small global root-mean-square error. Never extrapolate silently: mark proposals outside the training hull, expand the physical grid, or assign a discrepancy wide enough to express ignorance. Archive the training grid, preprocessing, emulator weights, and code version alongside posterior products.

13.5.2 Collisionally ionized and nonequilibrium plasma

Hot-plasma emissivities depend on electron temperature, ion fractions, abundances, and atomic rates. Databases such as CHIANTI provide versioned atomic data and software for optically thin plasma calculations (Dere et al., 1997; Del Zanna et al., 2021). Collisional ionization equilibrium assumes that ionization and recombination have had time to balance at one temperature. Cooling flows, shocks, interfaces, and rapidly heated gas can be in nonequilibrium; ion populations then remember thermal history. Resonant scattering, charge exchange, non-Maxwellian electrons, or optical depth can break an otherwise plausible one-temperature fit.

The observed X-ray count model must be folded through the response and include foreground absorption. A multi-temperature plasma fit by one temperature can bias abundances. Comparing multiple plasma codes or atomic-data releases is a practical model-systematics test, not an optional software preference.

13.5.3 From line emission to mass or rate

Every conversion should be written as a conditional model. Examples include $M_{\rm mol} = \alpha_{\rm CO} L'_{\rm CO}$, an ionized-gas mass proportional to $L_{\rm recomb}/(n_e \alpha_{\rm eff})$, and a star-formation calibration based on ionizing-photon production. The conversion coefficient depends on physical state, composition, stellar population, IMF, leakage, and geometry. Report the direct observable alongside the derived value so later work can replace the calibration without reverse engineering it (Kennicutt and Evans, 2012).

13.6 Validation: can the inference recover known truth?

Posterior predictive checks draw replicated data from the fitted model and compare features relevant to the claim: residual autocorrelation, line wings, doublet consistency, nondetection fraction, or ratio distribution. A good global χ^2 can coexist with a systematic failure in the diagnostic of interest. Simulation-based calibration tests whether Bayesian credible ranks are uniform when parameters and data are drawn from the stated prior and model. Coverage experiments test frequentist intervals across controlled truth. These checks address computation and calibration under the assumed model; they do not prove that the assumed model describes nature. Stress tests should therefore inject departures that are scientifically plausible: non-Gaussian line wings, spatially varying ratios, mismatched continua, correlated sky residuals, or a contaminating transition at an uncertain redshift.

Blind analysis can reduce experimenter-induced tuning for delicate detections. Negative controls, off-source apertures, inverted spectra, and line-free wavelengths reveal false positives. Split-data consistency is useful only when the split isolates plausible failure modes, such as detector, epoch, polarization, nod position, or array configuration. Repeatedly choosing the split that looks best creates another trials factor.

13.7 Provenance and machine-readable products

Reproducibility requires more than a code repository. Preserve immutable raw identifiers and checksums; calibration reference names; software and dependency versions; configuration files; random seeds; environment or container digest; commands; and links between inputs and outputs. Use stable object and line IDs, machine-readable units, explicit missing-value semantics, and documented coordinate and time frames. FITS remains central in astronomy, while ECSV, ASDF, HDF5, Parquet, and domain-specific models can preserve richer structure; the correct choice is the one accompanied by a schema and validation. The FAIR principles - findable, accessible, interoperable, and reusable - are a helpful publication target, but “accessible” need not mean unrestricted when privacy, proprietary periods, or facility policy apply (Wilkinson et al., 2016). Metadata and a documented route to authorized access can remain public while controlled data stay protected.

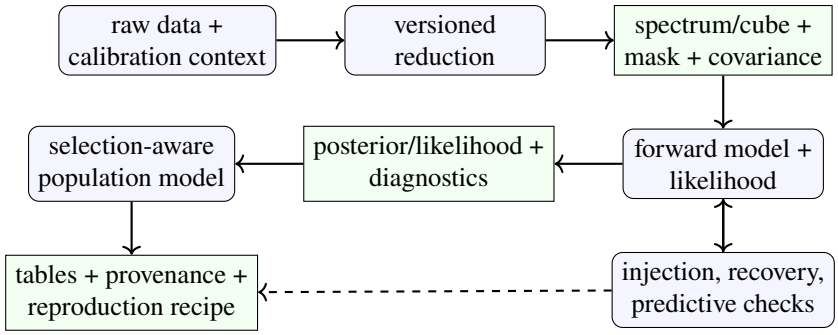


Figure 13.1: A reproducible line-analysis graph. Green artifacts are preserved interfaces; blue stages are deterministic or explicitly stochastic workflows. Validation is part of the inference loop, not a decorative final plot.

A line catalogue should minimally carry rest wavelength or frequency with source and medium, transition identifier, observed centroid and frame, integrated flux, signed estimator and uncertainty, width and its definition, continuum and equivalent width, aperture, response or resolution, covariance reference, detection procedure, quality mask, and provenance pointer. Store posterior samples or a reusable likelihood for nonlinear or multimodal results when practical. A symmetric error bar is not an adequate serialization of a bounded, skewed, or mixture posterior.

Continuous integration should rebuild representative reductions or synthetic tests, validate schemas and units, verify bibliography and documentation links, and compare scientific outputs within justified tolerances. Exact bitwise reproduction is desirable for deterministic stages but can be brittle across hardware; for stochastic inference, test seeds, distributional summaries, and coverage. Tag the release used for a publication and archive it with a durable identifier. A notebook can communicate exploration, but a parameterized, non-interactive pipeline should produce the claimed tables and figures from declared inputs.

A robust pipeline separates pure transformations from side effects. A command should accept declared inputs and configuration, write outputs to a new versioned location, and emit a manifest containing checksums and environment identity. Workflow engines can execute a directed acyclic graph, but the graph alone is not provenance if calibration files are downloaded by mutable URL or notebooks depend on hidden state. Cache keys should include scientific configuration and reference-data versions, not only source-code commits. Manual quality decisions belong in append-only, machine-readable annotations with reviewer, timestamp, reason, and supersession history.

13.8 A compact analysis contract

Before publication, write a one-page contract stating: the primary observable; the detector-space likelihood; response and background treatment; priors and nuisance parameters; selection function; censoring policy; atomic/model versions; uncertainty ledger; validation tests; and exact data/code release. This contract does not eliminate scientific judgment. It makes judgment visible, testable, and replaceable - the essential properties of a durable line measurement.

Part IV

Reference Atlas

Chapter 14

The Line-by-Line Atlas

The following reference pages are generated from the same source-cited catalogue used by the interactive website. Wavelength convention and precision are recorded for each entry whenever the source distinguishes air and vacuum values.

14.1 H I 21 cm

At a glance	
Rest coordinate	1420.405751768 MHz
Band	radio / L band
Transition	1s ground-state hyperfine F=1 -> 0
Mechanism	magnetic-dipole hyperfine transition
Primary tracers	neutral atomic hydrogen column; gas kinematics; rotation curves; spin temperature with absorption; large-scale structure via intensity mapping
Typical environments	cold and warm neutral medium; galaxy disks; high-velocity clouds; damped Ly-alpha systems in absorption; reionization-era IGM statistically
Indicative regime	CNM roughly 20-200 K; WNM roughly 5000-10000 K; detectable outside strict LTE
Density guidance	Not a conventional density diagnostic. Emission gives optically thin H I column; emission plus absorption constrains spin temperature and covering fraction.

H I 21 cm is produced by magnetic-dipole hyperfine transition and is encountered in cold and warm neutral medium; galaxy disks; high-velocity clouds; damped Ly-alpha systems in absorption; reionization-era IGM statistically. It is commonly used to trace neutral atomic hydrogen column; gas kinematics; rotation curves; spin temperature with absorption; large-scale structure via intensity mapping.

How to use it

Use this transition with its diagnostic companions to constrain neutral atomic hydrogen column; gas kinematics; rotation curves; spin temperature with absorption; large-scale structure via intensity mapping, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Rest frequency is exceptionally precise. State the radio/optical/relativistic velocity convention and reference frame. Self-absorption and optical depth can hide cold gas.

Useful companions

Ly-alpha 1215.6701 A; CO(1-0) 115.2712018 GHz; OH 1665/1667 MHz

Sources

NRAO_HI_OH; NIST_ASD; DRAINE_2011

14.2 OH 1665 MHz main line

At a glance

Rest coordinate	1665.4018 MHz
Band	radio / 18 cm
Transition	ground-state 2Pi3/2, J=3/2 hyperfine main-line transition
Mechanism	lambda-doubling plus hyperfine transition; often masing
Primary tracers	molecular gas magnetic field via Zeeman splitting; maser kinematics; pumped dense gas
Typical environments	massive star-forming regions; circumstellar envelopes; compact H II regions; megamaser hosts
Indicative regime	typically tens to hundreds of kelvin; non-LTE pumping controls masers
Density guidance	Population inversion depends on FIR pump, OH column and density; not a simple one-number density tracer.

OH 1665 MHz main line is produced by lambda-doubling plus hyperfine transition; often masing and is encountered in massive star-forming regions; circumstellar envelopes; compact H II regions; megamaser hosts. It is commonly used to trace molecular gas magnetic field via Zeeman splitting; maser kinematics; pumped dense gas.

How to use it

Use this transition with its diagnostic companions to constrain molecular gas magnetic field via Zeeman splitting; maser kinematics; pumped dense gas, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Compare 1665:1667 and satellite-line ratios to diagnose LTE departures, pumping and opacity. Strongly affected by radio-frequency interference in some bands.

Useful companions

OH 1667 MHz; OH satellite lines 1612 and 1720 MHz; H2O 22 GHz maser

Sources

`NRAO_HI_OH; JPL; SPLATALOGUE`

14.3 OH 1667 MHz main line

At a glance

Rest coordinate	1667.35903 MHz
Band	radio / 18 cm
Transition	ground-state 2Pi3/2, J=3/2 hyperfine main-line transition
Mechanism	lambda-doubling plus hyperfine transition; thermal absorption/emission or maser
Primary tracers	molecular gas; dust-obscured mergers; magnetic fields; line-of-sight velocity
Typical environments	diffuse molecular clouds; star-forming regions; OH megamaser galaxies; circumstellar envelopes
Indicative regime	typically 10-300 K, with excitation set by collisions and radiative pumping
Density guidance	Line ratios are excitation/opacity diagnostics rather than direct density meters.

OH 1667 MHz main line is produced by lambda-doubling plus hyperfine transition; thermal absorption/emission or maser and is encountered in diffuse molecular clouds; star-forming regions; OH megamaser galaxies; circumstellar envelopes. It is commonly used to trace molecular gas; dust-obscured mergers; magnetic fields; line-of-sight velocity.

How to use it

Use this transition with its diagnostic companions to constrain molecular gas; dust-obscured mergers; magnetic fields; line-of-sight velocity, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The 1667-MHz line usually dominates extragalactic OH megamasers. Hyperfine LTE expectations can fail spectacularly under FIR pumping.

Useful companions

OH 1665 MHz; OH 1612/1720 MHz; FIR OH pump transitions

Sources

`NRAO_HI_OH; JPL; SPLATALOGUE`

14.4 H2O 22 GHz maser

At a glance

Rest coordinate	22.23508 GHz
Band	microwave / K band
Transition	6_16 -> 5_23, ortho-H2O
Mechanism	collisionally pumped rotational maser
Primary tracers	sub-au kinematics; geometric distances; black-hole masses; dense shocked gas
Typical environments	shocked protostellar outflows; circumstellar envelopes; AGN accretion disks; nuclear winds
Indicative regime	commonly about 300-1000 K for strong collisional pumping
Density guidance	Maser inversion favors very dense gas, often n(H2) about 10^7-10^10 cm^-3, but geometry and velocity coherence dominate brightness.

H2O 22 GHz maser is produced by collisionally pumped rotational maser and is encountered in shocked protostellar outflows; circumstellar envelopes; AGN accretion disks; nuclear winds. It is commonly used to trace sub-au kinematics; geometric distances; black-hole masses; dense shocked gas.

How to use it

Use this transition with its diagnostic companions to constrain sub-au kinematics; geometric distances; black-hole masses; dense shocked gas, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Brightness temperature is non-thermal and cannot be interpreted as kinetic temperature. Narrow spikes vary and may be highly beamed.

Useful companions

SiO v=1 J=1-0; OH 1665/1667 MHz; thermal sub-mm H2O lines

Sources

`JPL; SPLATALOGUE; NRAO_HI_OH`

14.5 NH3 (1,1) inversion

At a glance

Rest coordinate	23.6944955 GHz
Band	microwave / K band
Transition	(J,K)=(1,1) inversion, resolved hyperfine complex
Mechanism	metastable inversion transition
Primary tracers	dense gas; kinetic temperature with NH3 (2,2); optical depth via satellites; velocity dispersion
Typical environments	cold dense cores; infrared dark clouds; molecular clouds; galactic nuclei
Indicative regime	especially effective near 8-40 K, but present in warmer dense gas
Density guidance	Effective excitation densities are typically around 10^3 cm^-3 and model dependent; hyperfine anomalies can affect opacity estimates.

NH3 (1,1) inversion is produced by metastable inversion transition and is encountered in cold dense cores; infrared dark clouds; molecular clouds; galactic nuclei. It is commonly used to trace dense gas; kinetic temperature with NH3 (2,2); optical depth via satellites; velocity dispersion.

How to use it

Use this transition with its diagnostic companions to constrain dense gas; kinetic temperature with NH3 (2,2); optical depth via satellites; velocity dispersion, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Fit the full hyperfine structure. The (2,2)/(1,1) rotational temperature is converted to kinetic temperature with a collisional model.

Useful companions

NH3 (2,2) 23.7226333 GHz; NH3 (3,3) 23.8701296 GHz; N2H+ (1-0)

Sources

`JPL; SPLATALOGUE; SHIRLEY_2015`

14.6 NH3 (2,2) inversion

At a glance

Rest coordinate	23.7226333 GHz
Band	microwave / K band
Transition	(J,K)=(2,2) inversion, resolved hyperfine complex
Mechanism	metastable inversion transition
Primary tracers	gas temperature with NH3 (1,1); warmer dense component; kinematics
Typical environments	dense molecular cores; massive star-forming clumps; galactic centers
Indicative regime	roughly 15-100 K for common applications; higher metastable lines extend the thermometer
Density guidance	Similar effective-density scale to (1,1), with excitation dependent on temperature and NH3 column.

NH3 (2,2) inversion is produced by metastable inversion transition and is encountered in dense molecular cores; massive star-forming clumps; galactic centers. It is commonly used to trace gas temperature with NH3 (1,1); warmer dense component; kinematics.

How to use it

Use this transition with its diagnostic companions to constrain gas temperature with NH3 (1,1); warmer dense component; kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Use matched beams and simultaneous calibration when deriving temperatures; (2,2) may miss the coldest component at low signal-to-noise.

Useful companions

NH3 (1,1) 23.6944955 GHz; NH3 (3,3) 23.8701296 GHz

Sources

`JPL; SPLATALOGUE; SHIRLEY_2015`

14.7 SiO v=1 J=1-0 maser

At a glance

Rest coordinate	43.122079 GHz
Band	millimetre / Q band
Transition	v=1, J=1 -> 0
Mechanism	vibrationally excited rotational maser
Primary tracers	stellar-envelope dynamics; pulsation; magnetic-field morphology; compact shocks
Typical environments	inner envelopes of oxygen-rich AGB stars; red supergiants; rare shocked star-forming gas
Indicative regime	warm to hot molecular gas, often about 1000-2000 K near evolved stars
Density guidance	Requires high density and an inversion; representative pumping solutions use n(H2) around 10^9-10^10 cm^-3.

SiO v=1 J=1-0 maser is produced by vibrationally excited rotational maser and is encountered in inner envelopes of oxygen-rich AGB stars; red supergiants; rare shocked star-forming gas. It is commonly used to trace stellar-envelope dynamics; pulsation; magnetic-field morphology; compact shocks.

How to use it

Use this transition with its diagnostic companions to constrain stellar-envelope dynamics; pulsation; magnetic-field morphology; compact shocks, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Non-LTE, polarized and time variable. Do not use maser flux as a linear mass estimator.

Useful companions

SiO v=2 J=1-0; SiO v=0 rotational lines; H2O 22 GHz maser

Sources

`CDMS; JPL; SPLATALOGUE`

14.8 HCN (1-0)

At a glance

Rest coordinate	88.631847 GHz
Band	millimetre / 3 mm
Transition	J=1 -> 0; F=2 -> 1 strongest hyperfine component
Mechanism	electric-dipole rotational transition
Primary tracers	dense molecular gas; star-forming gas fraction; infall via self-absorption with isotopologues
Typical environments	dense molecular cores; starburst nuclei; AGN circumnuclear disks; molecular outflows
Indicative regime	commonly 10-100 K; IR pumping can matter in luminous dusty nuclei
Density guidance	Optically thin critical density is of order 10^6 cm^-3 near 10-20 K, while trapping makes effective excitation densities commonly 1-2 dex lower.

HCN (1-0) is produced by electric-dipole rotational transition and is encountered in dense molecular cores; starburst nuclei; AGN circumnuclear disks; molecular outflows. It is commonly used to trace dense molecular gas; star-forming gas fraction; infall via self-absorption with isotopologues.

How to use it

Use this transition with its diagnostic companions to constrain dense molecular gas; star-forming gas fraction; infall via self-absorption with isotopologues, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Resolve or model three hyperfine components in narrow Galactic sources. Opacity, abundance, electron excitation and 14-micron pumping complicate the dense-gas interpretation.

Useful companions

HCO+ (1-0); HNC (1-0); H13CN (1-0); CN (1-0)

Sources

`CDMS; SPLATALOGUE; SHIRLEY_2015`

14.9 HCO+ (1-0)

At a glance

Rest coordinate	89.188526 GHz
Band	millimetre / 3 mm
Transition	J=1 -> 0
Mechanism	electric-dipole rotational transition of a molecular ion
Primary tracers	dense ionized-molecular chemistry; infall/outflow profiles; cosmic-ray ionization indirectly
Typical environments	dense molecular clouds; protostellar envelopes; galaxy nuclei; outflows
Indicative regime	about 10-100 K in most applications
Density guidance	Critical density is a few 10^5 cm^-3, but effective excitation density can be around 10^3-10^4 cm^-3 depending on opacity and column.

HCO+ (1-0) is produced by electric-dipole rotational transition of a molecular ion and is encountered in dense molecular clouds; protostellar envelopes; galaxy nuclei; outflows. It is commonly used to trace dense ionized-molecular chemistry; infall/outflow profiles; cosmic-ray ionization indirectly.

How to use it

Use this transition with its diagnostic companions to constrain dense ionized-molecular chemistry; infall/outflow profiles; cosmic-ray ionization indirectly, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Often optically thick and self-absorbed. Pair with H13CO+ to estimate opacity; abundance is sensitive to ionization and water/CO chemistry.

Useful companions

HCN (1-0); H13CO+ (1-0); CO isotopologues; N2H+ (1-0)

Sources

`CDMS; SPLATALOGUE; SHIRLEY_2015`

14.10 N2H+ (1-0)

At a glance	
Rest coordinate	93.173764 GHz
Band	millimetre / 3 mm
Transition	J=1 -> 0 hyperfine multiplet
Mechanism	electric-dipole rotational transition with nitrogen hyperfine structure
Primary tracers	cold dense gas; systemic velocity; optical depth from hyperfine fitting; CO freeze-out zones
Typical environments	cold prestellar cores; dense filaments; protostellar envelopes; CO-depleted gas
Indicative regime	especially useful around 7-30 K
Density guidance	Critical density is roughly 10^5 cm^-3; radiative trapping and hyperfine excitation lower the effective threshold.

N2H+ (1-0) is produced by electric-dipole rotational transition with nitrogen hyperfine structure and is encountered in cold prestellar cores; dense filaments; protostellar envelopes; CO-depleted gas. It is commonly used to trace cold dense gas; systemic velocity; optical depth from hyperfine fitting; CO freeze-out zones.

How to use it

Use this transition with its diagnostic companions to constrain cold dense gas; systemic velocity; optical depth from hyperfine fitting; CO freeze-out zones, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The quoted value is a conventional reference for the hyperfine complex; precision velocity work must adopt component-specific frequencies and a consistent catalog.

Useful companions

NH3 (1,1); HCO+ (1-0); C18O (1-0); N2D+

Sources

`CDMS; SPLATALOGUE; SHIRLEY_2015`

14.11 CS (2-1)

At a glance

Rest coordinate	97.980953 GHz
Band	millimetre / 3 mm
Transition	J=2 -> 1
Mechanism	electric-dipole rotational transition
Primary tracers	dense gas; velocity structure; sulfur chemistry
Typical environments	dense clumps; massive star-forming regions; galactic centers; shocked molecular gas
Indicative regime	about 10-100 K
Density guidance	Critical density is a few 10^5 cm^-3, with a substantially lower effective excitation density for optically thick bright columns.

CS (2-1) is produced by electric-dipole rotational transition and is encountered in dense clumps; massive star-forming regions; galactic centers; shocked molecular gas. It is commonly used to trace dense gas; velocity structure; sulfur chemistry.

How to use it

Use this transition with its diagnostic companions to constrain dense gas; velocity structure; sulfur chemistry, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

CS can deplete in cold cores and become optically thick; use isotopologues and multi-J excitation rather than a single conversion factor.

Useful companions

C34S (2-1); HCN (1-0); CS higher-J ladder

Sources

`CDMS; JPL; SHIRLEY_2015`

14.12 C18O (1-0)

At a glance

Rest coordinate	109.7821734 GHz
Band	millimetre / 3 mm
Transition	J=1 -> 0
Mechanism	electric-dipole rotational transition
Primary tracers	moderate-column molecular gas; CO optical-depth correction; gas kinematics
Typical environments	molecular clouds; dense cores; protoplanetary disks; galaxy molecular disks
Indicative regime	cold molecular gas, commonly 8-50 K
Density guidance	Critical density is around 10^3 cm^-3 and effective excitation density lower; abundance and shielding often dominate detectability.

C18O (1-0) is produced by electric-dipole rotational transition and is encountered in molecular clouds; dense cores; protoplanetary disks; galaxy molecular disks. It is commonly used to trace moderate-column molecular gas; CO optical-depth correction; gas kinematics.

How to use it

Use this transition with its diagnostic companions to constrain moderate-column molecular gas; CO optical-depth correction; gas kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Usually thinner than 13CO but not guaranteed optically thin. Selective photodissociation, fractionation and isotope-ratio gradients bias mass estimates.

Useful companions

13CO (1-0); 12CO (1-0); C17O (1-0); dust continuum

Sources

`CDMS; JPL; CARILLI_WALTER_2013`

14.13 13CO (1-0)

At a glance

Rest coordinate	110.2013543 GHz
Band	millimetre / 3 mm
Transition	J=1 -> 0
Mechanism	electric-dipole rotational transition
Primary tracers	molecular column; 12CO opacity; cloud structure; isotope chemistry
Typical environments	molecular clouds; galaxy disks; circumstellar envelopes; protoplanetary disks
Indicative regime	commonly 8-100 K
Density guidance	Critical density is around 10^3 cm^-3; photon trapping lowers the effective excitation threshold.

13CO (1-0) is produced by electric-dipole rotational transition and is encountered in molecular clouds; galaxy disks; circumstellar envelopes; protoplanetary disks. It is commonly used to trace molecular column; 12CO opacity; cloud structure; isotope chemistry.

How to use it

Use this transition with its diagnostic companions to constrain molecular column; 12CO opacity; cloud structure; isotope chemistry, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Often optically thick in cloud peaks. Carbon isotope gradients, chemical fractionation and selective photodissociation prevent a universal 12CO/13CO abundance ratio.

Useful companions

12CO (1-0); C18O (1-0); 13CO higher-J lines

Sources

`CDMS; JPL; CARILLI_WALTER_2013`

14.14 12CO (1-0)

At a glance

Rest coordinate	115.2712018 GHz
Band	millimetre / 2.6 mm
Transition	J=1 -> 0
Mechanism	electric-dipole rotational transition
Primary tracers	bulk molecular gas via CO-to-H2 conversion; kinematics; cloud temperature when thick
Typical environments	molecular clouds; galaxy disks; starbursts; circumstellar envelopes; molecular outflows
Indicative regime	cold molecular gas from about 5 K upward; upper-state energy is 5.53 K
Density guidance	Optically thin critical density is roughly 2x10^3 cm^-3, while trapping permits bright emission at a few 10^2 cm^-3 or below.

12CO (1-0) is produced by electric-dipole rotational transition and is encountered in molecular clouds; galaxy disks; starbursts; circumstellar envelopes; molecular outflows. It is commonly used to trace bulk molecular gas via CO-to-H2 conversion; kinematics; cloud temperature when thick.

How to use it

Use this transition with its diagnostic companions to constrain bulk molecular gas via CO-to-H2 conversion; kinematics; cloud temperature when thick, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Usually optically thick. X_CO depends on metallicity, dynamics, excitation, spatial scale and radiation field; CMB contrast matters at high redshift.

Useful companions

13CO (1-0); C18O (1-0); CO (2-1); dust continuum

Sources

`CDMS; JPL; CARILLI_WALTER_2013`

14.15 12CO (2-1)

At a glance

Rest coordinate	230.538 GHz
Band	submillimetre / 1.3 mm
Transition	J=2 -> 1
Mechanism	electric-dipole rotational transition
Primary tracers	molecular gas distribution; excitation with CO(1-0); kinematics
Typical environments	molecular clouds; nearby and high-redshift galaxies; disks; outflows
Indicative regime	cold to moderately warm gas; upper-state energy is 16.6 K
Density guidance	Critical density is around 10^4 cm^-3, but opacity makes the effective density much lower.

12CO (2-1) is produced by electric-dipole rotational transition and is encountered in molecular clouds; nearby and high-redshift galaxies; disks; outflows. It is commonly used to trace molecular gas distribution; excitation with CO(1-0); kinematics.

How to use it

Use this transition with its diagnostic companions to constrain molecular gas distribution; excitation with CO(1-0); kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Brightness-temperature ratio r21 is sensitive to excitation, beam filling and optical depth. Match angular resolution before interpreting line ratios.

Useful companions

CO (1-0); CO (3-2); 13CO (2-1)

Sources

`CDMS; JPL; CARILLI_WALTER_2013`

14.16 12CO (3-2)

At a glance

Rest coordinate	345.7959899 GHz
Band	submillimetre / 870 micron
Transition	J=3 -> 2
Mechanism	electric-dipole rotational transition
Primary tracers	warmer/denser molecular component; CO excitation ladder; kinematics
Typical environments	warm molecular clouds; star-forming clumps; starbursts; high-redshift galaxies
Indicative regime	commonly 20-100 K; upper-state energy is 33.2 K
Density guidance	Critical density is a few 10^4 cm^-3; radiative trapping lowers the effective density.

12CO (3-2) is produced by electric-dipole rotational transition and is encountered in warm molecular clouds; star-forming clumps; starbursts; high-redshift galaxies. It is commonly used to trace warmer/denser molecular component; CO excitation ladder; kinematics.

How to use it

Use this transition with its diagnostic companions to constrain warmer/denser molecular component; CO excitation ladder; kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Mid-J selection can bias samples toward excited gas. Use a CO SLED and consistent beam/uv coverage to infer multiple gas phases.

Useful companions

CO (2-1); CO (4-3); 13CO (3-2); [C I] 492 GHz

Sources

`CDMS; JPL; CARILLI_WALTER_2013`

14.17 [C I] 609 micron

At a glance

Rest coordinate	492.160651 GHz
Band	submillimetre / far-infrared
Transition	3P1 -> 3P0
Mechanism	magnetic-dipole fine-structure transition
Primary tracers	neutral carbon mass; molecular gas when calibrated; excitation with [C I](2-1)
Typical environments	molecular clouds; PDRs; high-redshift galaxies; CO-dark molecular interfaces
Indicative regime	cold molecular gas, commonly 15-60 K; upper level is 23.6 K above ground
Density guidance	Critical density with H2 is of order 5x10^2 cm^-3, weakly model dependent.

[C I] 609 micron is produced by magnetic-dipole fine-structure transition and is encountered in molecular clouds; PDRs; high-redshift galaxies; CO-dark molecular interfaces. It is commonly used to trace neutral carbon mass; molecular gas when calibrated; excitation with [C I](2-1).

How to use it

Use this transition with its diagnostic companions to constrain neutral carbon mass; molecular gas when calibrated; excitation with [C I](2-1), while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The two [C I] lines give an excitation temperature only under a common, often optically thin one-zone assumption. Carbon abundance is not universal.

Useful companions

[C I] 809 GHz; CO rotational ladder; [C II] 158 micron

Sources

`NIST_ASD; CDMS; CARILLI_WALTER_2013`

14.18 [C I] 370 micron

At a glance

Rest coordinate	809.34197 GHz
Band	submillimetre / far-infrared
Transition	3P2 -> 3P1
Mechanism	magnetic-dipole fine-structure transition
Primary tracers	neutral carbon excitation; molecular gas temperature; carbon mass
Typical environments	warm molecular clouds; PDRs; starbursts; high-redshift galaxies
Indicative regime	typically 25-100 K; upper level is 62.5 K above ground
Density guidance	Critical density with H2 is of order 10^3 cm^-3.

[C I] 370 micron is produced by magnetic-dipole fine-structure transition and is encountered in warm molecular clouds; PDRs; starbursts; high-redshift galaxies. It is commonly used to trace neutral carbon excitation; molecular gas temperature; carbon mass.

How to use it

Use this transition with its diagnostic companions to constrain neutral carbon excitation; molecular gas temperature; carbon mass, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The line sits close to CO(7-6) in frequency, useful for simultaneous setups but demanding correct identification at uncertain redshift.

Useful companions

[C I] 492 GHz; CO (7-6) nearby in frequency; [C II] 158 micron

Sources

`NIST_ASD; CDMS; CARILLI_WALTER_2013`

14.19 [C II] 158 micron

At a glance	
Rest coordinate	1900.5369 GHz
Band	far-infrared
Transition	2P3/2 -> 2P1/2
Mechanism	fine-structure transition
Primary tracers	gas cooling; star-forming ISM; dynamics and redshift; CO-dark molecular gas with modeling
Typical environments	PDR surfaces; diffuse neutral gas; CO-dark H2; diffuse ionized gas; high-redshift galaxies
Indicative regime	upper level corresponds to 91.2 K; excited in neutral and ionized gas from roughly 50 K to 10^4 K
Density guidance	Approximate critical densities: about 50 cm^-3 for electrons, 3x10^3 for H and 5x10^3 for H2, with temperature dependence.

[C II] 158 micron is produced by fine-structure transition and is encountered in PDR surfaces; diffuse neutral gas; CO-dark H2; diffuse ionized gas; high-redshift galaxies. It is commonly used to trace gas cooling; star-forming ISM; dynamics and redshift; CO-dark molecular gas with modeling.

How to use it

Use this transition with its diagnostic companions to constrain gas cooling; star-forming ISM; dynamics and redshift; CO-dark molecular gas with modeling, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

A multiphase line, not a pure PDR or SFR tracer. Optical depth, foreground absorption and the high-surface-brightness [C II] deficit affect interpretation.

Useful companions

[N II] 205 micron to isolate ionized contribution; [O I] 63 micron; CO rotational lines; FIR continuum

Sources

`NIST_ASD; CARILLI_WALTER_2013; HOLLENBACH_TIELENS_1999`

14.20 [N II] 205 micron

At a glance

Rest coordinate	1461.1319 GHz
Band	far-infrared
Transition	3P1 -> 3P0
Mechanism	fine-structure transition
Primary tracers	ionized-gas mass; massive-star ionization; ionized contribution to [C II]
Typical environments	diffuse ionized medium; H II regions; star-forming galaxies
Indicative regime	ionized gas near 6000-10000 K; excitation energy is only about 70 K
Density guidance	Electron critical density is roughly 40-50 cm^-3; the 122/205 ratio measures electron density over roughly 1-10^3 cm^-3.

[N II] 205 micron is produced by fine-structure transition and is encountered in diffuse ionized medium; H II regions; star-forming galaxies. It is commonly used to trace ionized-gas mass; massive-star ionization; ionized contribution to [C II].

How to use it

Use this transition with its diagnostic companions to constrain ionized-gas mass; massive-star ionization; ionized contribution to [C II], while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

N+ requires ionized gas because its formation threshold exceeds 13.6 eV. Low critical density makes 205 micron particularly sensitive to diffuse ionized material.

Useful companions

[N II] 122 micron; [C II] 158 micron; radio free-free emission

Sources

`NIST_NII; CARILLI_WALTER_2013; OSTERBROCK_FERLAND_2006`

14.21 [N II] 122 micron

At a glance

Rest coordinate	2459.38006 GHz
Band	far-infrared
Transition	3P2 -> 3P1
Mechanism	fine-structure transition
Primary tracers	electron density with 205 micron; ionized-gas cooling; massive-star photoionization
Typical environments	H II regions; diffuse ionized gas; star-forming galaxies
Indicative regime	ionized gas around 6000-10000 K
Density guidance	Electron critical density is about 300 cm^-3; 122/205 is a robust density ratio when both lines sample the same gas.

[N II] 122 micron is produced by fine-structure transition and is encountered in H II regions; diffuse ionized gas; star-forming galaxies. It is commonly used to trace electron density with 205 micron; ionized-gas cooling; massive-star photoionization.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 205 micron; ionized-gas cooling; massive-star photoionization, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Far-IR dust extinction is small, but aperture mismatch and multi-density structure can bias the ratio.

Useful companions

[N II] 205 micron; [O III] 88 micron; [C II] 158 micron

Sources

`NIST_NII; CARILLI_WALTER_2013; OSTERBROCK_FERLAND_2006`

14.22 [O I] 145 micron

At a glance

Rest coordinate	145.525 micron
Band	far-infrared
Transition	3P0 -> 3P1
Mechanism	fine-structure transition
Primary tracers	neutral-gas cooling; PDR density and radiation field with [O I] 63/[C II]; shock-heated gas
Typical environments	warm dense PDRs; shocks; neutral gas near H II regions
Indicative regime	warm neutral gas, commonly about 100-1000 K
Density guidance	Critical density for H/H2 collisions is of order 10^5 cm^-3, model dependent.

[O I] 145 micron is produced by fine-structure transition and is encountered in warm dense PDRs; shocks; neutral gas near H II regions. It is commonly used to trace neutral-gas cooling; PDR density and radiation field with [O I] 63/[C II]; shock-heated gas.

How to use it

Use this transition with its diagnostic companions to constrain neutral-gas cooling; PDR density and radiation field with [O I] 63/[C II]; shock-heated gas, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Usually weaker but less self-absorbed than the 63-micron line. The 63/145 ratio can reveal opacity, foreground absorption, temperature and density.

Useful companions

[O I] 63 micron; [C II] 158 micron; high-J CO

Sources

`NIST_ASD; HOLLENBACH_TIELENS_1999; CARILLI_WALTER_2013`

14.23 [O I] 63 micron

At a glance

Rest coordinate	63.1837 micron
Band	far-infrared
Transition	3P1 -> 3P2
Mechanism	fine-structure transition
Primary tracers	dominant neutral-gas cooling in warm dense regions; FUV field; shock power
Typical environments	dense warm PDRs; dissociative and non-dissociative shocks; protostellar outflows
Indicative regime	upper level is about 228 K; strongest in gas roughly 100-1000 K
Density guidance	Critical density for collisions with H is roughly 5x10^5 cm^-3; optical depth often matters first.

[O I] 63 micron is produced by fine-structure transition and is encountered in dense warm PDRs; dissociative and non-dissociative shocks; protostellar outflows. It is commonly used to trace dominant neutral-gas cooling in warm dense regions; FUV field; shock power.

How to use it

Use this transition with its diagnostic companions to constrain dominant neutral-gas cooling in warm dense regions; FUV field; shock power, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Can be optically thick and self-absorbed by colder foreground O I; a line deficit does not uniquely imply weak PDR heating.

Useful companions

[O I] 145 micron; [C II] 158 micron; H2 rotational lines

Sources

`NIST_ASD; HOLLENBACH_TIELENS_1999; CARILLI_WALTER_2013`

14.24 [O III] 88 micron

At a glance	
Rest coordinate	3393.006 GHz
Band	far-infrared
Transition	3P1 -> 3P0
Mechanism	fine-structure transition
Primary tracers	hard massive-star radiation; ionized gas; electron density with 52 micron; high-redshift systemic redshift
Typical environments	H II regions; low-metallicity star-forming galaxies; planetary nebulae; reionization-era galaxies
Indicative regime	photoionized gas around 7000-20000 K; low excitation energy minimizes temperature sensitivity
Density guidance	Electron critical density is about 500 cm^-3; 52/88 measures densities from roughly 10^2 to 10^4 cm^-3.

[O III] 88 micron is produced by fine-structure transition and is encountered in H II regions; low-metallicity star-forming galaxies; planetary nebulae; reionization-era galaxies. It is commonly used to trace hard massive-star radiation; ionized gas; electron density with 52 micron; high-redshift systemic redshift.

How to use it

Use this transition with its diagnostic companions to constrain hard massive-star radiation; ionized gas; electron density with 52 micron; high-redshift systemic redshift, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Bright in hard, low-metallicity H II regions and relatively extinction-free. Abundance, ionization parameter and density all affect line-to-SFR conversions.

Useful companions

[O III] 52 micron; [N II] 122 micron; optical [O III] 5008 A

Sources

`NIST_ASD; CARILLI_WALTER_2013; OSTERBROCK_FERLAND_2006`

14.25 [O III] 52 micron

At a glance

Rest coordinate	51.8145 micron
Band	far-infrared
Transition	3P2 -> 3P1
Mechanism	fine-structure transition
Primary tracers	electron density with 88 micron; hard ionizing field; ionized-gas cooling
Typical environments	compact H II regions; planetary nebulae; starbursts
Indicative regime	photoionized gas around 7000-20000 K
Density guidance	Electron critical density is roughly 3.6x10^3 cm^-3; 52/88 is a standard far-IR density diagnostic.

[O III] 52 micron is produced by fine-structure transition and is encountered in compact H II regions; planetary nebulae; starbursts. It is commonly used to trace electron density with 88 micron; hard ionizing field; ionized-gas cooling.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 88 micron; hard ionizing field; ionized-gas cooling, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Use matched apertures. At high density both lines approach LTE and the ratio saturates; abundance and ionization structure remain separate uncertainties.

Useful companions

[O III] 88 micron; optical [O III] 5008 A; [N III] 57 micron

Sources

`MPE_FINE_STRUCTURE; CARILLI_WALTER_2013; OSTERBROCK_FERLAND_2006`

14.26 [Si II] 34.8 micron

At a glance

Rest coordinate	34.814 micron
Band	mid-infrared
Transition	2P3/2 -> 2P1/2
Mechanism	fine-structure transition
Primary tracers	gas cooling; silicon released from grains; PDR/XDR/shock conditions
Typical environments	PDRs; diffuse ionized gas; shocks; X-ray-dominated regions
Indicative regime	warm neutral or ionized gas, about 100-10000 K
Density guidance	Critical density is roughly 10^3 cm^-3 for electrons and a few 10^5 cm^-3 for H, with temperature dependence.

[Si II] 34.8 micron is produced by fine-structure transition and is encountered in PDRs; diffuse ionized gas; shocks; X-ray-dominated regions. It is commonly used to trace gas cooling; silicon released from grains; PDR/XDR/shock conditions.

How to use it

Use this transition with its diagnostic companions to constrain gas cooling; silicon released from grains; PDR/XDR/shock conditions, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Silicon depletion and grain destruction dominate absolute calibration; strong emission may arise from multiple phases along the same sightline.

Useful companions

[Fe II] 26 micron; [S III] 33.5 micron; [C II] 158 micron

Sources

`ISO_SWS; MPE_FINE_STRUCTURE; HOLLENBACH_TIELENS_1999`

14.27 [S III] 33.5 micron

At a glance	
Rest coordinate	33.481 micron
Band	mid-infrared
Transition	3P1 -> 3P0
Mechanism	fine-structure transition
Primary tracers	ionized gas; electron density with 18.7 micron; sulfur abundance with modeling
Typical environments	H II regions; starburst galaxies; planetary nebulae
Indicative regime	photoionized gas around 5000-15000 K
Density guidance	Electron critical density is of order 10^3 cm^-3; the 18.7/33.5 ratio spans roughly 10^2-10^4 cm^-3.

[S III] 33.5 micron is produced by fine-structure transition and is encountered in H II regions; starburst galaxies; planetary nebulae. It is commonly used to trace ionized gas; electron density with 18.7 micron; sulfur abundance with modeling.

How to use it

Use this transition with its diagnostic companions to constrain ionized gas; electron density with 18.7 micron; sulfur abundance with modeling, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The widely separated pair is susceptible to aperture and extinction mismatch; use the correct low-density limit for the adopted atomic data.

Useful companions

[S III] 18.7 micron; optical [S III] 9071/9533 A; [Ne II] 12.8 micron

Sources

`ISO_SWS; MPE_FINE_STRUCTURE; OSTERBROCK_FERLAND_2006`

14.28 [S III] 18.7 micron

At a glance

Rest coordinate	18.713 micron
Band	mid-infrared
Transition	3P2 -> 3P1
Mechanism	fine-structure transition
Primary tracers	electron density with 33.5 micron; ionized gas; obscured star formation
Typical environments	H II regions; compact starbursts; planetary nebulae
Indicative regime	photoionized gas around 5000-15000 K
Density guidance	Electron critical density is roughly 2x10^4 cm^-3; 18.7/33.5 rises with density before saturating.

[S III] 18.7 micron is produced by fine-structure transition and is encountered in H II regions; compact starbursts; planetary nebulae. It is commonly used to trace electron density with 33.5 micron; ionized gas; obscured star formation.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 33.5 micron; ionized gas; obscured star formation, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Less affected by dust than optical lines but can suffer silicate-feature continuum systematics and aperture differences.

Useful companions

[S III] 33.5 micron; [Ne III] 15.6 micron; optical [S II] density pair

Sources

`ISO_SWS; MPE_FINE_STRUCTURE; OSTERBROCK_FERLAND_2006`

14.29 [Ne II] 12.8 micron

At a glance	
Rest coordinate	12.814 micron
Band	mid-infrared
Transition	2P1/2 -> 2P3/2
Mechanism	fine-structure transition
Primary tracers	obscured massive-star ionization; ionized-gas kinematics; soft-to-moderate radiation fields
Typical environments	H II regions; embedded star formation; protoplanetary disk surfaces; AGN narrow-line regions
Indicative regime	photoionized gas around 5000-15000 K
Density guidance	Electron critical density is of order 6x10^5 cm^-3; usually in the low-density limit for extended H II gas.

[Ne II] 12.8 micron is produced by fine-structure transition and is encountered in H II regions; embedded star formation; protoplanetary disk surfaces; AGN narrow-line regions. It is commonly used to trace obscured massive-star ionization; ionized-gas kinematics; soft-to-moderate radiation fields.

How to use it

Use this transition with its diagnostic companions to constrain obscured massive-star ionization; ionized-gas kinematics; soft-to-moderate radiation fields, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

[Ne III]/[Ne II] is sensitive to ionizing hardness and ionization parameter, but also geometry, metallicity and stellar population age.

Useful companions

[Ne III] 15.6 micron; hydrogen recombination lines; [S IV] 10.5 micron

Sources

`ISO_SWS; MPE_FINE_STRUCTURE; KENNICUTT_EVANS_2012`

14.30 [Ne III] 15.6 micron

At a glance

Rest coordinate	15.5551 micron
Band	mid-infrared
Transition	3P1 -> 3P2
Mechanism	fine-structure transition
Primary tracers	hard ionizing radiation; obscured star formation; ionization parameter with other neon lines
Typical environments	high-excitation H II regions; low-metallicity starbursts; planetary nebulae; AGN narrow-line regions
Indicative regime	photoionized gas around 7000-20000 K
Density guidance	Electron critical density is about 10^5 cm^-3; typically low-density in integrated galaxies.

[Ne III] 15.6 micron is produced by fine-structure transition and is encountered in high-excitation H II regions; low-metallicity starbursts; planetary nebulae; AGN narrow-line regions. It is commonly used to trace hard ionizing radiation; obscured star formation; ionization parameter with other neon lines.

How to use it

Use this transition with its diagnostic companions to constrain hard ionizing radiation; obscured star formation; ionization parameter with other neon lines, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Neon is only weakly depleted, but the ratio to [Ne II] is not a one-parameter hardness indicator.

Useful companions

[Ne II] 12.8 micron; [Ne V] 14.3 micron; optical [Ne III] 3869.9 A

Sources

`ISO_SWS; MPE_FINE_STRUCTURE; KENNICUTT_EVANS_2012`

14.31 H2 0-0 S(0)

At a glance	
Rest coordinate	28.21883 micron
Band	mid-infrared
Transition	v=0, J=2 -> 0 (para-H2)
Mechanism	electric-quadrupole pure rotational transition
Primary tracers	warm molecular hydrogen mass; low-temperature rotational excitation; ortho-to-para ratio with other lines
Typical environments	warm molecular clouds; PDRs; shocks; galaxy disks
Indicative regime	sensitive to roughly 80-300 K gas; upper-state energy is about 510 K
Density guidance	Critical density depends on H, H2 and He collision rates, typically around 10^2-10^3 cm^-3 at warm temperatures.

H2 0-0 S(0) is produced by electric-quadrupole pure rotational transition and is encountered in warm molecular clouds; PDRs; shocks; galaxy disks. It is commonly used to trace warm molecular hydrogen mass; low-temperature rotational excitation; ortho-to-para ratio with other lines.

How to use it

Use this transition with its diagnostic companions to constrain warm molecular hydrogen mass; low-temperature rotational excitation; ortho-to-para ratio with other lines, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Quadrupole emission is optically thin but weak. Warm-H2 mass is exponentially sensitive to assumed temperature and ortho-to-para ratio.

Useful companions

H2 0-0 S(1) 17.03484 micron; higher H2 rotational lines; CO rotational ladder

Sources

`MPE_H2; DRAINE_2011; HOLLENBACH_TIELENS_1999`

14.32 H2 0-0 S(1)

At a glance

Rest coordinate	17.03484 micron
Band	mid-infrared
Transition	v=0, J=3 -> 1 (ortho-H2)
Mechanism	electric-quadrupole pure rotational transition
Primary tracers	warm H2; shock/PDR excitation; rotational temperature
Typical environments	PDRs; molecular shocks; turbulent galaxy disks; AGN-heated molecular gas
Indicative regime	commonly traces about 150-700 K gas; upper-state energy is about 1015 K
Density guidance	Critical density is typically around 10^3-10^4 cm^-3, depending strongly on collider and temperature.

H2 0-0 S(1) is produced by electric-quadrupole pure rotational transition and is encountered in PDRs; molecular shocks; turbulent galaxy disks; AGN-heated molecular gas. It is commonly used to trace warm H2; shock/PDR excitation; rotational temperature.

How to use it

Use this transition with its diagnostic companions to constrain warm H2; shock/PDR excitation; rotational temperature, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Multiple-temperature excitation diagrams are usually required; a single Boltzmann slope can hide non-LTE or mixed gas.

Useful companions

H2 0-0 S(0) 28.21883 micron; H2 0-0 S(2)/S(3); [Fe II] shock lines

Sources

`MPE_H2; DRAINE_2011; HOLLENBACH_TIELENS_1999`

14.33 H2 1-0 S(1)

At a glance

Rest coordinate	2.12183 micron
Band	near-infrared / K band
Transition	v=1, J=3 -> v=0, J=1
Mechanism	rovibrational electric-quadrupole transition
Primary tracers	hot molecular gas; shock fronts; fluorescent versus thermal excitation with line ratios
Typical environments	molecular shocks; PDRs; protostellar jets; AGN and starburst nuclei
Indicative regime	roughly 1000-3000 K for thermal emission; UV fluorescence can populate it in cooler gas
Density guidance	Thermalization requires densities roughly 10^4-10^6 cm^-3 depending on collider and temperature; ratios carry the diagnostic power.

H2 1-0 S(1) is produced by rovibrational electric-quadrupole transition and is encountered in molecular shocks; PDRs; protostellar jets; AGN and starburst nuclei. It is commonly used to trace hot molecular gas; shock fronts; fluorescent versus thermal excitation with line ratios.

How to use it

Use this transition with its diagnostic companions to constrain hot molecular gas; shock fronts; fluorescent versus thermal excitation with line ratios, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

2-1 S(1)/1-0 S(1) helps distinguish fluorescence from thermal shocks, but mixed excitation and differential extinction matter.

Useful companions

H2 2-1 S(1) 2.2477 micron; Br-gamma 2.1661 micron; [Fe II] 1.644 micron

Sources

`MPE_H2; DRAINE_2011; HOLLENBACH_TIELENS_1999`

14.34 H I Brackett-gamma

At a glance	
Rest coordinate	2.16612 micron
Band	near-infrared / K band
Transition	n=7 -> 4
Mechanism	hydrogen recombination transition
Primary tracers	ionizing-photon rate; obscured star formation; accretion/outflow kinematics
Typical environments	dust-obscured H II regions; young stellar objects; stellar winds; AGN broad/narrow-line regions
Indicative regime	photoionized gas near 5000-20000 K
Density guidance	Case-B emissivity has modest density dependence over ordinary nebular conditions; high densities and optical depth alter decrements.

H I Brackett-gamma is produced by hydrogen recombination transition and is encountered in dust-obscured H II regions; young stellar objects; stellar winds; AGN broad/narrow-line regions. It is commonly used to trace ionizing-photon rate; obscured star formation; accretion/outflow kinematics.

How to use it

Use this transition with its diagnostic companions to constrain ionizing-photon rate; obscured star formation; accretion/outflow kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Much less extinguished than H-alpha but fainter. Correct for underlying stellar absorption, extinction and broad/narrow component mixing.

Useful companions

Pa-alpha 1.8756 micron; H-alpha/H-beta; H2 2.1218 micron

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006; KENNICUTT_EVANS_2012`

14.35 H I Paschen-alpha

At a glance

Rest coordinate	1.87561 micron
Band	near-infrared
Transition	n=4 -> 3
Mechanism	hydrogen recombination transition
Primary tracers	ionizing-photon rate; dust-obscured star formation; extinction through recombination decrements
Typical environments	embedded H II regions; dusty starbursts; AGN line regions
Indicative regime	photoionized gas near 5000-20000 K
Density guidance	Weak density dependence in Case B; departures arise at high density, high optical depth or strong collisional excitation.

H I Paschen-alpha is produced by hydrogen recombination transition and is encountered in embedded H II regions; dusty starbursts; AGN line regions. It is commonly used to trace ionizing-photon rate; dust-obscured star formation; extinction through recombination decrements.

How to use it

Use this transition with its diagnostic companions to constrain ionizing-photon rate; dust-obscured star formation; extinction through recombination decrements, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Ground-based access is difficult because of telluric absorption; space observations are powerful for heavily obscured star formation.

Useful companions

Br-gamma 2.1661 micron; H-alpha 6564.6 A; Pa-beta 1.2818 micron

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006; KENNICUTT_EVANS_2012`

14.36 [Fe II] 1.644 micron

At a glance

Rest coordinate	1.644 micron
Band	near-infrared / H band
Transition	a4D7/2 -> a4F9/2
Mechanism	forbidden fine-structure transition
Primary tracers	shocked/partially ionized gas; grain destruction; electron density with other [Fe II] lines
Typical environments	radiative shocks; supernova remnants; jets; partially ionized zones; AGN narrow-line regions
Indicative regime	typically several thousand to about 15000 K
Density guidance	Critical density is of order 10^4-10^5 cm^-3; ratios among lines from different upper levels constrain density.

[Fe II] 1.644 micron is produced by forbidden fine-structure transition and is encountered in radiative shocks; supernova remnants; jets; partially ionized zones; AGN narrow-line regions. It is commonly used to trace shocked/partially ionized gas; grain destruction; electron density with other [Fe II] lines.

How to use it

Use this transition with its diagnostic companions to constrain shocked/partially ionized gas; grain destruction; electron density with other [Fe II] lines, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The 1.257/1.644 ratio shares an upper level and is useful for extinction, but published A-values introduce systematic uncertainty.

Useful companions

[Fe II] 1.257 micron; H2 2.1218 micron; Br-gamma

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006; DRAINE_2011`

14.37 He I 10830

At a glance	
Rest coordinate	10833.3 angstrom
Band	near-infrared / Y-J boundary
Transition	2s 3S -> 2p 3P multiplet
Mechanism	permitted transition from the metastable triplet state
Primary tracers	outflow velocity and covering fraction; metastable-helium column; hard EUV irradiation
Typical environments	stellar chromospheres; winds and accretion funnels; H II regions; AGN outflows; exoplanet escaping atmospheres
Indicative regime	photoionized/recombining gas from roughly 5000 K upward; metastable population is radiation and density dependent
Density guidance	Metastable 2s 3S population responds to electron density, ionization and radiative transfer; not a simple critical-density line.

He I 10830 is produced by permitted transition from the metastable triplet state and is encountered in stellar chromospheres; winds and accretion funnels; H II regions; AGN outflows; exoplanet escaping atmospheres. It is commonly used to trace outflow velocity and covering fraction; metastable-helium column; hard EUV irradiation.

How to use it

Use this transition with its diagnostic companions to constrain outflow velocity and covering fraction; metastable-helium column; hard EUV irradiation, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

A multiplet with close components. Absorption can be saturated with partial covering; telluric correction and air/vacuum convention are essential.

Useful companions

He I 5877.25 A; He I 2.058 micron; H-alpha; He II 4687 A

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.38 [S III] 9533

At a glance	
Rest coordinate	9533.2 angstrom
Band	optical / near-infrared
Transition	1D2 -> 3P2
Mechanism	forbidden collisionally excited transition
Primary tracers	S++ abundance; ionization/excitation; electron temperature with auroral [S III] 6312
Typical environments	H II regions; planetary nebulae; star-forming galaxies
Indicative regime	photoionized gas around 5000-15000 K
Density guidance	Critical density is high enough that normal H II regions are usually in the low-density regime; IR fine-structure lines provide stronger density leverage.

[S III] 9533 is produced by forbidden collisionally excited transition and is encountered in H II regions; planetary nebulae; star-forming galaxies. It is commonly used to trace S++ abundance; ionization/excitation; electron temperature with auroral [S III] 6312.

How to use it

Use this transition with its diagnostic companions to constrain S++ abundance; ionization/excitation; electron temperature with auroral [S III] 6312, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Intrinsic 9533/9071 branching ratio is near 2.44 and can check calibration. Strong telluric absorption/fringing often dominates uncertainty.

Useful companions

[S III] 9071 A; [S III] 6312 A; [S II] 6718/6733 A

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.39 [S III] 9071

At a glance

Rest coordinate	9071.1 angstrom
Band	optical / near-infrared
Transition	1D2 -> 3P1
Mechanism	forbidden collisionally excited transition
Primary tracers	S++ abundance; ionized gas; electron temperature with [S III] 6312
Typical environments	H II regions; planetary nebulae; star-forming galaxies
Indicative regime	photoionized gas around 5000-15000 K
Density guidance	Usually below its critical density in ordinary nebulae; ratio to 9533 is fixed mainly by A-values.

[S III] 9071 is produced by forbidden collisionally excited transition and is encountered in H II regions; planetary nebulae; star-forming galaxies. It is commonly used to trace S++ abundance; ionized gas; electron temperature with [S III] 6312.

How to use it

Use this transition with its diagnostic companions to constrain S++ abundance; ionized gas; electron temperature with [S III] 6312, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value shown; the familiar air wavelength is about 9068.6 A. Correct for telluric absorption and use the theoretical branching ratio.

Useful companions

[S III] 9533 A; [S III] 6312 A; [O II] 3727/3729 A

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.40 H-alpha

At a glance	
Rest coordinate	6564.61 angstrom
Band	optical / red
Transition	Balmer n=3 -> 2
Mechanism	hydrogen recombination with possible collisional contribution
Primary tracers	recent massive-star formation; ionized-gas morphology and kinematics; recombination rate
Typical environments	H II regions; diffuse ionized gas; star-forming galaxies; AGN broad/narrow-line regions; radiative shocks
Indicative regime	photoionized gas near 5000-20000 K
Density guidance	Case-B emissivity scales approximately with n_e n_p but the H-alpha/H-beta decrement is mainly extinction/transfer sensitive over ordinary nebular densities.

H-alpha is produced by hydrogen recombination with possible collisional contribution and is encountered in H II regions; diffuse ionized gas; star-forming galaxies; AGN broad/narrow-line regions; radiative shocks. It is commonly used to trace recent massive-star formation; ionized-gas morphology and kinematics; recombination rate.

How to use it

Use this transition with its diagnostic companions to constrain recent massive-star formation; ionized-gas morphology and kinematics; recombination rate, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value shown; the traditional air wavelength is 6562.8 A. Dust, [N II] blending, stellar absorption, leakage and stochastic IMF sampling affect SFR estimates.

Useful companions

H-beta 4862.68 A for Balmer decrement; [N II] 6585 A and [S II] 6718/6733 A for BPT/shock diagnostics; Pa-alpha/Br-gamma

Sources

`NIST_ASD; KENNICUTT_EVANS_2012; OSTERBROCK_FERLAND_2006`

14.41 [N II] 6585

At a glance

Rest coordinate	6585.27 angstrom
Band	optical / red
Transition	1D2 -> 3P2
Mechanism	forbidden collisionally excited transition
Primary tracers	gas-phase metallicity through calibrated ratios; excitation source with H-alpha; N+ abundance
Typical environments	H II regions; diffuse ionized gas; planetary nebulae; AGN narrow-line regions; shocks
Indicative regime	photoionized or shocked gas around 6000-20000 K
Density guidance	Critical density is roughly 10^5 cm^-3; usually in the low-density regime for normal H II regions.

[N II] 6585 is produced by forbidden collisionally excited transition and is encountered in H II regions; diffuse ionized gas; planetary nebulae; AGN narrow-line regions; shocks. It is commonly used to trace gas-phase metallicity through calibrated ratios; excitation source with H-alpha; N+ abundance.

How to use it

Use this transition with its diagnostic companions to constrain gas-phase metallicity through calibrated ratios; excitation source with H-alpha; N+ abundance, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 6583.45 A. N/O variations and ionization parameter limit one-dimensional metallicity calibrations.

Useful companions

[N II] 6549.86 A fixed branching partner; H-alpha for N2/BPT; [N II] 5756 A for electron temperature

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.42 [S II] 6718

At a glance	
Rest coordinate	6718.29 angstrom
Band	optical / red
Transition	2D5/2 -> 4S3/2
Mechanism	forbidden collisionally excited transition
Primary tracers	electron density with 6733; low-ionization/shock zones; excitation source with H-alpha
Typical environments	H II regions; diffuse ionized gas; shock fronts; AGN narrow-line regions; supernova remnants
Indicative regime	roughly 5000-20000 K ionized or partially ionized gas
Density guidance	Critical density near 10^4 K is about 1.5x10^3 cm^-3; the doublet ratio is most useful around 10^2-10^4 cm^-3.

[S II] 6718 is produced by forbidden collisionally excited transition and is encountered in H II regions; diffuse ionized gas; shock fronts; AGN narrow-line regions; supernova remnants. It is commonly used to trace electron density with 6733; low-ionization/shock zones; excitation source with H-alpha.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 6733; low-ionization/shock zones; excitation source with H-alpha, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 6716.44 A. Ratio saturates at low and high density and can mix unresolved density phases.

Useful companions

[S II] 6733 A; H-alpha for BPT/shock ratios; [O I] 6302 A

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.43 [S II] 6733

At a glance

Rest coordinate	6732.67 angstrom
Band	optical / red
Transition	2D3/2 -> 4S3/2
Mechanism	forbidden collisionally excited transition
Primary tracers	electron density with 6718; low-ionization gas; shock excitation
Typical environments	H II regions; diffuse ionized gas; shock fronts; AGN narrow-line regions; supernova remnants
Indicative regime	roughly 5000-20000 K
Density guidance	Critical density near 10^4 K is about 4x10^3 cm^-3; paired ratio carries the density information.

[S II] 6733 is produced by forbidden collisionally excited transition and is encountered in H II regions; diffuse ionized gas; shock fronts; AGN narrow-line regions; supernova remnants. It is commonly used to trace electron density with 6718; low-ionization gas; shock excitation.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 6718; low-ionization gas; shock excitation, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 6730.82 A. Deblend both lines from sky residuals and broad H-alpha wings in complex nuclei.

Useful companions

[S II] 6718 A; H-alpha; [O II] 3727/3729 A

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.44 [O I] 6302

At a glance

Rest coordinate	6302.05 angstrom
Band	optical / red
Transition	1D2 -> 3P2
Mechanism	forbidden collisionally excited transition
Primary tracers	hard ionizing spectra; shock excitation; neutral/ionized interface
Typical environments	partially ionized zones; shocks; AGN narrow-line regions; nebular ionization fronts; night-sky airglow
Indicative regime	warm neutral or partially ionized gas around several thousand to 20000 K
Density guidance	Electron critical density is of order 2x10^6 cm^-3; collisional quenching matters in dense gas.

[O I] 6302 is produced by forbidden collisionally excited transition and is encountered in partially ionized zones; shocks; AGN narrow-line regions; nebular ionization fronts; night-sky airglow. It is commonly used to trace hard ionizing spectra; shock excitation; neutral/ionized interface.

How to use it

Use this transition with its diagnostic companions to constrain hard ionizing spectra; shock excitation; neutral/ionized interface, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 6300.30 A. Bright terrestrial airglow demands careful sky subtraction, especially at low redshift.

Useful companions

[O I] 6365.54 A fixed branching partner; H-alpha for BPT; [S II] doublet

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.45 He I 5877

At a glance	
Rest coordinate	5877.25 angstrom
Band	optical / yellow
Transition	3d 3D -> 2p 3P multiplet
Mechanism	helium recombination transition with collisional and radiative-transfer corrections
Primary tracers	He+/H+ abundance; hardness above 24.6 eV; nebular physical conditions with other He I lines
Typical environments	H II regions; planetary nebulae; hot-star winds; AGN line regions
Indicative regime	photoionized gas around 5000-20000 K
Density guidance	He I line ratios respond to electron density, optical depth in triplet lines and underlying absorption; multi-line fitting is required.

He I 5877 is produced by helium recombination transition with collisional and radiative-transfer corrections and is encountered in H II regions; planetary nebulae; hot-star winds; AGN line regions. It is commonly used to trace He+/H+ abundance; hardness above 24.6 eV; nebular physical conditions with other He I lines.

How to use it

Use this transition with its diagnostic companions to constrain He+/H+ abundance; hardness above 24.6 eV; nebular physical conditions with other He I lines, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is about 5875.62 A. Correct for stellar absorption, Na sky features and triplet optical-depth effects.

Useful companions

He I 6679 A; He I 7067 A; He I 10833 A; H-beta

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.46 [O III] 5008

At a glance	
Rest coordinate	5008.24 angstrom
Band	optical / green
Transition	1D2 -> 3P2
Mechanism	forbidden collisionally excited transition
Primary tracers	high-excitation ionized gas; AGN power statistically; metallicity/ionization with other lines; kinematics
Typical environments	H II regions; planetary nebulae; AGN narrow-line regions; ionized outflows
Indicative regime	photoionized gas about 7000-20000 K; shocks can extend higher
Density guidance	Critical density is about 7x10^5 cm^-3; the nebular-to-auroral ratio with 4364 A is strongly temperature sensitive.

[O III] 5008 is produced by forbidden collisionally excited transition and is encountered in H II regions; planetary nebulae; AGN narrow-line regions; ionized outflows. It is commonly used to trace high-excitation ionized gas; AGN power statistically; metallicity/ionization with other lines; kinematics.

How to use it

Use this transition with its diagnostic companions to constrain high-excitation ionized gas; AGN power statistically; metallicity/ionization with other lines; kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 5006.84 A. Use the theoretical 5008/4960 ratio near 2.98 to test fits and saturation.

Useful companions

[O III] 4960 A fixed branching partner; [O III] 4364 A temperature line; H-beta for BPT/R3; [O II] for O32/R23

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.47 [O III] 4960

At a glance	
Rest coordinate	4960.3 angstrom
Band	optical / blue-green
Transition	1D2 -> 3P1
Mechanism	forbidden collisionally excited transition
Primary tracers	O++ abundance; high-excitation gas; quality control with 5008
Typical environments	H II regions; planetary nebulae; AGN narrow-line regions
Indicative regime	photoionized gas about 7000-20000 K
Density guidance	Same upper level and critical-density behavior as 5008 A.

[O III] 4960 is produced by forbidden collisionally excited transition and is encountered in H II regions; planetary nebulae; AGN narrow-line regions. It is commonly used to trace O++ abundance; high-excitation gas; quality control with 5008.

How to use it

Use this transition with its diagnostic companions to constrain O++ abundance; high-excitation gas; quality control with 5008, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 4958.91 A. Tie centroid, width and theoretical flux ratio to 5008 in multi-component fits when justified.

Useful companions

[O III] 5008 A; [O III] 4364 A; H-beta

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.48 H-beta

At a glance

Rest coordinate	4862.68 angstrom
Band	optical / blue
Transition	Balmer n=4 -> 2
Mechanism	hydrogen recombination with possible collisional contribution
Primary tracers	recombination rate; dust reddening with H-alpha; normalization for excitation and abundance ratios
Typical environments	H II regions; star-forming galaxies; planetary nebulae; AGN line regions
Indicative regime	photoionized gas near 5000-20000 K
Density guidance	Case-B ratios depend weakly on density over normal nebular conditions but change with temperature, optical depth and collisions.

H-beta is produced by hydrogen recombination with possible collisional contribution and is encountered in H II regions; star-forming galaxies; planetary nebulae; AGN line regions. It is commonly used to trace recombination rate; dust reddening with H-alpha; normalization for excitation and abundance ratios.

How to use it

Use this transition with its diagnostic companions to constrain recombination rate; dust reddening with H-alpha; normalization for excitation and abundance ratios, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 4861.33 A. Stellar Balmer absorption can severely bias weak emission and extinction estimates.

Useful companions

H-alpha Balmer decrement; [O III] 5008/H-beta; H-gamma absorption/emission correction

Sources

`NIST_ASD; KEWLEY_2006; OSTERBROCK_FERLAND_2006`

14.49 He II 4687

At a glance	
Rest coordinate	4687.02 angstrom
Band	optical / blue
Transition	hydrogenic n=4 -> 3
Mechanism	recombination transition of He+
Primary tracers	photons above 54.4 eV; very hard ionizing spectra; He++ abundance
Typical environments	planetary nebulae; Wolf-Rayet populations; X-ray binaries; AGN; very low-metallicity star-forming regions; fast shocks
Indicative regime	photoionized gas commonly 10000-30000 K; shock zones can be hotter
Density guidance	Recombination emissivity rather than a critical-density diagnostic; line width separates nebular and stellar components.

He II 4687 is produced by recombination transition of He+ and is encountered in planetary nebulae; Wolf-Rayet populations; X-ray binaries; AGN; very low-metallicity star-forming regions; fast shocks. It is commonly used to trace photons above 54.4 eV; very hard ionizing spectra; He++ abundance.

How to use it

Use this transition with its diagnostic companions to constrain photons above 54.4 eV; very hard ionizing spectra; He++ abundance, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 4685.7 A. Broad stellar and narrow nebular He II require decomposition before inferring ionizing hardness.

Useful companions

He II 1640 A; H-beta; [Ne V] or high-ionization lines; Wolf-Rayet blue bump

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.50 [O II] 3727

At a glance

Rest coordinate	3727.09 angstrom
Band	optical / near-UV
Transition	2D3/2 -> 4S3/2 (traditional 3726 component)
Mechanism	forbidden collisionally excited transition
Primary tracers	electron density with 3729; O+ abundance; star formation after calibration; ionization parameter with [O III]
Typical environments	H II regions; star-forming galaxies; diffuse ionized gas; AGN narrow-line regions
Indicative regime	photoionized gas roughly 7000-20000 K
Density guidance	Critical density is roughly 1.4x10^4 cm^-3 near 10^4 K; doublet ratio spans common nebular densities.

[O II] 3727 is produced by forbidden collisionally excited transition and is encountered in H II regions; star-forming galaxies; diffuse ionized gas; AGN narrow-line regions. It is commonly used to trace electron density with 3729; O+ abundance; star formation after calibration; ionization parameter with [O III].

How to use it

Use this transition with its diagnostic companions to constrain electron density with 3729; O+ abundance; star formation after calibration; ionization parameter with [O III], while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; traditional air wavelength is 3726.03 A. Unresolved surveys often quote [O II] 3727 for the sum—record that convention explicitly.

Useful companions

[O II] 3729.88 A; [O III] 5008 A for O32/R23; [O II] 7320/7330 A temperature lines

Sources

`NIST_ASD; KENNICUTT_EVANS_2012; OSTERBROCK_FERLAND_2006`

14.51 [O II] 3729

At a glance	
Rest coordinate	3729.88 angstrom
Band	optical / near-UV
Transition	2D5/2 -> 4S3/2 (traditional 3729 component)
Mechanism	forbidden collisionally excited transition
Primary tracers	electron density with 3727; O+ abundance; low-ionization gas
Typical environments	H II regions; star-forming galaxies; diffuse ionized gas; AGN narrow-line regions
Indicative regime	photoionized gas roughly 7000-20000 K
Density guidance	Critical density is roughly 3x10^3 cm^-3 near 10^4 K; ratio to 3727 saturates outside its useful density range.

[O II] 3729 is produced by forbidden collisionally excited transition and is encountered in H II regions; star-forming galaxies; diffuse ionized gas; AGN narrow-line regions. It is commonly used to trace electron density with 3727; O+ abundance; low-ionization gas.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 3727; O+ abundance; low-ionization gas, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; traditional air wavelength is 3728.82 A. Instrumental resolution and stellar continuum modeling determine whether the doublet yields useful densities.

Useful companions

[O II] 3727.09 A; [O III] 5008 A; [S II] 6718/6733 A

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.52 [Ne III] 3869

At a glance

Rest coordinate	3869.86 angstrom
Band	optical / violet
Transition	1D2 -> 3P2
Mechanism	forbidden collisionally excited transition
Primary tracers	hard ionizing spectrum; Ne++ abundance; ionization parameter with [O II] or [Ne II]
Typical environments	high-excitation H II regions; planetary nebulae; AGN narrow-line regions
Indicative regime	photoionized gas around 8000-20000 K
Density guidance	Critical density is high, of order 10^7 cm^-3; generally in the low-density regime for H II regions.

[Ne III] 3869 is produced by forbidden collisionally excited transition and is encountered in high-excitation H II regions; planetary nebulae; AGN narrow-line regions. It is commonly used to trace hard ionizing spectrum; Ne++ abundance; ionization parameter with [O II] or [Ne II].

How to use it

Use this transition with its diagnostic companions to constrain hard ionizing spectrum; Ne++ abundance; ionization parameter with [O II] or [Ne II], while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Vacuum value; familiar air wavelength is 3868.76 A. The weaker 3968-A partner blends with H-epsilon/Ca H in many spectra.

Useful companions

[Ne III] 3968 A branching partner; [O II] 3727+3729; [Ne III] 15.6 micron

Sources

`NIST_ASD; OSTERBROCK_FERLAND_2006`

14.53 Mg II 2796

At a glance	
Rest coordinate	2796.352 angstrom
Band	near-ultraviolet
Transition	3s 2S1/2 -> 3p 2P3/2
Mechanism	permitted resonance transition
Primary tracers	cool photoionized gas; kinematics and covering fraction; outflows/inflows; magnesium column with saturation modeling
Typical environments	cool CGM; intervening absorbers; galactic winds; stellar chromospheres; quasar broad-line regions
Indicative regime	commonly about 10^4 K photoionized gas; not a unique thermometer
Density guidance	Resonance absorption is column/ionization sensitive, not a critical-density diagnostic; emission is shaped by resonant scattering.

Mg II 2796 is produced by permitted resonance transition and is encountered in cool CGM; intervening absorbers; galactic winds; stellar chromospheres; quasar broad-line regions. It is commonly used to trace cool photoionized gas; kinematics and covering fraction; outflows/inflows; magnesium column with saturation modeling.

How to use it

Use this transition with its diagnostic companions to constrain cool photoionized gas; kinematics and covering fraction; outflows/inflows; magnesium column with saturation modeling, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Stronger doublet member with oscillator-strength ratio near 2. Saturation and partial covering drive the observed doublet ratio toward unity.

Useful companions

Mg II 2803.530 A; Mg I 2852.96 A; Fe II UV lines

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.54 Mg II 2803

At a glance	
Rest coordinate	2803.53 angstrom
Band	near-ultraviolet
Transition	3s 2S1/2 -> 3p 2P1/2
Mechanism	permitted resonance transition
Primary tracers	cool gas column and kinematics; saturation/covering with 2796; outflows and inflows
Typical environments	cool CGM; intervening absorbers; galactic winds; stellar chromospheres; quasar broad-line regions
Indicative regime	commonly about 10^4 K photoionized gas
Density guidance	Not a direct density diagnostic; its value comes from doublet curve-of-growth and covering analysis.

Mg II 2803 is produced by permitted resonance transition and is encountered in cool CGM; intervening absorbers; galactic winds; stellar chromospheres; quasar broad-line regions. It is commonly used to trace cool gas column and kinematics; saturation/covering with 2796; outflows and inflows.

How to use it

Use this transition with its diagnostic companions to constrain cool gas column and kinematics; saturation/-covering with 2796; outflows and inflows, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Weaker doublet member is essential for diagnosing saturation. Resonant emission infill can bias absorption equivalent widths.

Useful companions

Mg II 2796.352 A; Mg I 2852.96 A; Fe II 2600.17 A

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.55 C III] 1907

At a glance	
Rest coordinate	1906.683 angstrom
Band	ultraviolet
Transition	2s2p 3P2 -> 2s2 1S0
Mechanism	semi-forbidden intercombination transition
Primary tracers	electron density with 1909; carbon abundance and ionization; hard young stellar populations
Typical environments	H II regions; high-redshift star-forming galaxies; AGN narrow/broad-line regions; planetary nebulae
Indicative regime	photoionized gas around 10000-30000 K
Density guidance	The 1907/1909 ratio is density sensitive from low nebular densities into roughly 10^5-10^6 cm^-3, with isotope/atomic-data nuances.

C III] 1907 is produced by semi-forbidden intercombination transition and is encountered in H II regions; high-redshift star-forming galaxies; AGN narrow/broad-line regions; planetary nebulae. It is commonly used to trace electron density with 1909; carbon abundance and ionization; hard young stellar populations.

How to use it

Use this transition with its diagnostic companions to constrain electron density with 1909; carbon abundance and ionization; hard young stellar populations, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Usually unresolved at modest resolution and reported as C III] 1908. AGN broad-line density can suppress or alter the doublet.

Useful companions

C III] 1908.734 A; O III] 1661/1666 A; He II 1640 A; C IV 1548/1551 A

Sources

`NIST_ASD; MORTON_2003; OSTERBROCK_FERLAND_2006`

14.56 C III] 1909

At a glance

Rest coordinate	1908.734 angstrom
Band	ultraviolet
Transition	2s2p 3P1 -> 2s2 1S0
Mechanism	semi-forbidden intercombination transition
Primary tracers	electron density; carbon abundance; ionizing conditions
Typical environments	H II regions; high-redshift galaxies; AGN line regions; planetary nebulae
Indicative regime	photoionized gas around 10000-30000 K
Density guidance	Pair with 1907; the ratio approaches atomic low-density and high-density limits and must not be extrapolated beyond them.

C III] 1909 is produced by semi-forbidden intercombination transition and is encountered in H II regions; high-redshift galaxies; AGN line regions; planetary nebulae. It is commonly used to trace electron density; carbon abundance; ionizing conditions.

How to use it

Use this transition with its diagnostic companions to constrain electron density; carbon abundance; ionizing conditions, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Blends with 1907 at common survey resolution; density, C/O, ionization parameter and stellar/AGN continuum are degenerate in equivalent width.

Useful companions

C III] 1906.683 A; Si III] 1883/1892 A; O III] 1666 A

Sources

`NIST_ASD; MORTON_2003; OSTERBROCK_FERLAND_2006`

14.57 C IV 1548

At a glance	
Rest coordinate	1548.204 angstrom
Band	far-ultraviolet
Transition	2s 2S1/2 -> 2p 2P3/2
Mechanism	permitted resonance transition
Primary tracers	high-ionization gas; metal-enriched CGM; outflow kinematics; AGN/stars when in emission
Typical environments	warm CGM/IGM; quasar absorbers; AGN broad-line regions; hot-star winds; transition-temperature plasma
Indicative regime	CIE abundance peaks near 10^5 K; photoionized C IV commonly occurs near 10^4 K
Density guidance	Absorption constrains ion column and ionization, not density alone; emission can be resonantly scattered and collisionally excited.

C IV 1548 is produced by permitted resonance transition and is encountered in warm CGM/IGM; quasar absorbers; AGN broad-line regions; hot-star winds; transition-temperature plasma. It is commonly used to trace high-ionization gas; metal-enriched CGM; outflow kinematics; AGN/stars when in emission.

How to use it

Use this transition with its diagnostic companions to constrain high-ionization gas; metal-enriched CGM; outflow kinematics; AGN/stars when in emission, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Stronger doublet member. Saturation, unresolved components and partial covering are major column-density systematics; stellar P-Cygni and nebular/AGN emission can blend.

Useful companions

C IV 1550.781 A; Si IV 1394/1403 A; N V 1239/1243 A; O VI 1032/1038 A

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.58 C IV 1551

At a glance

Rest coordinate	1550.781 angstrom
Band	far-ultraviolet
Transition	2s 2S1/2 -> 2p 2P1/2
Mechanism	permitted resonance transition
Primary tracers	doublet saturation and covering; high-ionization gas kinematics; metal enrichment
Typical environments	warm CGM/IGM; quasar absorbers; AGN; hot-star winds
Indicative regime	CIE peak near 10^5 K; also common in cooler photoionized gas
Density guidance	No simple critical-density diagnostic in absorption; line ratio tests saturation and radiative transfer.

C IV 1551 is produced by permitted resonance transition and is encountered in warm CGM/IGM; quasar absorbers; AGN; hot-star winds. It is commonly used to trace doublet saturation and covering; high-ionization gas kinematics; metal enrichment.

How to use it

Use this transition with its diagnostic companions to constrain doublet saturation and covering; high-ionization gas kinematics; metal enrichment, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Weaker member has about half the oscillator strength of 1548. Intrinsic AGN doublet blending depends on line width.

Useful companions

C IV 1548.204 A; Si IV doublet; O VI doublet

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.59 Si IV 1394

At a glance	
Rest coordinate	1393.755 angstrom
Band	far-ultraviolet
Transition	3s 2S1/2 -> 3p 2P3/2
Mechanism	permitted resonance transition
Primary tracers	intermediate/high-ionization gas; outflow kinematics; silicon enrichment; stellar-wind profiles
Typical environments	photoionized CGM; galactic winds; IGM absorbers; hot-star winds; transition-temperature interfaces
Indicative regime	CIE abundance peaks around 6-8x10^4 K; photoionized gas can be near 10^4 K
Density guidance	Primarily column/ionization sensitive; the doublet tests saturation and covering fraction.

Si IV 1394 is produced by permitted resonance transition and is encountered in photoionized CGM; galactic winds; IGM absorbers; hot-star winds; transition-temperature interfaces. It is commonly used to trace intermediate/high-ionization gas; outflow kinematics; silicon enrichment; stellar-wind profiles.

How to use it

Use this transition with its diagnostic companions to constrain intermediate/high-ionization gas; outflow kinematics; silicon enrichment; stellar-wind profiles, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Stronger doublet member; stellar-wind contamination and unresolved saturation are common in galaxy spectra.

Useful companions

Si IV 1402.770 A; C IV doublet; Si II 1260/1527 A

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.60 Si IV 1403

At a glance

Rest coordinate	1402.77 angstrom
Band	far-ultraviolet
Transition	3s 2S1/2 -> 3p 2P1/2
Mechanism	permitted resonance transition
Primary tracers	doublet saturation; ionized-gas kinematics; covering fraction
Typical environments	photoionized CGM; galactic winds; IGM absorbers; hot-star winds
Indicative regime	CIE peak around 6-8x10^4 K, or cooler photoionized gas
Density guidance	Not a direct density probe; doublet ratio and profile constrain opacity and covering.

Si IV 1403 is produced by permitted resonance transition and is encountered in photoionized CGM; galactic winds; IGM absorbers; hot-star winds. It is commonly used to trace doublet saturation; ionized-gas kinematics; covering fraction.

How to use it

Use this transition with its diagnostic companions to constrain doublet saturation; ionized-gas kinematics; covering fraction, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Weaker member helps reveal saturation; nearby interstellar/stellar features must be modeled in integrated spectra.

Useful companions

Si IV 1393.755 A; C IV doublet; O VI doublet

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.61 C II 1334

At a glance

Rest coordinate	1334.5323 angstrom
Band	far-ultraviolet
Transition	2s2 2p 2P1/2 -> 2s2 3s 2S1/2
Mechanism	permitted resonance transition
Primary tracers	low-ionization gas column and kinematics; metallicity with ionization/depletion corrections; outflow covering fraction
Typical environments	neutral and weakly ionized ISM; cool CGM; damped Ly-alpha systems; galactic outflows
Indicative regime	typically cool/warm gas around 10^2-10^4 K
Density guidance	Ground-state absorption is column sensitive; excited C II* 1335 lines provide density/cooling constraints when separately measured.

C II 1334 is produced by permitted resonance transition and is encountered in neutral and weakly ionized ISM; cool CGM; damped Ly-alpha systems; galactic outflows. It is commonly used to trace low-ionization gas column and kinematics; metallicity with ionization/depletion corrections; outflow covering fraction.

How to use it

Use this transition with its diagnostic companions to constrain low-ionization gas column and kinematics; metallicity with ionization/depletion corrections; outflow covering fraction, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Often saturated even when not black because of unresolved velocity structure or partial covering; emission infill can reduce apparent absorption.

Useful companions

C II* 1335.66/1335.71 A; Si II 1260 A; O I 1302 A; [C II] 158 micron

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.62 H I Ly-alpha

At a glance

Rest coordinate	1215.6701 angstrom
Band	far-ultraviolet
Transition	Lyman n=2 -> 1 (fine-structure blend)
Mechanism	permitted resonant transition; recombination/collisional emission and absorption
Primary tracers	neutral-hydrogen column; IGM density field; reionization topology statistically; gas kinematics through radiative-transfer modeling
Typical environments	H II regions and galaxies; CGM; Ly-alpha forest IGM; damped Ly-alpha systems; AGN
Indicative regime	neutral/photoionized gas around 10^4 K is common, but absorption spans broad thermal histories
Density guidance	Not a direct density line; forest statistics and damping wings constrain density/column after ionization and radiative-transfer modeling.

H I Ly-alpha is produced by permitted resonant transition; recombination/collisional emission and absorption and is encountered in H II regions and galaxies; CGM; Ly-alpha forest IGM; damped Ly-alpha systems; AGN. It is commonly used to trace neutral-hydrogen column; IGM density field; reionization topology statistically; gas kinematics through radiative-transfer modeling.

How to use it

Use this transition with its diagnostic companions to constrain neutral-hydrogen column; IGM density field; reionization topology statistically; gas kinematics through radiative-transfer modeling, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Resonant scattering, dust and IGM absorption decouple observed flux/profile from intrinsic production. Geocoronal Ly-alpha contaminates low-redshift work.

Useful companions

Ly-beta 1025.72 A; H I Lyman limit; metal absorption lines; H-alpha under recombination assumptions

Sources

`NIST_ASD; DRAINE_2011; TUMLINSON_2017`

14.63 O VI 1032

At a glance	
Rest coordinate	1031.9261 angstrom
Band	far-ultraviolet
Transition	2s 2S1/2 -> 2p 2P3/2
Mechanism	permitted resonance transition
Primary tracers	highly ionized oxygen; transition-temperature gas; feedback and halo baryons; outflow kinematics
Typical environments	warm-hot CGM/IGM; interfaces and mixing layers; radiative shocks; galactic winds; hot-star winds
Indicative regime	CIE abundance peaks near 3x10^5 K; photoionized O VI may occur at much lower density and temperature
Density guidance	Absorption gives column rather than density; combine with H I, path length, aligned ions and models. Emission scales strongly with density.

O VI 1032 is produced by permitted resonance transition and is encountered in warm-hot CGM/IGM; interfaces and mixing layers; radiative shocks; galactic winds; hot-star winds. It is commonly used to trace highly ionized oxygen; transition-temperature gas; feedback and halo baryons; outflow kinematics.

How to use it

Use this transition with its diagnostic companions to constrain highly ionized oxygen; transition-temperature gas; feedback and halo baryons; outflow kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Stronger member. H2 and interstellar blends, saturation, non-equilibrium cooling and multiphase alignment complicate temperature inference.

Useful companions

O VI 1037.6167 A; H I Ly-beta nearby; C IV/Ne VIII; X-ray O VII

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.64 O VI 1038

At a glance	
Rest coordinate	1037.6167 angstrom
Band	far-ultraviolet
Transition	2s 2S1/2 -> 2p 2P1/2
Mechanism	permitted resonance transition
Primary tracers	O VI saturation/covering; transition-temperature gas; high-ionization kinematics
Typical environments	warm-hot CGM/IGM; interfaces; shocks; winds
Indicative regime	CIE peak near 3x10^5 K, with viable photoionized alternatives
Density guidance	No direct density measurement; doublet consistency reveals saturation and blends.

O VI 1038 is produced by permitted resonance transition and is encountered in warm-hot CGM/IGM; interfaces; shocks; winds. It is commonly used to trace O VI saturation/covering; transition-temperature gas; high-ionization kinematics.

How to use it

Use this transition with its diagnostic companions to constrain O VI saturation/covering; transition-temperature gas; high-ionization kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Weaker member is frequently blended with C II 1036, C II* and Galactic H2; fit all contaminants jointly.

Useful companions

O VI 1031.9261 A; C II 1036.34 A; H2 absorption lines; Ne VIII doublet

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.65 Ne VIII 770

At a glance	
Rest coordinate	770.409 angstrom
Band	extreme/far ultraviolet
Transition	2s 2S1/2 -> 2p 2P3/2
Mechanism	permitted resonance transition
Primary tracers	hotter ionized gas than O VI; warm-hot baryons; multiphase halo interfaces
Typical environments	warm-hot CGM/IGM; quasar absorbers; conductive/turbulent interfaces; AGN outflows
Indicative regime	CIE abundance peaks near 6-7x10^5 K; low-density photoionized solutions require long path lengths
Density guidance	Column/ionization diagnostic, not a direct density line; combine with O VI, H I and adjacent ions.

Ne VIII 770 is produced by permitted resonance transition and is encountered in warm-hot CGM/IGM; quasar absorbers; conductive/turbulent interfaces; AGN outflows. It is commonly used to trace hotter ionized gas than O VI; warm-hot baryons; multiphase halo interfaces.

How to use it

Use this transition with its diagnostic companions to constrain hotter ionized gas than O VI; warm-hot baryons; multiphase halo interfaces, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Usually accessible only after redshifting above the Lyman limit/instrument cutoff. Forest blending and continuum placement dominate weak detections.

Useful companions

Ne VIII 780.324 A; O VI 1032/1038 A; Mg X 610/625 A; X-ray O VII

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.66 Ne VIII 780

At a glance	
Rest coordinate	780.324 angstrom
Band	extreme/far ultraviolet
Transition	2s 2S1/2 -> 2p 2P1/2
Mechanism	permitted resonance transition
Primary tracers	doublet confirmation; warm-hot ionized gas; saturation
Typical environments	warm-hot CGM/IGM; quasar absorbers; interfaces; AGN outflows
Indicative regime	CIE peak near 6-7x10^5 K; photoionized alternatives possible
Density guidance	Not direct; weaker doublet member constrains saturation and false-positive blends.

Ne VIII 780 is produced by permitted resonance transition and is encountered in warm-hot CGM/IGM; quasar absorbers; interfaces; AGN outflows. It is commonly used to trace doublet confirmation; warm-hot ionized gas; saturation.

How to use it

Use this transition with its diagnostic companions to constrain doublet confirmation; warm-hot ionized gas; saturation, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Require velocity-coincident detection with the stronger member when possible; the far-UV forest yields a significant interloper rate.

Useful companions

Ne VIII 770.409 A; O VI doublet; X-ray O VII/O VIII

Sources

`NIST_ASD; MORTON_2003; TUMLINSON_2017`

14.67 O VII He-alpha resonance (w)

At a glance	
Rest coordinate	21.6019 angstrom
Band	soft X-ray
Transition	1s2 1S0 -> 1s2p 1P1 resonance line
Mechanism	He-like resonance absorption/emission, usually collisional or photoionized plasma
Primary tracers	million-kelvin plasma; missing baryons; hot-halo column; temperature/ionization with O VIII
Typical environments	hot CGM/IGM; galaxy groups and clusters; stellar coronae; supernova remnants; AGN warm absorbers
Indicative regime	CIE abundance peaks around 2x10^6 K; photoionized O VII can occur at lower kinetic temperature
Density guidance	The resonance line alone is not a density meter; the forbidden/intercombination ratio in the full He-like triplet is density and UV-field sensitive.

O VII He-alpha resonance (w) is produced by He-like resonance absorption/emission, usually collisional or photoionized plasma and is encountered in hot CGM/IGM; galaxy groups and clusters; stellar coronae; supernova remnants; AGN warm absorbers. It is commonly used to trace million-kelvin plasma; missing baryons; hot-halo column; temperature/ionization with O VIII.

How to use it

Use this transition with its diagnostic companions to constrain million-kelvin plasma; missing baryons; hot-halo column; temperature/ionization with O VIII, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Weak absorption requires high spectral resolution and careful treatment of instrumental features, Galactic foreground, look-elsewhere effects and wavelength calibration.

Useful companions

O VII intercombination near 21.80 A; O VII forbidden 22.10 A; O VIII 18.967 A; O VI UV doublet

Sources

`ATOMDB; NIST_ASD; TUMLINSON_2017`

14.68 O VIII Ly-alpha1

At a glance	
Rest coordinate	18.96711 angstrom
Band	soft X-ray
Transition	2p 2P3/2 -> 1s 2S1/2
Mechanism	hydrogenic resonance transition
Primary tracers	several-million-kelvin gas; ionization temperature with O VII; hot-gas kinematics
Typical environments	hot CGM/IGM; galaxy groups and clusters; stellar coronae; supernova remnants; AGN warm absorbers
Indicative regime	CIE abundance peaks around 3x10^6 K; photoionized gas is also common
Density guidance	Primarily temperature/ionization and column sensitive; density requires metastable lines, emission measure or photoionization modeling.

O VIII Ly-alpha1 is produced by hydrogenic resonance transition and is encountered in hot CGM/IGM; galaxy groups and clusters; stellar coronae; supernova remnants; AGN warm absorbers. It is commonly used to trace several-million-kelvin gas; ionization temperature with O VII; hot-gas kinematics.

How to use it

Use this transition with its diagnostic companions to constrain several-million-kelvin gas; ionization temperature with O VII; hot-gas kinematics, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The Ly-alpha doublet is often unresolved. Resonant scattering can suppress central emission in optically thick hot atmospheres.

Useful companions

O VIII Ly-alpha2 18.9725 A; O VII triplet near 21.6-22.1 A; O VIII Ly-beta

Sources

`ATOMDB; NIST_ASD`

14.69 Fe XVII 15.01

At a glance	
Rest coordinate	15.014 angstrom
Band	soft X-ray
Transition	2p6 1S0 -> 2p5 3d 1P1 (3C resonance line)
Mechanism	collisionally excited resonance emission
Primary tracers	roughly 5 MK plasma; resonant scattering/turbulence using Fe XVII ratios; cooling gas
Typical environments	stellar coronae; elliptical-galaxy atmospheres; cool-core clusters; supernova remnants
Indicative regime	CIE emissivity is strong around 4-7x10^6 K
Density guidance	The 15.01/17.05-17.10 complex is more sensitive to resonant scattering and atomic physics than ordinary low-density plasma density; some Fe XVII ratios gain density sensitivity at high densities.

Fe XVII 15.01 is produced by collisionally excited resonance emission and is encountered in stellar coronae; elliptical-galaxy atmospheres; cool-core clusters; supernova remnants. It is commonly used to trace roughly 5 MK plasma; resonant scattering/turbulence using Fe XVII ratios; cooling gas.

How to use it

Use this transition with its diagnostic companions to constrain roughly 5 MK plasma; resonant scattering/turbulence using Fe XVII ratios; cooling gas, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Resonant scattering lowers the central 15.01-A intensity and can constrain turbulence, but plasma-code and line-blend uncertainties must be propagated.

Useful companions

Fe XVII 15.26 A; Fe XVII 16.78/17.05/17.10 A; O VIII 18.97 A

Sources

`ATOMDB; NIST_ASD`

14.70 Ne IX He-alpha resonance (w)

At a glance	
Rest coordinate	13.4473 angstrom
Band	soft X-ray
Transition	1s2 1S0 -> 1s2p 1P1
Mechanism	He-like resonance transition
Primary tracers	few-million-kelvin plasma; ionization state with Ne X; plasma excitation with triplet ratios
Typical environments	stellar coronae; supernova remnants; hot galactic gas; AGN photoionized plasma
Indicative regime	CIE abundance/emissivity is strong around 4x10^6 K
Density guidance	Use the forbidden/intercombination components of the Ne IX triplet for density/UV-field sensitivity; w alone is not sufficient.

Ne IX He-alpha resonance (w) is produced by He-like resonance transition and is encountered in stellar coronae; supernova remnants; hot galactic gas; AGN photoionized plasma. It is commonly used to trace few-million-kelvin plasma; ionization state with Ne X; plasma excitation with triplet ratios.

How to use it

Use this transition with its diagnostic companions to constrain few-million-kelvin plasma; ionization state with Ne X; plasma excitation with triplet ratios, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Fe XIX-XXI blends can contaminate the triplet in hot plasmas; use a full plasma model at finite spectral resolution.

Useful companions

Ne IX intercombination near 13.55 A; Ne IX forbidden 13.70 A; Ne X Ly-alpha 12.13 A

Sources

`ATOMDB; NIST_ASD`

14.71 Fe XXV He-alpha resonance (w)

At a glance	
Rest coordinate	1.8504 angstrom
Band	hard X-ray / Fe K complex
Transition	1s2 1S0 -> 1s2p 1P1
Mechanism	He-like resonance transition in highly ionized plasma
Primary tracers	very hot thermal plasma; bulk/turbulent velocity; ionization temperature; resonant scattering
Typical environments	galaxy clusters; accreting compact objects; solar/stellar flares; supernova remnants; AGN winds
Indicative regime	CIE line emissivity is strong at several 10^7 K
Density guidance	The resolved He-like complex and satellite ratios diagnose temperature, ionization age and at high density density; CCD-blended centroids are model dependent.

Fe XXV He-alpha resonance (w) is produced by He-like resonance transition in highly ionized plasma and is encountered in galaxy clusters; accreting compact objects; solar/stellar flares; supernova remnants; AGN winds. It is commonly used to trace very hot thermal plasma; bulk/turbulent velocity; ionization temperature; resonant scattering.

How to use it

Use this transition with its diagnostic companions to constrain very hot thermal plasma; bulk/turbulent velocity; ionization temperature; resonant scattering, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Quote whether energy refers to the w line or an unresolved Fe XXV complex. Gain calibration and plasma-code choices set velocity/temperature systematics.

Useful companions

Fe XXV x,y,z components; dielectronic satellite lines; Fe XXVI Ly-alpha; neutral Fe K-alpha

Sources

`ATOMDB; NIST_ASD`

14.72 Fe XXVI Ly-alpha1

At a glance	
Rest coordinate	1.778 angstrom
Band	hard X-ray / Fe K complex
Transition	2p 2P3/2 -> 1s 2S1/2
Mechanism	hydrogenic resonance transition
Primary tracers	extreme ionization; very hot plasma; ultrafast outflow velocity; ionization temperature with Fe XXV
Typical environments	hottest cluster plasma; accreting black-hole and neutron-star winds; solar/stellar flares; photoionized AGN
Indicative regime	CIE fraction becomes important around 10^8 K; photoionized Fe XXVI also occurs
Density guidance	Primarily ionization/temperature and column sensitive; density comes from broader photoionization or time-dependent modeling.

Fe XXVI Ly-alpha1 is produced by hydrogenic resonance transition and is encountered in hottest cluster plasma; accreting black-hole and neutron-star winds; solar/stellar flares; photoionized AGN. It is commonly used to trace extreme ionization; very hot plasma; ultrafast outflow velocity; ionization temperature with Fe XXV.

How to use it

Use this transition with its diagnostic companions to constrain extreme ionization; very hot plasma; ultrafast outflow velocity; ionization temperature with Fe XXV, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Often unresolved as a doublet; line identification in fast AGN winds can be degenerate with Fe XXV at a different velocity.

Useful companions

Fe XXVI Ly-alpha2 1.7835 A; Fe XXV 1.8504 A; Fe XXVI Ly-beta

Sources

`ATOMDB; NIST_ASD`

14.73 neutral Fe K-alpha1

At a glance	
Rest coordinate	6.4040062 keV
Band	hard X-ray
Transition	K-L3 fluorescence, 2p3/2 -> 1s vacancy
Mechanism	K-shell fluorescence following photoionization or energetic-particle impact
Primary tracers	cold reflection; reprocessing geometry; black-hole spin through relativistic broadening with a disk model; past X-ray illumination
Typical environments	AGN tori and accretion disks; X-ray binaries; molecular clouds irradiated by X-rays; stellar flares
Indicative regime	cold or weakly ionized material illuminated by photons above the Fe K edge; not a thermal-excitation temperature line
Density guidance	Equivalent width and Compton shoulder constrain column, abundance and geometry rather than local density uniquely.

neutral Fe K-alpha1 is produced by K-shell fluorescence following photoionization or energetic-particle impact and is encountered in AGN tori and accretion disks; X-ray binaries; molecular clouds irradiated by X-rays; stellar flares. It is commonly used to trace cold reflection; reprocessing geometry; black-hole spin through relativistic broadening with a disk model; past X-ray illumination.

How to use it

Use this transition with its diagnostic companions to constrain cold reflection; reprocessing geometry; black-hole spin through relativistic broadening with a disk model; past X-ray illumination, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The common '6.4 keV line' is a K-alpha doublet and shifts with ionization. Broad profiles require continuum, absorption and relativistic-reflection systematics to be modeled jointly.

Useful companions

Fe K-alpha2 6.391026 keV; Fe K-beta near 7.06 keV; Fe K edge; Compton reflection hump; Fe XXV/XXVI

Sources

`NIST_FE_K; NIST_XRAY`

14.74 electron-positron annihilation line

At a glance	
Rest coordinate	511 keV
Band	gamma ray
Transition	two-photon annihilation at the electron rest-mass energy
Mechanism	direct annihilation and para-positronium two-photon decay
Primary tracers	positron production and transport; annihilation phase of the ISM; compact-source activity statistically
Typical environments	Galactic bulge and disk ISM; compact-object environments; radioactive-ejecta environments
Indicative regime	line width and positronium fraction probe warm neutral/ionized annihilation media; production is not tied to gas temperature
Density guidance	No critical density; annihilation timescale and positronium fraction depend on phase, ionization fraction, grains and density.

electron-positron annihilation line is produced by direct annihilation and para-positronium two-photon decay and is encountered in Galactic bulge and disk ISM; compact-object environments; radioactive-ejecta environments. It is commonly used to trace positron production and transport; annihilation phase of the ISM; compact-source activity statistically.

How to use it

Use this transition with its diagnostic companions to constrain positron production and transport; annihilation phase of the ISM; compact-source activity statistically, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

The centroid is fixed by rest mass, while astrophysical broadening and the positronium continuum constrain thermalization/annihilation conditions. Diffuse-background modeling is central.

Useful companions

three-photon positronium continuum below 511 keV; 26Al 1808.65 keV; 60Fe 1173/1333 keV

Sources

`NASA_511`

14.75 26Al decay gamma-ray line

At a glance	
Rest coordinate	1808.65 keV
Band	gamma ray
Transition	de-excitation of the 1808.65-keV state of 26Mg
Mechanism	radioactive beta-plus/electron-capture decay followed by nuclear gamma de-excitation
Primary tracers	ongoing nucleosynthesis; recent massive-star activity over Myr timescales; hot-ejecta kinematics and Galactic rotation
Typical environments	Galactic plane; massive-star associations; superbubbles; core-collapse supernova and Wolf-Rayet ejecta
Indicative regime	decay is independent of thermal excitation; emitting isotope is transported through hot superbubbles and ISM phases
Density guidance	No critical density; intensity traces isotope mass and decay, while centroid/width trace bulk motion and turbulence.

26Al decay gamma-ray line is produced by radioactive beta-plus/electron-capture decay followed by nuclear gamma de-excitation and is encountered in Galactic plane; massive-star associations; superbubbles; core-collapse supernova and Wolf-Rayet ejecta. It is commonly used to trace ongoing nucleosynthesis; recent massive-star activity over Myr timescales; hot-ejecta kinematics and Galactic rotation.

How to use it

Use this transition with its diagnostic companions to constrain ongoing nucleosynthesis; recent massive-star activity over Myr timescales; hot-ejecta kinematics and Galactic rotation, while fitting the continuum, line-spread function, and relevant blends.

Interpretive cautions

Laboratory energy is 1808.65 +/- 0.07 keV. Instrumental background, broad response and diffuse-sky modeling dominate; the mean lifetime is about 1.04 Myr.

Useful companions

60Fe 1173 and 1333 keV; 511 keV annihilation; massive-star tracers and free-free emission

Sources

`SAMWORTH_1972; DIEHL_2021`

Appendix A

Constants and Conversions

Quantity	Symbol	Value
Speed of light	c	$2.997\,924\,58 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Planck constant	h	$6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,s}$
Boltzmann constant	k_{B}	$1.380\,649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Electron volt	eV	$1.602\,176\,634 \times 10^{-19}\,\mathrm{J}$
Parsec	pc	$3.085\,677\,581 \times 10^{16}\,\mathrm{m}$

Useful photon conversions are

$$E = h\nu = \frac{hc}{\lambda}, \tag{A.1}$$

$$E\,[\mathrm{eV}] \simeq \frac{12398.42}{\lambda\,[\mathrm{\AA}]}, \tag{A.2}$$

$$\lambda_{\mathrm{obs}} = (1+z)\lambda_{\mathrm{rest}}. \tag{A.3}$$

Appendix B

Diagnostic Recipes and Sanity Checks

1. Confirm vacuum versus air wavelength convention before matching lines.
2. Propagate continuum placement and covariance, not only per-pixel noise.
3. Correct for instrumental broadening before interpreting velocity width.
4. Inspect doublet ratios for saturation, blending, or radiative-transfer effects.
5. Treat a temperature label as a regime, not a unique origin condition.
6. Compare multiple ionization stages before adopting an ionization correction.
7. Report atomic data, abundance scale, attenuation law, and model grid version.

Appendix C

Glossary

Critical density The collider density at which collisional de-excitation competes with spontaneous radiative decay for a specified transition and temperature.

Equivalent width
The continuum-normalized integrated strength of an emission or absorption feature, with sign convention stated explicitly.

Ionization parameter
A dimensionless ratio of ionizing-photon density to hydrogen density; conventions differ by geometry and definition.

Oscillator strength
A dimensionless measure of a bound–bound transition probability, commonly appearing as f or gf.

Resonance line A transition connected to the ground state, often strong and susceptible to absorption and multiple scattering.

Ritz wavelength A wavelength derived from optimized upper and lower energy levels rather than directly from a measured line centroid.

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